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TENTH STANDARD

PART-II



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POLYNOMIALS

9

9.1 Introduction

In Class IX, you have studied polynomials in one variable and their degrees. Recall that if $p(x)$ is a polynomial in x , the highest power of x in $p(x)$ is called **the degree of the polynomial** $p(x)$. For example, $4x + 2$ is a polynomial in the variable x of degree 1, $2y^2 - 3y + 4$ is a polynomial in the variable y of degree 2, $5x^3 - 4x^2 + x - \sqrt{2}$ is a polynomial in the variable x of degree 3 and $7u^6 - \frac{3}{2}u^4 + 4u^2 + u - 8$ is a polynomial

in the variable u of degree 6. Expressions like $\frac{1}{x-1}$, $\sqrt{x} + 2$, $\frac{1}{x^2 + 2x + 3}$ etc., are not polynomials.

A polynomial of degree 1 is called a **linear polynomial**. For example, $2x - 3$, $\sqrt{3}x + 5$, $y + \sqrt{2}$, $x - \frac{2}{11}$, $3z + 4$, $\frac{2}{3}u + 1$, etc., are all linear polynomials. Polynomials such as $2x + 5 - x^2$, $x^3 + 1$, etc., are not linear polynomials.

A polynomial of degree 2 is called a **quadratic polynomial**. The name 'quadratic' has been derived from the word 'quadrate', which means 'square'. $2x^2 + 3x - \frac{2}{5}$,

$y^2 - 2$, $2 - x^2 + \sqrt{3}x$, $\frac{u}{3} - 2u^2 + 5$, $\sqrt{5}v^2 - \frac{2}{3}v$, $4z^2 + \frac{1}{7}$ are some examples of quadratic polynomials (whose coefficients are real numbers). More generally, any quadratic polynomial in x is of the form $ax^2 + bx + c$, where a , b , c are real numbers and $a \neq 0$. A polynomial of degree 3 is called a **cubic polynomial**. Some examples of

a cubic polynomial are $2 - x^3$, x^3 , $\sqrt{2}x^3$, $3 - x^2 + x^3$, $3x^3 - 2x^2 + x - 1$. In fact, the most general form of a cubic polynomial is

$$ax^3 + bx^2 + cx + d,$$

where, a, b, c, d are real numbers and $a \neq 0$.

Now consider the polynomial $p(x) = x^2 - 3x - 4$. Then, putting $x = 2$ in the polynomial, we get $p(2) = 2^2 - 3 \times 2 - 4 = -6$. The value ‘-6’, obtained by replacing x by 2 in $x^2 - 3x - 4$, is the value of $x^2 - 3x - 4$ at $x = 2$. Similarly, $p(0)$ is the value of $p(x)$ at $x = 0$, which is -4.

If $p(x)$ is a polynomial in x , and if k is any real number, then the value obtained by replacing x by k in $p(x)$, is called **the value of $p(x)$ at $x = k$** , and is denoted by $p(k)$.

What is the value of $p(x) = x^2 - 3x - 4$ at $x = -1$? We have :

$$p(-1) = (-1)^2 - \{3 \times (-1)\} - 4 = 0$$

Also, note that $p(4) = 4^2 - (3 \times 4) - 4 = 0$.

As $p(-1) = 0$ and $p(4) = 0$, -1 and 4 are called the zeroes of the quadratic polynomial $x^2 - 3x - 4$. More generally, a real number k is said to be a **zero of a polynomial $p(x)$** , if $p(k) = 0$.

You have already studied in Class IX, how to find the zeroes of a linear polynomial. For example, if k is a zero of $p(x) = 2x + 3$, then $p(k) = 0$ gives us $2k + 3 = 0$, i.e., $k = -\frac{3}{2}$.

In general, if k is a zero of $p(x) = ax + b$, then $p(k) = ak + b = 0$, i.e., $k = \frac{-b}{a}$.

So, the zero of the linear polynomial $ax + b$ is $\frac{-b}{a} = \frac{-(\text{Constant term})}{\text{Coefficient of } x}$.

Thus, the zero of a linear polynomial is related to its coefficients. Does this happen in the case of other polynomials too? For example, are the zeroes of a quadratic polynomial also related to its coefficients?

In this chapter, we will try to answer these questions. We will also study the division algorithm for polynomials.

9.2 Geometrical Meaning of the Zeroes of a Polynomial

You know that a real number k is a zero of the polynomial $p(x)$ if $p(k) = 0$. But why are the zeroes of a polynomial so important? To answer this, first we will see the **geometrical** representations of linear and quadratic polynomials and the geometrical meaning of their zeroes.

Consider first a linear polynomial $ax + b$, $a \neq 0$. You have studied in Class IX that the graph of $y = ax + b$ is a straight line. For example, the graph of $y = 2x + 3$ is a straight line passing through the points $(-2, -1)$ and $(2, 7)$.

x	-2	2
$y = 2x + 3$	-1	7

From Fig. 9.1, you can see that the graph of $y = 2x + 3$ intersects the x -axis mid-way between $x = -1$ and $x = -2$, that is, at the point $\left(-\frac{3}{2}, 0\right)$.

You also know that the zero of $2x + 3$ is $-\frac{3}{2}$. Thus, the zero of the polynomial $2x + 3$ is the x -coordinate of the point where the graph of $y = 2x + 3$ intersects the x -axis.

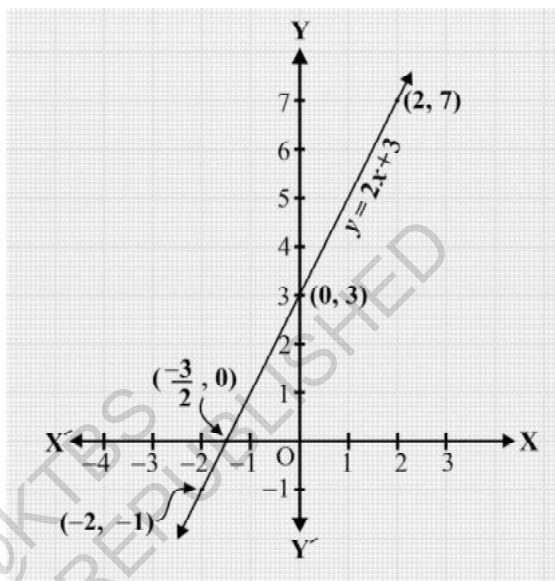


Fig. 9.1

In general, for a linear polynomial $ax + b$, $a \neq 0$, the graph of $y = ax + b$ is a straight line which intersects the x -axis at exactly one point, namely, $\left(\frac{-b}{a}, 0\right)$.

Therefore, the linear polynomial $ax + b$, $a \neq 0$, has exactly one zero, namely, the x -coordinate of the point where the graph of $y = ax + b$ intersects the x -axis.

Now, let us look for the geometrical meaning of a zero of a quadratic polynomial. Consider the quadratic polynomial $x^2 - 3x - 4$. Let us see what the graph* of $y = x^2 - 3x - 4$ looks like. Let us list a few values of $y = x^2 - 3x - 4$ corresponding to a few values for x as given in Table 9.1.

* Plotting of graphs of quadratic or cubic polynomials is not meant to be done by the students, nor is to be evaluated.

Table 9.1

x	-2	-1	0	1	2	3	4	5
$y = x^2 - 3x - 4$	6	0	-4	-6	-6	-4	0	6

If we locate the points listed above on a graph paper and draw the graph, it will actually look like the one given in Fig. 9.2.

In fact, for any quadratic polynomial $ax^2 + bx + c$, $a \neq 0$, the graph of the corresponding equation $y = ax^2 + bx + c$ has one of the two shapes either open upwards like $\cup$ or open downwards like $\cap$ depending on whether $a > 0$ or $a < 0$. (These curves are called **parabolas**.)

You can see from Table 9.1 that -1 and 4 are zeroes of the quadratic polynomial. Also note from Fig. 9.2 that -1 and 4 are the x -coordinates of the points where the graph of $y = x^2 - 3x - 4$ intersects the x -axis. Thus, the zeroes of the quadratic polynomial $x^2 - 3x - 4$ are x -coordinates of the points where the graph of $y = x^2 - 3x - 4$ intersects the x -axis.

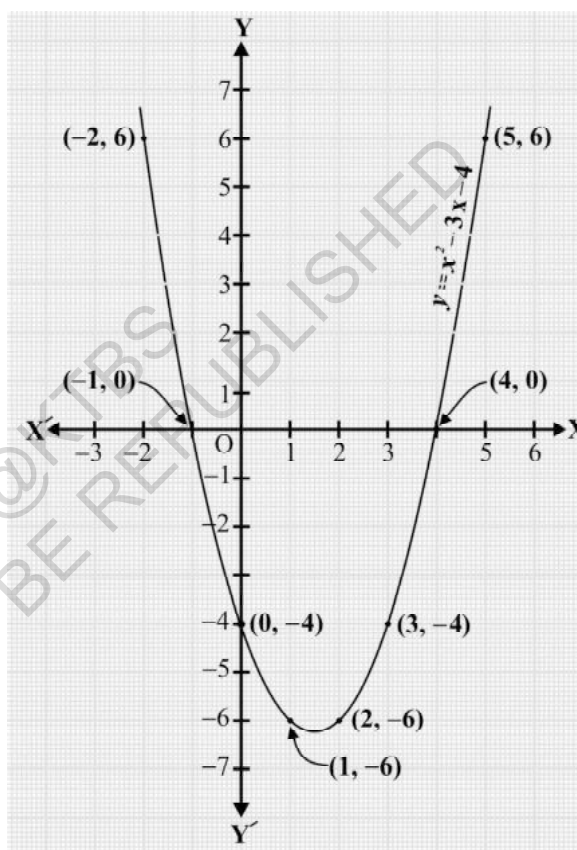


Fig. 9.2

This fact is true for any quadratic polynomial, i.e., the zeroes of a quadratic polynomial $ax^2 + bx + c$, $a \neq 0$, are precisely the x -coordinates of the points where the parabola representing $y = ax^2 + bx + c$ intersects the x -axis.

From our observation earlier about the shape of the graph of $y = ax^2 + bx + c$, the following three cases can happen:

Case (i) : Here, the graph cuts x -axis at two distinct points A and A'.

The x -coordinates of A and A' are the **two zeroes** of the quadratic polynomial $ax^2 + bx + c$ in this case (see Fig. 9.3).

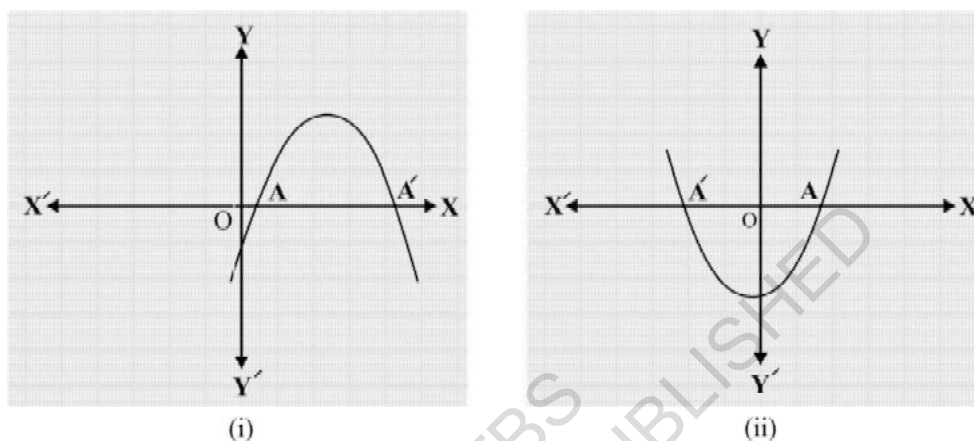


Fig. 9.3

Case (ii) : Here, the graph cuts the x -axis at exactly one point, i.e., at two coincident points. So, the two points A and A' of Case (i) coincide here to become one point A (see Fig. 9.4).

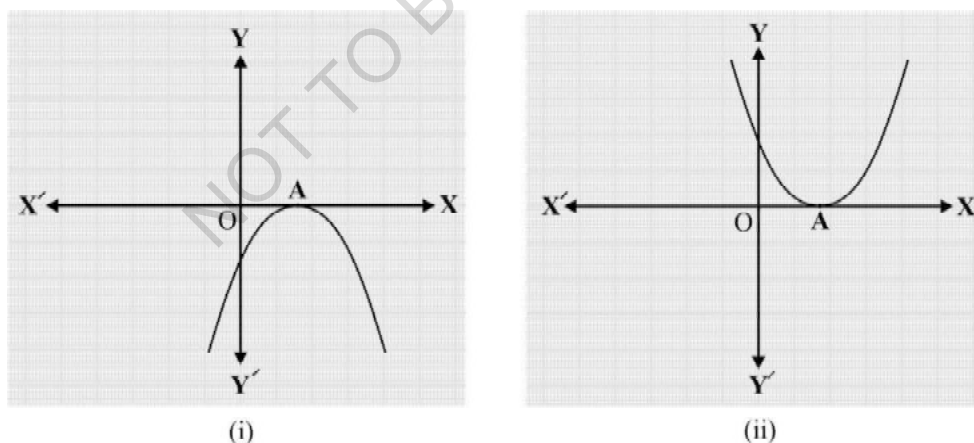


Fig. 9.4

The x -coordinate of A is the **only zero** for the quadratic polynomial $ax^2 + bx + c$ in this case.

Case (iii) : Here, the graph is either completely above the x -axis or completely below the x -axis. So, it does not cut the x -axis at any point (see Fig. 9.5).

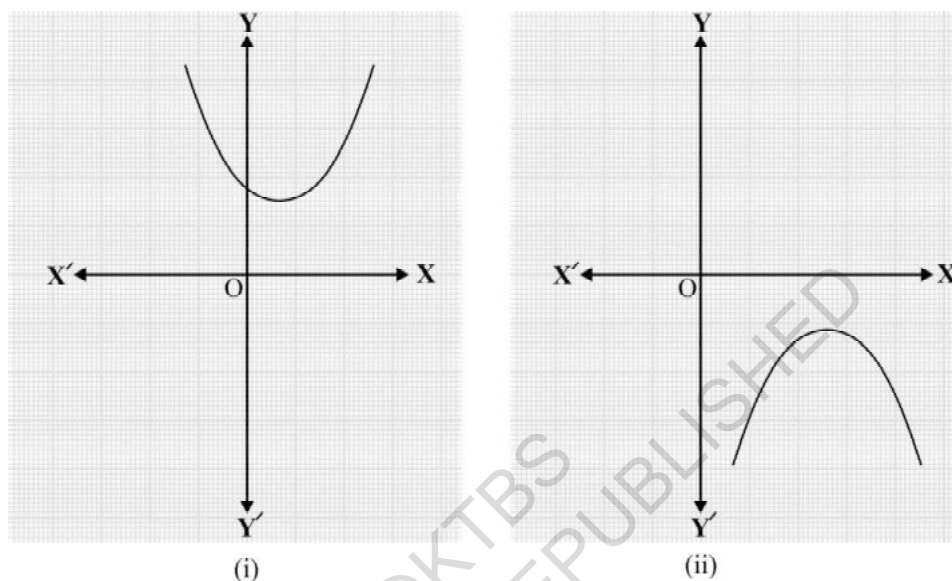


Fig. 9.5

So, the quadratic polynomial $ax^2 + bx + c$ has **no zero** in this case.

So, you can see geometrically that a quadratic polynomial can have either two distinct zeroes or two equal zeroes (i.e., one zero), or no zero. This also means that a polynomial of degree 2 has atmost two zeroes.

Now, what do you expect the geometrical meaning of the zeroes of a cubic polynomial to be? Let us find out. Consider the cubic polynomial $x^3 - 4x$. To see what the graph of $y = x^3 - 4x$ looks like, let us list a few values of y corresponding to a few values for x as shown in Table 9.2.

Table 9.2

x	-2	-1	0	1	2
$y = x^3 - 4x$	0	3	0	-3	0

Locating the points of the table on a graph paper and drawing the graph, we see that the graph of $y = x^3 - 4x$ actually looks like the one given in Fig. 9.6.

We see from the table above that -2 , 0 and 2 are zeroes of the cubic polynomial $x^3 - 4x$. Observe that -2 , 0 and 2 are, in fact, the x -coordinates of the only points where the graph of $y = x^3 - 4x$ intersects the x -axis. Since the curve meets the x -axis in only these 3 points, their x -coordinates are the only zeroes of the polynomial.

Let us take a few more examples. Consider the cubic polynomials x^3 and $x^3 - x^2$. We draw the graphs of $y = x^3$ and $y = x^3 - x^2$ in Fig. 9.7 and Fig. 9.8 respectively.

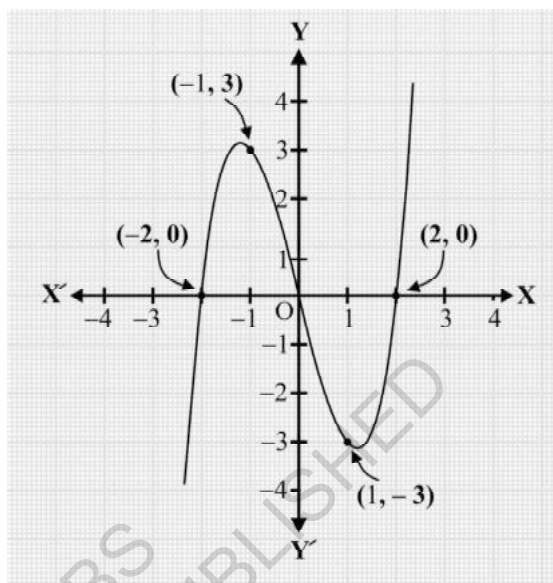


Fig. 9.6

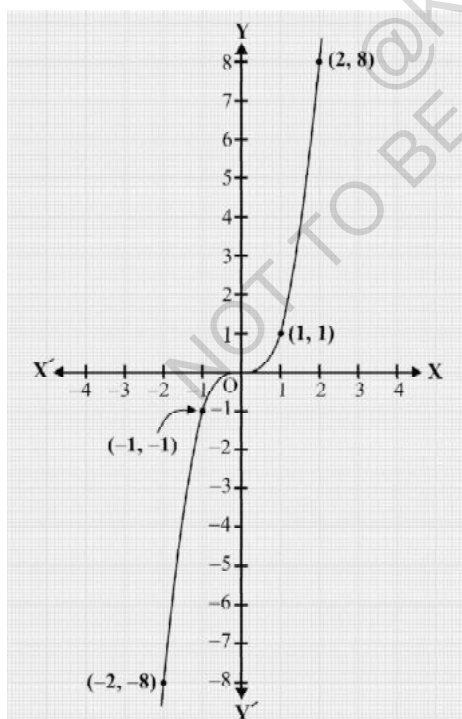


Fig. 9.7

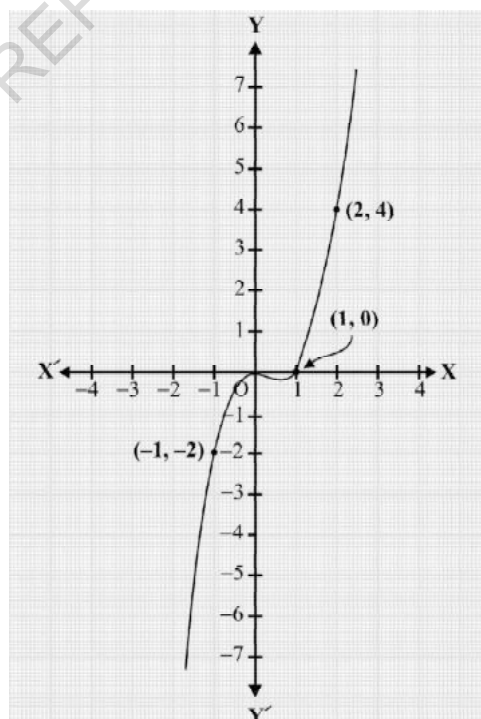


Fig. 9.8

Note that 0 is the only zero of the polynomial x^3 . Also, from Fig. 9.7, you can see that 0 is the x -coordinate of the only point where the graph of $y = x^3$ intersects the x -axis. Similarly, since $x^3 - x^2 = x^2(x - 1)$, 0 and 1 are the only zeroes of the polynomial $x^3 - x^2$. Also, from Fig. 9.8, these values are the x -coordinates of the only points where the graph of $y = x^3 - x^2$ intersects the x -axis.

From the examples above, we see that there are at most 3 zeroes for any cubic polynomial. In other words, any polynomial of degree 3 can have at most three zeroes.

Remark : In general, given a polynomial $p(x)$ of degree n , the graph of $y = p(x)$ intersects the x -axis at at most n points. Therefore, a polynomial $p(x)$ of degree n has at most n zeroes.

Example 1 : Look at the graphs in Fig. 9.9 given below. Each is the graph of $y = p(x)$, where $p(x)$ is a polynomial. For each of the graphs, find the number of zeroes of $p(x)$.

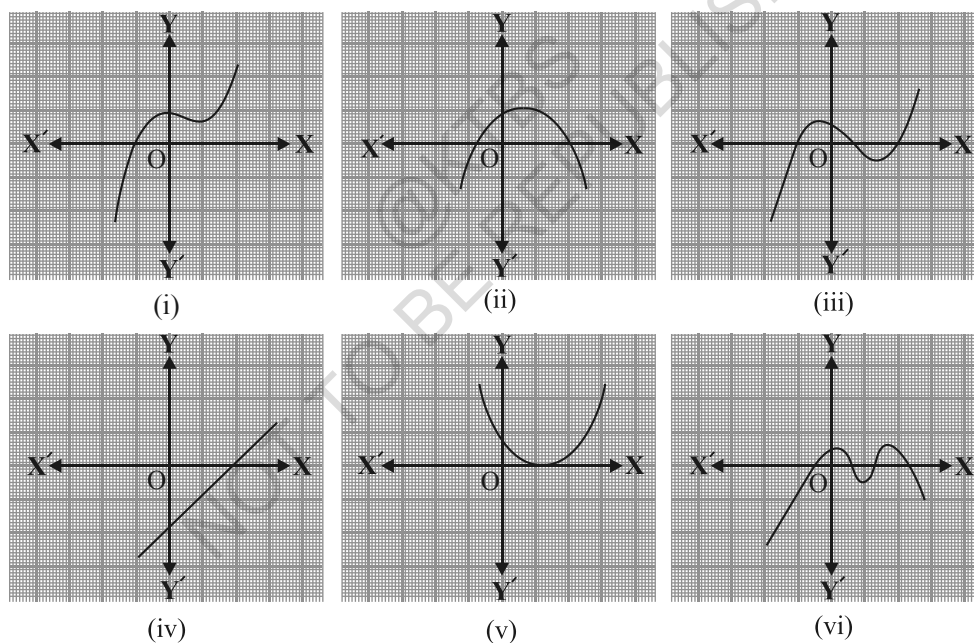


Fig. 9.9

Solution :

- (i) The number of zeroes is 1 as the graph intersects the x -axis at one point only.
- (ii) The number of zeroes is 2 as the graph intersects the x -axis at two points.
- (iii) The number of zeroes is 3. (Why?)

- (iv) The number of zeroes is 1. (Why?)
- (v) The number of zeroes is 1. (Why?)
- (vi) The number of zeroes is 4. (Why?)

EXERCISE 9.1

- The graphs of $y = p(x)$ are given in Fig. 9.10 below, for some polynomials $p(x)$. Find the number of zeroes of $p(x)$, in each case.

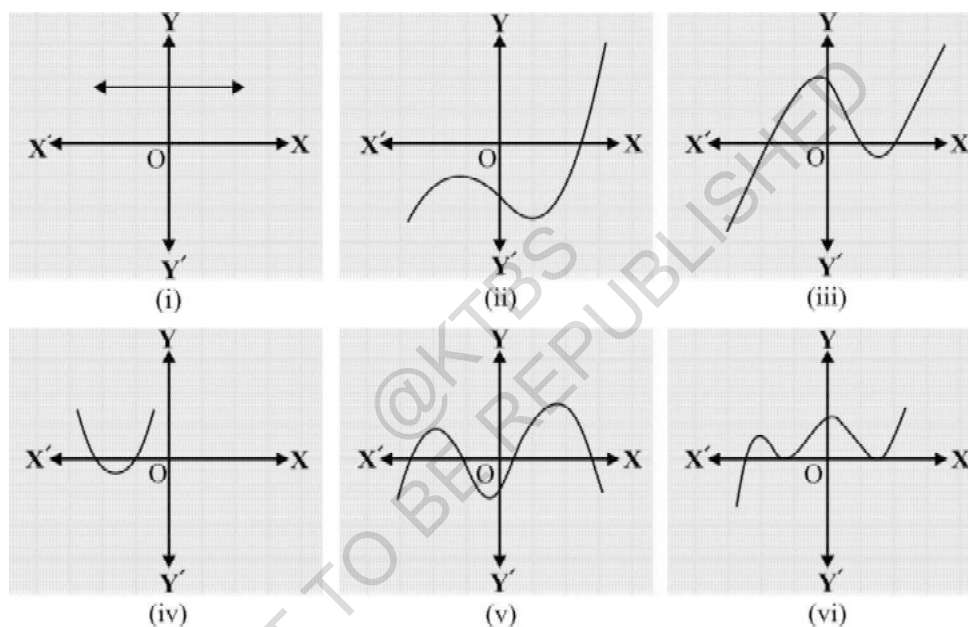


Fig. 9.10

9.3 Relationship between Zeroes and Coefficients of a Polynomial

You have already seen that zero of a linear polynomial $ax + b$ is $-\frac{b}{a}$. We will now try to answer the question raised in Section 9.1 regarding the relationship between zeroes and coefficients of a quadratic polynomial. For this, let us take a quadratic polynomial, say $p(x) = 2x^2 - 8x + 6$. In Class IX, you have learnt how to factorise quadratic polynomials by splitting the middle term. So, here we need to split the middle term $-8x$ as a sum of two terms, whose product is $6 \times 2x^2 = 12x^2$. So, we write

$$\begin{aligned} 2x^2 - 8x + 6 &= 2x^2 - 6x - 2x + 6 = 2x(x - 3) - 2(x - 3) \\ &= (2x - 2)(x - 3) = 2(x - 1)(x - 3) \end{aligned}$$

So, the value of $p(x) = 2x^2 - 8x + 6$ is zero when $x - 1 = 0$ or $x - 3 = 0$, i.e., when $x = 1$ or $x = 3$. So, the zeroes of $2x^2 - 8x + 6$ are 1 and 3. Observe that :

$$\text{Sum of its zeroes} = 1 + 3 = 4 = \frac{-(-8)}{2} = \frac{-(\text{Coefficient of } x)}{\text{Coefficient of } x^2}$$

$$\text{Product of its zeroes} = 1 \times 3 = 3 = \frac{6}{2} = \frac{\text{Constant term}}{\text{Coefficient of } x^2}$$

Let us take one more quadratic polynomial, say, $p(x) = 3x^2 + 5x - 2$. By the method of splitting the middle term,

$$\begin{aligned} 3x^2 + 5x - 2 &= 3x^2 + 6x - x - 2 = 3x(x + 2) - 1(x + 2) \\ &= (3x - 1)(x + 2) \end{aligned}$$

Hence, the value of $3x^2 + 5x - 2$ is zero when either $3x - 1 = 0$ or $x + 2 = 0$, i.e., when $x = \frac{1}{3}$ or $x = -2$. So, the zeroes of $3x^2 + 5x - 2$ are $\frac{1}{3}$ and -2 . Observe that :

$$\text{Sum of its zeroes} = \frac{1}{3} + (-2) = \frac{-5}{3} = \frac{-(\text{Coefficient of } x)}{\text{Coefficient of } x^2}$$

$$\text{Product of its zeroes} = \frac{1}{3} \times (-2) = \frac{-2}{3} = \frac{\text{Constant term}}{\text{Coefficient of } x^2}$$

In general, if α^* and β^* are the zeroes of the quadratic polynomial $p(x) = ax^2 + bx + c$, $a \neq 0$, then you know that $x - \alpha$ and $x - \beta$ are the factors of $p(x)$. Therefore,

$$\begin{aligned} ax^2 + bx + c &= k(x - \alpha)(x - \beta), \text{ where } k \text{ is a constant} \\ &= k[x^2 - (\alpha + \beta)x + \alpha\beta] \\ &= kx^2 - k(\alpha + \beta)x + k\alpha\beta \end{aligned}$$

Comparing the coefficients of x^2 , x and constant terms on both the sides, we get

$$a = k, b = -k(\alpha + \beta) \text{ and } c = k\alpha\beta.$$

This gives $\alpha + \beta = \frac{-b}{a},$

$$\alpha\beta = \frac{c}{a}$$

* α, β are Greek letters pronounced as 'alpha' and 'beta' respectively. We will use later one more letter ' γ ' pronounced as 'gamma'.

i.e.,
$$\text{sum of zeroes} = \alpha + \beta = -\frac{b}{a} = \frac{-(\text{Coefficient of } x)}{\text{Coefficient of } x^2},$$

$$\text{product of zeroes} = \alpha\beta = \frac{c}{a} = \frac{\text{Constant term}}{\text{Coefficient of } x^2}.$$

Let us consider some examples.

Example 2 : Find the zeroes of the quadratic polynomial $x^2 + 7x + 10$, and verify the relationship between the zeroes and the coefficients.

Solution : We have

$$x^2 + 7x + 10 = (x + 2)(x + 5)$$

So, the value of $x^2 + 7x + 10$ is zero when $x + 2 = 0$ or $x + 5 = 0$, i.e., when $x = -2$ or $x = -5$. Therefore, the zeroes of $x^2 + 7x + 10$ are -2 and -5 . Now,

$$\text{sum of zeroes} = -2 + (-5) = -(7) = \frac{-(7)}{1} = \frac{-(\text{Coefficient of } x)}{\text{Coefficient of } x^2},$$

$$\text{product of zeroes} = (-2) \times (-5) = 10 = \frac{10}{1} = \frac{\text{Constant term}}{\text{Coefficient of } x^2}.$$

Example 3 : Find the zeroes of the polynomial $x^2 - 3$ and verify the relationship between the zeroes and the coefficients.

Solution : Recall the identity $a^2 - b^2 = (a - b)(a + b)$. Using it, we can write:

$$x^2 - 3 = (x - \sqrt{3})(x + \sqrt{3})$$

So, the value of $x^2 - 3$ is zero when $x = \sqrt{3}$ or $x = -\sqrt{3}$.

Therefore, the zeroes of $x^2 - 3$ are $\sqrt{3}$ and $-\sqrt{3}$.

Now,

$$\text{sum of zeroes} = \sqrt{3} - \sqrt{3} = 0 = \frac{-(\text{Coefficient of } x)}{\text{Coefficient of } x^2},$$

$$\text{product of zeroes} = (\sqrt{3})(-\sqrt{3}) = -3 = \frac{-3}{1} = \frac{\text{Constant term}}{\text{Coefficient of } x^2}.$$

Example 4 : Find a quadratic polynomial, the sum and product of whose zeroes are -3 and 2 , respectively.

Solution : Let the quadratic polynomial be $ax^2 + bx + c$, and its zeroes be α and β . We have

$$\alpha + \beta = -3 = \frac{-b}{a},$$

and
$$\alpha\beta = 2 = \frac{c}{a}.$$

If $a = 1$, then $b = 3$ and $c = 2$.

So, one quadratic polynomial which fits the given conditions is $x^2 + 3x + 2$.

You can check that any other quadratic polynomial that fits these conditions will be of the form $k(x^2 + 3x + 2)$, where k is real.

Let us now look at cubic polynomials. Do you think a similar relation holds between the zeroes of a cubic polynomial and its coefficients?

Let us consider $p(x) = 2x^3 - 5x^2 - 14x + 8$.

You can check that $p(x) = 0$ for $x = 4, -2, \frac{1}{2}$. Since $p(x)$ can have at most three zeroes, these are the zeroes of $2x^3 - 5x^2 - 14x + 8$. Now,

$$\text{sum of the zeroes} = 4 + (-2) + \frac{1}{2} = \frac{5}{2} = \frac{-(-5)}{2} = \frac{-(\text{Coefficient of } x^2)}{\text{Coefficient of } x^3},$$

$$\text{product of the zeroes} = 4 \times (-2) \times \frac{1}{2} = -4 = \frac{-8}{2} = \frac{-\text{Constant term}}{\text{Coefficient of } x^3}.$$

However, there is one more relationship here. Consider the sum of the products of the zeroes taken two at a time. We have

$$\begin{aligned} & \{4 \times (-2)\} + \left\{(-2) \times \frac{1}{2}\right\} + \left\{\frac{1}{2} \times 4\right\} \\ &= -8 - 1 + 2 = -7 = \frac{-14}{2} = \frac{\text{Coefficient of } x}{\text{Coefficient of } x^3}. \end{aligned}$$

In general, it can be proved that if α, β, γ are the zeroes of the cubic polynomial $ax^3 + bx^2 + cx + d$, then

$$\begin{aligned}\alpha + \beta + \gamma &= \frac{-b}{a}, \\ \alpha\beta + \beta\gamma + \gamma\alpha &= \frac{c}{a}, \\ \alpha\beta\gamma &= \frac{-d}{a}.\end{aligned}$$

Let us consider an example.

Example 5* : Verify that 3, -1, $-\frac{1}{3}$ are the zeroes of the cubic polynomial $p(x) = 3x^3 - 5x^2 - 11x - 3$, and then verify the relationship between the zeroes and the coefficients.

Solution : Comparing the given polynomial with $ax^3 + bx^2 + cx + d$, we get

$a = 3, b = -5, c = -11, d = -3$. Further

$$p(3) = 3 \times 3^3 - (5 \times 3^2) - (11 \times 3) - 3 = 81 - 45 - 33 - 3 = 0,$$

$$p(-1) = 3 \times (-1)^3 - 5 \times (-1)^2 - 11 \times (-1) - 3 = -3 - 5 + 11 - 3 = 0,$$

$$\begin{aligned}p\left(-\frac{1}{3}\right) &= 3 \times \left(-\frac{1}{3}\right)^3 - 5 \times \left(-\frac{1}{3}\right)^2 - 11 \times \left(-\frac{1}{3}\right) - 3, \\ &= -\frac{1}{9} - \frac{5}{9} + \frac{11}{3} - 3 = -\frac{2}{3} + \frac{2}{3} = 0\end{aligned}$$

Therefore, 3, -1 and $-\frac{1}{3}$ are the zeroes of $3x^3 - 5x^2 - 11x - 3$.

So, we take $\alpha = 3, \beta = -1$ and $\gamma = -\frac{1}{3}$.

Now,

$$\alpha + \beta + \gamma = 3 + (-1) + \left(-\frac{1}{3}\right) = 2 - \frac{1}{3} = \frac{5}{3} = \frac{-(-5)}{3} = \frac{-b}{a},$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = 3 \times (-1) + (-1) \times \left(-\frac{1}{3}\right) + \left(-\frac{1}{3}\right) \times 3 = -3 + \frac{1}{3} - 1 = \frac{-11}{3} = \frac{c}{a},$$

$$\alpha\beta\gamma = 3 \times (-1) \times \left(-\frac{1}{3}\right) = 1 = \frac{-(-3)}{3} = \frac{-d}{a}.$$

* Not from the examination point of view.

EXERCISE 9.2

1. Find the zeroes of the following quadratic polynomials and verify the relationship between the zeroes and the coefficients.

(i) $x^2 - 2x - 8$

(ii) $4s^2 - 4s + 1$

(iii) $6x^2 - 3 - 7x$

(iv) $4u^2 + 8u$

(v) $t^2 - 15$

(vi) $3x^2 - x - 4$

2. Find a quadratic polynomial each with the given numbers as the sum and product of its zeroes respectively.

(i) $\frac{1}{4}, -1$

(ii) $\sqrt{2}, \frac{1}{3}$

(iii) $0, \sqrt{5}$

(iv) $1, 1$

(v) $-\frac{1}{4}, \frac{1}{4}$

(vi) $4, 1$

9.4 Division Algorithm for Polynomials

You know that a cubic polynomial has at most three zeroes. However, if you are given only one zero, can you find the other two? For this, let us consider the cubic polynomial $x^3 - 3x^2 - x + 3$. If we tell you that one of its zeroes is 1, then you know that $x - 1$ is a factor of $x^3 - 3x^2 - x + 3$. So, you can divide $x^3 - 3x^2 - x + 3$ by $x - 1$, as you have learnt in Class IX, to get the quotient $x^2 - 2x - 3$.

Next, you could get the factors of $x^2 - 2x - 3$, by splitting the middle term, as $(x + 1)(x - 3)$. This would give you

$$\begin{aligned} x^3 - 3x^2 - x + 3 &= (x - 1)(x^2 - 2x - 3) \\ &= (x - 1)(x + 1)(x - 3) \end{aligned}$$

So, all the three zeroes of the cubic polynomial are now known to you as 1, -1, 3.

Let us discuss the method of dividing one polynomial by another in some detail. Before noting the steps formally, consider an example.

Example 6 : Divide $2x^2 + 3x + 1$ by $x + 2$.

Solution : Note that we stop the division process when either the remainder is zero or its degree is less than the degree of the divisor. So, here the quotient is $2x - 1$ and the remainder is 3. Also,

$$\begin{aligned} (2x - 1)(x + 2) + 3 &= 2x^2 + 3x - 2 + 3 = 2x^2 + 3x + 1 \\ \text{i.e., } 2x^2 + 3x + 1 &= (x + 2)(2x - 1) + 3 \end{aligned}$$

Therefore, Dividend = Divisor $\times$ Quotient + Remainder

Let us now extend this process to divide a polynomial by a quadratic polynomial.

$$\begin{array}{r} \overline{2x-1} \\ x+2 \overline{) 2x^2+3x+1} \\ \underline{2x^2+4x} \\ -x+1 \\ \underline{+x+2} \\ 3 \end{array}$$

Example 7 : Divide $3x^3 + x^2 + 2x + 5$ by $1 + 2x + x^2$.

Solution : We first arrange the terms of the dividend and the divisor in the decreasing order of their degrees. Recall that arranging the terms in this order is called writing the polynomials in standard form. In this example, the dividend is already in standard form, and the divisor, in standard form, is $x^2 + 2x + 1$.

$$\begin{array}{r}
 3x - 5 \\
 x^2 + 2x + 1 \overline{) 3x^3 + x^2 + 2x + 5} \\
 \underline{3x^3 + 6x^2 + 3x} \\
 -5x^2 - x + 5 \\
 \underline{-5x^2 - 10x - 5} \\
 9x + 10
 \end{array}$$

Step 1 : To obtain the first term of the quotient, divide the highest degree term of the dividend (i.e., $3x^3$) by the highest degree term of the divisor (i.e., x^2). This is $3x$. Then carry out the division process. What remains is $-5x^2 - x + 5$.

Step 2 : Now, to obtain the second term of the quotient, divide the highest degree term of the new dividend (i.e., $-5x^2$) by the highest degree term of the divisor (i.e., x^2). This gives -5 . Again carry out the division process with $-5x^2 - x + 5$.

Step 3 : What remains is $9x + 10$. Now, the degree of $9x + 10$ is less than the degree of the divisor $x^2 + 2x + 1$. So, we cannot continue the division any further.

So, the quotient is $3x - 5$ and the remainder is $9x + 10$. Also,

$$\begin{aligned}
 (x^2 + 2x + 1) \times (3x - 5) + (9x + 10) &= 3x^3 + 6x^2 + 3x - 5x^2 - 10x - 5 + 9x + 10 \\
 &= 3x^3 + x^2 + 2x + 5
 \end{aligned}$$

Here again, we see that

$$\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder}$$

What we are applying here is an algorithm which is similar to Euclid's division algorithm that you studied in Chapter 1.

This says that

If $p(x)$ and $g(x)$ are any two polynomials with $g(x) \neq 0$, then we can find polynomials $q(x)$ and $r(x)$ such that

$$p(x) = g(x) \times q(x) + r(x),$$

where $r(x) = 0$ or degree of $r(x) < \text{degree of } g(x)$.

This result is known as the **Division Algorithm** for polynomials.

Let us now take some examples to illustrate its use.

Example 8 : Divide $3x^2 - x^3 - 3x + 5$ by $x - 1 - x^2$, and verify the division algorithm.

Solution : Note that the given polynomials are not in standard form. To carry out division, we first write both the dividend and divisor in decreasing orders of their degrees.

So, dividend = $-x^3 + 3x^2 - 3x + 5$ and divisor = $-x^2 + x - 1$.

Division process is shown on the right side.

We stop here since degree (3) = 0 < 2 = degree ($-x^2 + x - 1$).

So, quotient = $x - 2$, remainder = 3.

Now,

Divisor $\times$ Quotient + Remainder

$$\begin{aligned} &= (-x^2 + x - 1)(x - 2) + 3 \\ &= -x^3 + x^2 - x + 2x^2 - 2x + 2 + 3 \\ &= -x^3 + 3x^2 - 3x + 5 \\ &= \text{Dividend} \end{aligned}$$

In this way, the division algorithm is verified.

Example 9 : Find all the zeroes of $2x^4 - 3x^3 - 3x^2 + 6x - 2$, if you know that two of its zeroes are $\sqrt{2}$ and $-\sqrt{2}$.

Solution : Since two zeroes are $\sqrt{2}$ and $-\sqrt{2}$, $(x - \sqrt{2})(x + \sqrt{2}) = x^2 - 2$ is a factor of the given polynomial. Now, we divide the given polynomial by $x^2 - 2$.

$$\begin{array}{r} 2x^2 - 3x + 1 \\ x^2 - 2 \overline{) 2x^4 - 3x^3 - 3x^2 + 6x - 2} \\ \underline{2x^4 - 4x^2} \\ -3x^3 + x^2 + 6x - 2 \\ \underline{-3x^3 + 6x} \\ x^2 - 2 \\ \underline{x^2 - 2} \\ 0 \end{array}$$

First term of quotient is $\frac{2x^4}{x^2} = 2x^2$

Second term of quotient is $\frac{-3x^3}{x^2} = -3x$

Third term of quotient is $\frac{x^2}{x^2} = 1$

So, $2x^4 - 3x^3 - 3x^2 + 6x - 2 = (x^2 - 2)(2x^2 - 3x + 1)$.

Now, by splitting $-3x$, we factorise $2x^2 - 3x + 1$ as $(2x - 1)(x - 1)$. So, its zeroes are given by $x = \frac{1}{2}$ and $x = 1$. Therefore, the zeroes of the given polynomial are $\sqrt{2}$, $-\sqrt{2}$, $\frac{1}{2}$, and 1 .

EXERCISE 9.3

1. Divide the polynomial $p(x)$ by the polynomial $g(x)$ and find the quotient and remainder in each of the following :
 - (i) $p(x) = x^3 - 3x^2 + 5x - 3$, $g(x) = x^2 - 2$
 - (ii) $p(x) = x^4 - 3x^2 + 4x + 5$, $g(x) = x^2 + 1 - x$
 - (iii) $p(x) = x^4 - 5x + 6$, $g(x) = 2 - x^2$
2. Check whether the first polynomial is a factor of the second polynomial by dividing the second polynomial by the first polynomial:
 - (i) $t^2 - 3$, $2t^4 + 3t^3 - 2t^2 - 9t - 12$
 - (ii) $x^2 + 3x + 1$, $3x^4 + 5x^3 - 7x^2 + 2x + 2$
 - (iii) $x^3 - 3x + 1$, $x^5 - 4x^3 + x^2 + 3x + 1$
3. Obtain all other zeroes of $3x^4 + 6x^3 - 2x^2 - 10x - 5$, if two of its zeroes are $\sqrt{\frac{5}{3}}$ and $-\sqrt{\frac{5}{3}}$.
4. On dividing $x^3 - 3x^2 + x + 2$ by a polynomial $g(x)$, the quotient and remainder were $x - 2$ and $-2x + 4$, respectively. Find $g(x)$.
5. Give examples of polynomials $p(x)$, $g(x)$, $q(x)$ and $r(x)$, which satisfy the division algorithm and
 - (i) $\deg p(x) = \deg q(x)$
 - (ii) $\deg q(x) = \deg r(x)$
 - (iii) $\deg r(x) = 0$

EXERCISE 9.4 (Optional)*

1. Verify that the numbers given alongside of the cubic polynomials below are their zeroes. Also verify the relationship between the zeroes and the coefficients in each case:
 - (i) $2x^3 + x^2 - 5x + 2$; $\frac{1}{2}$, 1 , -2
 - (ii) $x^3 - 4x^2 + 5x - 2$; 2 , 1 , 1
2. Find a cubic polynomial with the sum, sum of the product of its zeroes taken two at a time, and the product of its zeroes as 2 , -7 , -14 respectively.

*These exercises are not from the examination point of view.

3. If the zeroes of the polynomial $x^3 - 3x^2 + x + 1$ are $a - b$, a , $a + b$, find a and b .
4. If two zeroes of the polynomial $x^4 - 6x^3 - 26x^2 + 138x - 35$ are $2 \pm \sqrt{3}$, find other zeroes.
5. If the polynomial $x^4 - 6x^3 + 16x^2 - 25x + 10$ is divided by another polynomial $x^2 - 2x + k$, the remainder comes out to be $x + a$, find k and a .

9.5 Summary

In this chapter, you have studied the following points:

1. Polynomials of degrees 1, 2 and 3 are called linear, quadratic and cubic polynomials respectively.
2. A quadratic polynomial in x with real coefficients is of the form $ax^2 + bx + c$, where a, b, c are real numbers with $a \neq 0$.
3. The zeroes of a polynomial $p(x)$ are precisely the x -coordinates of the points, where the graph of $y = p(x)$ intersects the x -axis.
4. A quadratic polynomial can have at most 2 zeroes and a cubic polynomial can have at most 3 zeroes.
5. If α and β are the zeroes of the quadratic polynomial $ax^2 + bx + c$, then

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}.$$

6. If α, β, γ are the zeroes of the cubic polynomial $ax^3 + bx^2 + cx + d$, then

$$\alpha + \beta + \gamma = -\frac{b}{a},$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a},$$

$$\text{and} \quad \alpha\beta\gamma = -\frac{d}{a}.$$

7. The division algorithm states that given any polynomial $p(x)$ and any non-zero polynomial $g(x)$, there are polynomials $q(x)$ and $r(x)$ such that

$$p(x) = g(x) q(x) + r(x),$$

where $r(x) = 0$ or $\text{degree } r(x) < \text{degree } g(x)$.



QUADRATIC EQUATIONS

10

10.1 Introduction

In Chapter 2, you have studied different types of polynomials. One type was the quadratic polynomial of the form $ax^2 + bx + c$, $a \neq 0$. When we equate this polynomial to zero, we get a quadratic equation. Quadratic equations come up when we deal with many real-life situations. For instance, suppose a charity trust decides to build a prayer hall having a carpet area of 300 square metres with its length one metre more than twice its breadth. What should be the length and breadth of the hall? Suppose the breadth of the hall is x metres. Then, its length should be $(2x + 1)$ metres. We can depict this information pictorially as shown in Fig. 10.1.

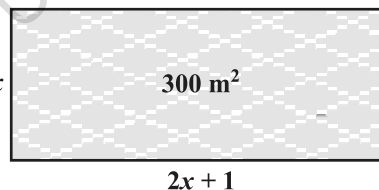


Fig. 10.1

Now, area of the hall = $(2x + 1) \cdot x \text{ m}^2 = (2x^2 + x) \text{ m}^2$

So, $2x^2 + x = 300$ (Given)

Therefore, $2x^2 + x - 300 = 0$

So, the breadth of the hall should satisfy the equation $2x^2 + x - 300 = 0$ which is a quadratic equation.

Many people believe that Babylonians were the first to solve quadratic equations. For instance, they knew how to find two positive numbers with a given positive sum and a given positive product, and this problem is equivalent to solving a quadratic equation of the form $x^2 - px + q = 0$. Greek mathematician Euclid developed a geometrical approach for finding out lengths which, in our present day terminology, are solutions of quadratic equations. Solving of quadratic equations, in general form, is often credited to ancient Indian mathematicians. In fact, Brahmagupta (C.E.598–665) gave an explicit formula to solve a quadratic equation of the form $ax^2 + bx = c$. Later,

Sridharacharya (C.E. 1025) derived a formula, now known as the quadratic formula, (as quoted by Bhaskara II) for solving a quadratic equation by the method of completing the square. An Arab mathematician Al-Khwarizmi (about C.E. 800) also studied quadratic equations of different types. Abraham bar Hiyya Ha-Nasi, in his book 'Liber embadorum' published in Europe in C.E. 1145 gave complete solutions of different quadratic equations.

In this chapter, you will study quadratic equations, and various ways of finding their roots. You will also see some applications of quadratic equations in daily life situations.

10.2 Quadratic Equations

A quadratic equation in the variable x is an equation of the form $ax^2 + bx + c = 0$, where a, b, c are real numbers, $a \neq 0$. For example, $2x^2 + x - 300 = 0$ is a quadratic equation. Similarly, $2x^2 - 3x + 1 = 0$, $4x - 3x^2 + 2 = 0$ and $1 - x^2 + 300 = 0$ are also quadratic equations.

In fact, any equation of the form $p(x) = 0$, where $p(x)$ is a polynomial of degree 2, is a quadratic equation. But when we write the terms of $p(x)$ in descending order of their degrees, then we get the standard form of the equation. That is, $ax^2 + bx + c = 0$, $a \neq 0$ is called the **standard form of a quadratic equation**.

Quadratic equations arise in several situations in the world around us and in different fields of mathematics. Let us consider a few examples.

Example 1 : Represent the following situations mathematically:

- (i) John and Jivanti together have 45 marbles. Both of them lost 5 marbles each, and the product of the number of marbles they now have is 124. We would like to find out how many marbles they had to start with.
- (ii) A cottage industry produces a certain number of toys in a day. The cost of production of each toy (in rupees) was found to be 55 minus the number of toys produced in a day. On a particular day, the total cost of production was ₹ 750. We would like to find out the number of toys produced on that day.

Solution :

- (i) Let the number of marbles John had be x .

Then the number of marbles Jivanti had = $45 - x$ (Why?).

The number of marbles left with John, when he lost 5 marbles = $x - 5$

The number of marbles left with Jivanti, when she lost 5 marbles = $45 - x - 5$
 $= 40 - x$

$$\begin{aligned}\text{Therefore, their product} &= (x - 5)(40 - x) \\ &= 40x - x^2 - 200 + 5x \\ &= -x^2 + 45x - 200\end{aligned}$$

$$\text{So, } -x^2 + 45x - 200 = 124 \quad (\text{Given that product} = 124)$$

$$\text{i.e., } -x^2 + 45x - 324 = 0$$

$$\text{i.e., } x^2 - 45x + 324 = 0$$

Therefore, the number of marbles John had, satisfies the quadratic equation

$$x^2 - 45x + 324 = 0$$

which is the required representation of the problem mathematically.

(ii) Let the number of toys produced on that day be x .

Therefore, the cost of production (in rupees) of each toy that day = $55 - x$

So, the total cost of production (in rupees) that day = $x(55 - x)$

$$\text{Therefore, } x(55 - x) = 750$$

$$\text{i.e., } 55x - x^2 = 750$$

$$\text{i.e., } -x^2 + 55x - 750 = 0$$

$$\text{i.e., } x^2 - 55x + 750 = 0$$

Therefore, the number of toys produced that day satisfies the quadratic equation

$$x^2 - 55x + 750 = 0$$

which is the required representation of the problem mathematically.

Example 2 : Check whether the following are quadratic equations:

$$(i) (x - 2)^2 + 1 = 2x - 3$$

$$(ii) x(x + 1) + 8 = (x + 2)(x - 2)$$

$$(iii) x(2x + 3) = x^2 + 1$$

$$(iv) (x + 2)^3 = x^3 - 4$$

Solution :

$$(i) \text{ LHS} = (x - 2)^2 + 1 = x^2 - 4x + 4 + 1 = x^2 - 4x + 5$$

Therefore, $(x - 2)^2 + 1 = 2x - 3$ can be rewritten as

$$x^2 - 4x + 5 = 2x - 3$$

$$\text{i.e., } x^2 - 6x + 8 = 0$$

It is of the form $ax^2 + bx + c = 0$.

Therefore, the given equation is a quadratic equation.

- (ii) Since $x(x + 1) + 8 = x^2 + x + 8$ and $(x + 2)(x - 2) = x^2 - 4$

Therefore, $x^2 + x + 8 = x^2 - 4$

i.e., $x + 12 = 0$

It is not of the form $ax^2 + bx + c = 0$.

Therefore, the given equation is not a quadratic equation.

- (iii) Here, LHS = $x(2x + 3) = 2x^2 + 3x$

So, $x(2x + 3) = x^2 + 1$ can be rewritten as

$$2x^2 + 3x = x^2 + 1$$

Therefore, we get $x^2 + 3x - 1 = 0$

It is of the form $ax^2 + bx + c = 0$.

So, the given equation is a quadratic equation.

- (iv) Here, LHS = $(x + 2)^3 = x^3 + 6x^2 + 12x + 8$

Therefore, $(x + 2)^3 = x^3 - 4$ can be rewritten as

$$x^3 + 6x^2 + 12x + 8 = x^3 - 4$$

i.e., $6x^2 + 12x + 12 = 0$ or, $x^2 + 2x + 2 = 0$

It is of the form $ax^2 + bx + c = 0$.

So, the given equation is a quadratic equation.

Remark : Be careful! In (ii) above, the given equation appears to be a quadratic equation, but it is not a quadratic equation.

In (iv) above, the given equation appears to be a cubic equation (an equation of degree 3) and not a quadratic equation. But it turns out to be a quadratic equation. As you can see, often we need to simplify the given equation before deciding whether it is quadratic or not.

EXERCISE 10.1

1. Check whether the following are quadratic equations :

(i) $(x + 1)^2 = 2(x - 3)$

(ii) $x^2 - 2x = (-2)(3 - x)$

(iii) $(x - 2)(x + 1) = (x - 1)(x + 3)$

(iv) $(x - 3)(2x + 1) = x(x + 5)$

(v) $(2x - 1)(x - 3) = (x + 5)(x - 1)$

(vi) $x^2 + 3x + 1 = (x - 2)^2$

(vii) $(x + 2)^3 = 2x(x^2 - 1)$

(viii) $x^3 - 4x^2 - x + 1 = (x - 2)^3$

2. Represent the following situations in the form of quadratic equations :

- (i) The area of a rectangular plot is 528 m^2 . The length of the plot (in metres) is one more than twice its breadth. We need to find the length and breadth of the plot.

- (ii) The product of two consecutive positive integers is 306. We need to find the integers.
- (iii) Rohan's mother is 26 years older than him. The product of their ages (in years) 3 years from now will be 360. We would like to find Rohan's present age.
- (iv) A train travels a distance of 480 km at a uniform speed. If the speed had been 8 km/h less, then it would have taken 3 hours more to cover the same distance. We need to find the speed of the train.

10.3 Solution of a Quadratic Equation by Factorisation

Consider the quadratic equation $2x^2 - 3x + 1 = 0$. If we replace x by 1 on the LHS of this equation, we get $(2 \times 1^2) - (3 \times 1) + 1 = 0 = \text{RHS}$ of the equation. We say that 1 is a root of the quadratic equation $2x^2 - 3x + 1 = 0$. This also means that 1 is a zero of the quadratic polynomial $2x^2 - 3x + 1$.

In general, a real number α is called a root of the quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$ if $a\alpha^2 + b\alpha + c = 0$. We also say that $x = \alpha$ is a **solution of the quadratic equation**, or that α **satisfies the quadratic equation**. Note that **the zeroes of the quadratic polynomial $ax^2 + bx + c$ and the roots of the quadratic equation $ax^2 + bx + c = 0$ are the same**.

You have observed, in Chapter 9, that a quadratic polynomial can have at most two zeroes. So, any quadratic equation can have at most two roots.

You have learnt in Class IX, how to factorise quadratic polynomials by splitting their middle terms. We shall use this knowledge for finding the roots of a quadratic equation. Let us see how.

Example 3 : Find the roots of the equation $2x^2 - 5x + 3 = 0$, by factorisation.

Solution : Let us first split the middle term $-5x$ as $-2x - 3x$ [because $(-2x) \times (-3x) = 6x^2 = (2x^2) \times 3$].

$$\text{So, } 2x^2 - 5x + 3 = 2x^2 - 2x - 3x + 3 = 2x(x - 1) - 3(x - 1) = (2x - 3)(x - 1)$$

$$\text{Now, } 2x^2 - 5x + 3 = 0 \text{ can be rewritten as } (2x - 3)(x - 1) = 0.$$

So, the values of x for which $2x^2 - 5x + 3 = 0$ are the same for which $(2x - 3)(x - 1) = 0$, i.e., either $2x - 3 = 0$ or $x - 1 = 0$.

$$\text{Now, } 2x - 3 = 0 \text{ gives } x = \frac{3}{2} \text{ and } x - 1 = 0 \text{ gives } x = 1.$$

$$\text{So, } x = \frac{3}{2} \text{ and } x = 1 \text{ are the solutions of the equation.}$$

$$\text{In other words, } 1 \text{ and } \frac{3}{2} \text{ are the roots of the equation } 2x^2 - 5x + 3 = 0.$$

Verify that these are the roots of the given equation.

Note that we have found the roots of $2x^2 - 5x + 3 = 0$ by factorising $2x^2 - 5x + 3$ into two linear factors and equating each factor to zero.

Example 4 : Find the roots of the quadratic equation $6x^2 - x - 2 = 0$.

Solution : We have

$$\begin{aligned} 6x^2 - x - 2 &= 6x^2 + 3x - 4x - 2 \\ &= 3x(2x + 1) - 2(2x + 1) \\ &= (3x - 2)(2x + 1) \end{aligned}$$

The roots of $6x^2 - x - 2 = 0$ are the values of x for which $(3x - 2)(2x + 1) = 0$

Therefore, $3x - 2 = 0$ or $2x + 1 = 0$,

i.e.,
$$x = \frac{2}{3} \quad \text{or} \quad x = -\frac{1}{2}$$

Therefore, the roots of $6x^2 - x - 2 = 0$ are $\frac{2}{3}$ and $-\frac{1}{2}$.

We verify the roots, by checking that $\frac{2}{3}$ and $-\frac{1}{2}$ satisfy $6x^2 - x - 2 = 0$.

Example 5 : Find the roots of the quadratic equation $3x^2 - 2\sqrt{6}x + 2 = 0$.

Solution :

$$\begin{aligned} 3x^2 - 2\sqrt{6}x + 2 &= 3x^2 - \sqrt{6}x - \sqrt{6}x + 2 \\ &= \sqrt{3}x(\sqrt{3}x - \sqrt{2}) - \sqrt{2}(\sqrt{3}x - \sqrt{2}) \\ &= (\sqrt{3}x - \sqrt{2})(\sqrt{3}x - \sqrt{2}) \end{aligned}$$

So, the roots of the equation are the values of x for which

$$(\sqrt{3}x - \sqrt{2})(\sqrt{3}x - \sqrt{2}) = 0$$

Now, $\sqrt{3}x - \sqrt{2} = 0$ for $x = \sqrt{\frac{2}{3}}$.

So, this root is repeated twice, one for each repeated factor $\sqrt{3}x - \sqrt{2}$.

Therefore, the roots of $3x^2 - 2\sqrt{6}x + 2 = 0$ are $\sqrt{\frac{2}{3}}, \sqrt{\frac{2}{3}}$.

Example 6 : Find the dimensions of the prayer hall discussed in Section 10.1.

Solution : In Section 10.1, we found that if the breadth of the hall is x m, then x satisfies the equation $2x^2 + x - 300 = 0$. Applying the factorisation method, we write this equation as

$$2x^2 - 24x + 25x - 300 = 0$$

$$2x(x - 12) + 25(x - 12) = 0$$

i.e., $(x - 12)(2x + 25) = 0$

So, the roots of the given equation are $x = 12$ or $x = -12.5$. Since x is the breadth of the hall, it cannot be negative.

Thus, the breadth of the hall is 12 m. Its length $= 2x + 1 = 25$ m.

EXERCISE 10.2

- Find the roots of the following quadratic equations by factorisation:

(i) $x^2 - 3x - 10 = 0$

(ii) $2x^2 + x - 6 = 0$

(iii) $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$

(iv) $2x^2 - x + \frac{1}{8} = 0$

(v) $100x^2 - 20x + 1 = 0$

- Solve the problems given in Example 1.
- Find two numbers whose sum is 27 and product is 182.
- Find two consecutive positive integers, sum of whose squares is 365.
- The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, find the other two sides.
- A cottage industry produces a certain number of pottery articles in a day. It was observed on a particular day that the cost of production of each article (in rupees) was 3 more than twice the number of articles produced on that day. If the total cost of production on that day was ₹ 90, find the number of articles produced and the cost of each article.

10.4 Solution of a Quadratic Equation by Completing the Square

In the previous section, you have learnt one method of obtaining the roots of a quadratic equation. In this section, we shall study another method.

Consider the following situation:

The product of Sunita's age (in years) two years ago and her age four years from now is one more than twice her present age. What is her present age?

To answer this, let her present age (in years) be x . Then the product of her ages two years ago and four years from now is $(x - 2)(x + 4)$.

Therefore, $(x - 2)(x + 4) = 2x + 1$

i.e., $x^2 + 2x - 8 = 2x + 1$

i.e., $x^2 - 9 = 0$

So, Sunita's present age satisfies the quadratic equation $x^2 - 9 = 0$.

We can write this as $x^2 = 9$. Taking square roots, we get $x = 3$ or $x = -3$. Since the age is a positive number, $x = 3$.

So, Sunita's present age is 3 years.

Now consider the quadratic equation $(x + 2)^2 - 9 = 0$. To solve it, we can write it as $(x + 2)^2 = 9$. Taking square roots, we get $x + 2 = 3$ or $x + 2 = -3$.

Therefore, $x = 1$ or $x = -5$

So, the roots of the equation $(x + 2)^2 - 9 = 0$ are 1 and -5 .

In both the examples above, the term containing x is completely inside a square, and we found the roots easily by taking the square roots. But, what happens if we are asked to solve the equation $x^2 + 4x - 5 = 0$? We would probably apply factorisation to do so, unless we realise (somehow!) that $x^2 + 4x - 5 = (x + 2)^2 - 9$.

So, solving $x^2 + 4x - 5 = 0$ is equivalent to solving $(x + 2)^2 - 9 = 0$, which we have seen is very quick to do. In fact, we can convert any quadratic equation to the form $(x + a)^2 - b^2 = 0$ and then we can easily find its roots. Let us see if this is possible. Look at Fig. 10.2.

In this figure, we can see how $x^2 + 4x$ is being converted to $(x + 2)^2 - 4$.

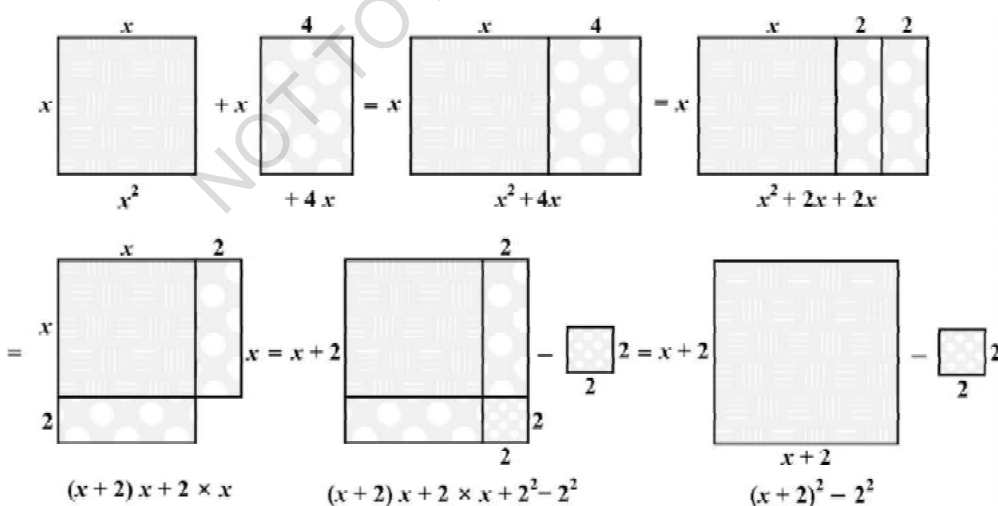


Fig. 10.2

The process is as follows:

$$\begin{aligned}
 x^2 + 4x &= \left(x^2 + \frac{4}{2}x\right) + \frac{4}{2}x \\
 &= x^2 + 2x + 2x \\
 &= (x + 2)x + 2 \times x \\
 &= (x + 2)x + 2 \times x + 2 \times 2 - 2 \times 2 \\
 &= (x + 2)x + (x + 2) \times 2 - 2 \times 2 \\
 &= (x + 2)(x + 2) - 2^2 \\
 &= (x + 2)^2 - 4
 \end{aligned}$$

So, $x^2 + 4x - 5 = (x + 2)^2 - 4 - 5 = (x + 2)^2 - 9$

So, $x^2 + 4x - 5 = 0$ can be written as $(x + 2)^2 - 9 = 0$ by this process of completing the square. This is known as the **method of completing the square**.

In brief, this can be shown as follows:

$$x^2 + 4x = \left(x + \frac{4}{2}\right)^2 - \left(\frac{4}{2}\right)^2 = \left(x + \frac{4}{2}\right)^2 - 4$$

So, $x^2 + 4x - 5 = 0$ can be rewritten as

$$\left(x + \frac{4}{2}\right)^2 - 4 - 5 = 0$$

i.e., $(x + 2)^2 - 9 = 0$

Consider now the equation $3x^2 - 5x + 2 = 0$. Note that the coefficient of x^2 is not a perfect square. So, we multiply the equation throughout by 3 to get

$$9x^2 - 15x + 6 = 0$$

Now, $9x^2 - 15x + 6 = (3x)^2 - 2 \times 3x \times \frac{5}{2} + 6$

$$= (3x)^2 - 2 \times 3x \times \frac{5}{2} + \left(\frac{5}{2}\right)^2 - \left(\frac{5}{2}\right)^2 + 6$$

$$= \left(3x - \frac{5}{2}\right)^2 - \frac{25}{4} + 6 = \left(3x - \frac{5}{2}\right)^2 - \frac{1}{4}$$

So, $9x^2 - 15x + 6 = 0$ can be written as

$$\left(3x - \frac{5}{2}\right)^2 - \frac{1}{4} = 0$$

i.e.,
$$\left(3x - \frac{5}{2}\right)^2 = \frac{1}{4}$$

So, the solutions of $9x^2 - 15x + 6 = 0$ are the same as those of $\left(3x - \frac{5}{2}\right)^2 = \frac{1}{4}$.

i.e.,
$$3x - \frac{5}{2} = \frac{1}{2} \quad \text{or} \quad 3x - \frac{5}{2} = -\frac{1}{2}$$

(We can also write this as $3x - \frac{5}{2} = \pm \frac{1}{2}$, where ‘ $\pm$ ’ denotes ‘plus minus’.)

Thus,
$$3x = \frac{5}{2} + \frac{1}{2} \quad \text{or} \quad 3x = \frac{5}{2} - \frac{1}{2}$$

So,
$$x = \frac{5}{6} + \frac{1}{6} \quad \text{or} \quad x = \frac{5}{6} - \frac{1}{6}$$

Therefore,
$$x = 1 \quad \text{or} \quad x = \frac{4}{6}$$

i.e.,
$$x = 1 \quad \text{or} \quad x = \frac{2}{3}$$

Therefore, the roots of the given equation are 1 and $\frac{2}{3}$.

Remark : Another way of showing this process is as follows :

The equation
$$3x^2 - 5x + 2 = 0$$

is the same as

$$x^2 - \frac{5}{3}x + \frac{2}{3} = 0$$

Now,
$$x^2 - \frac{5}{3}x + \frac{2}{3} = \left\{x - \frac{1}{2}\left(\frac{5}{3}\right)\right\}^2 - \left\{\frac{1}{2}\left(\frac{5}{3}\right)\right\}^2 + \frac{2}{3}$$

$$= \left(x - \frac{5}{6}\right)^2 + \frac{2}{3} - \frac{25}{36}$$

$$= \left(x - \frac{5}{6}\right)^2 - \frac{1}{36} = \left(x - \frac{5}{6}\right)^2 - \left(\frac{1}{6}\right)^2$$

So, the solutions of $3x^2 - 5x + 2 = 0$ are the same as those of $\left(x - \frac{5}{6}\right)^2 - \left(\frac{1}{6}\right)^2 = 0$,

which are $x - \frac{5}{6} = \pm \frac{1}{6}$, i.e., $x = \frac{5}{6} + \frac{1}{6} = 1$ and $x = \frac{5}{6} - \frac{1}{6} = \frac{2}{3}$.

Let us consider some examples to illustrate the above process.

Example 7 : Solve the equation given in Example 3 by the method of completing the square.

Solution : The equation $2x^2 - 5x + 3 = 0$ is the same as $x^2 - \frac{5}{2}x + \frac{3}{2} = 0$.

Now,

$$x^2 - \frac{5}{2}x + \frac{3}{2} = \left(x - \frac{5}{4}\right)^2 - \left(\frac{5}{4}\right)^2 + \frac{3}{2} = \left(x - \frac{5}{4}\right)^2 - \frac{1}{16}$$

Therefore, $2x^2 - 5x + 3 = 0$ can be written as $\left(x - \frac{5}{4}\right)^2 - \frac{1}{16} = 0$.

So, the roots of the equation $2x^2 - 5x + 3 = 0$ are exactly the same as those of

$$\left(x - \frac{5}{4}\right)^2 - \frac{1}{16} = 0. \text{ Now, } \left(x - \frac{5}{4}\right)^2 - \frac{1}{16} = 0 \text{ is the same as } \left(x - \frac{5}{4}\right)^2 = \frac{1}{16}$$

Therefore,

$$x - \frac{5}{4} = \pm \frac{1}{4}$$

i.e.,

$$x = \frac{5}{4} \pm \frac{1}{4}$$

i.e.,

$$x = \frac{5}{4} + \frac{1}{4} \text{ or } x = \frac{5}{4} - \frac{1}{4}$$

i.e.,

$$x = \frac{3}{2} \text{ or } x = 1$$

Therefore, the solutions of the equations are $x = \frac{3}{2}$ and 1.

Let us **verify** our solutions.

Putting $x = \frac{3}{2}$ in $2x^2 - 5x + 3 = 0$, we get $2\left(\frac{3}{2}\right)^2 - 5\left(\frac{3}{2}\right) + 3 = 0$, which is correct. Similarly, you can verify that $x = 1$ also satisfies the given equation.

In Example 7, we divided the equation $2x^2 - 5x + 3 = 0$ throughout by 2 to get $x^2 - \frac{5}{2}x + \frac{3}{2} = 0$ to make the first term a perfect square and then completed the square. Instead, we can multiply throughout by 2 to make the first term as $4x^2 = (2x)^2$ and then complete the square.

This method is illustrated in the next example.

Example 8 : Find the roots of the equation $5x^2 - 6x - 2 = 0$ by the method of completing the square.

Solution : Multiplying the equation throughout by 5, we get

$$25x^2 - 30x - 10 = 0$$

This is the same as

$$(5x)^2 - 2 \times (5x) \times 3 + 3^2 - 3^2 - 10 = 0$$

$$\text{i.e.,} \quad (5x - 3)^2 - 9 - 10 = 0$$

$$\text{i.e.,} \quad (5x - 3)^2 - 19 = 0$$

$$\text{i.e.,} \quad (5x - 3)^2 = 19$$

$$\text{i.e.,} \quad 5x - 3 = \pm\sqrt{19}$$

$$\text{i.e.,} \quad 5x = 3 \pm \sqrt{19}$$

$$\text{So,} \quad x = \frac{3 \pm \sqrt{19}}{5}$$

Therefore, the roots are $\frac{3 + \sqrt{19}}{5}$ and $\frac{3 - \sqrt{19}}{5}$.

Verify that the roots are $\frac{3 + \sqrt{19}}{5}$ and $\frac{3 - \sqrt{19}}{5}$.

Example 9 : Find the roots of $4x^2 + 3x + 5 = 0$ by the method of completing the square.

Solution : Note that $4x^2 + 3x + 5 = 0$ is the same as

$$(2x)^2 + 2 \times (2x) \times \frac{3}{4} + \left(\frac{3}{4}\right)^2 - \left(\frac{3}{4}\right)^2 + 5 = 0$$

$$\text{i.e.,} \quad \left(2x + \frac{3}{4}\right)^2 - \frac{9}{16} + 5 = 0$$

$$\text{i.e.,} \quad \left(2x + \frac{3}{4}\right)^2 + \frac{71}{16} = 0$$

$$\text{i.e.,} \quad \left(2x + \frac{3}{4}\right)^2 = \frac{-17}{16} < 0$$

But $\left(2x + \frac{3}{4}\right)^2$ cannot be negative for any real value of x (Why?). So, there is no real value of x satisfying the given equation. Therefore, the given equation has no *real roots*.

Now, you have seen several examples of the use of the method of completing the square. So, let us give this method in general.

Consider the quadratic equation $ax^2 + bx + c = 0$ ($a \neq 0$). Dividing throughout by a , we get

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0$$

$$\text{This is the same as} \quad \left(x + \frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2 + \frac{c}{a} = 0$$

$$\text{i.e.,} \quad \left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a^2} = 0$$

So, the roots of the given equation are the same as those of

$$\left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a^2} = 0, \text{ i.e., those of } \left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2} \quad (1)$$

If $b^2 - 4ac \geq 0$, then by taking the square roots in (1), we get

$$x + \frac{b}{2a} = \frac{\pm \sqrt{b^2 - 4ac}}{2a}$$

Therefore,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

So, the roots of $ax^2 + bx + c = 0$ are $\frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $\frac{-b - \sqrt{b^2 - 4ac}}{2a}$, if $b^2 - 4ac \geq 0$. If $b^2 - 4ac < 0$, the equation will have no real roots. (Why?)

Thus, if $b^2 - 4ac \geq 0$, then the roots of the quadratic equation $ax^2 + bx + c = 0$ are given by $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

This formula for finding the roots of a quadratic equation is known as the **quadratic formula**.

Let us consider some examples for illustrating the use of the quadratic formula.

Example 10 : Solve Q. 2(i) of Exercise 10.1 by using the quadratic formula.

Solution : Let the breadth of the plot be x metres. Then the length is $(2x + 1)$ metres. Then we are given that $x(2x + 1) = 528$, i.e., $2x^2 + x - 528 = 0$.

This is of the form $ax^2 + bx + c = 0$, where $a = 2$, $b = 1$, $c = -528$.

So, the quadratic formula gives us the solution as

$$x = \frac{-1 \pm \sqrt{1 + 4(2)(528)}}{4} = \frac{-1 \pm \sqrt{4225}}{4} = \frac{-1 \pm 65}{4}$$

$$\text{i.e., } x = \frac{64}{4} \text{ or } x = \frac{-66}{4}$$

$$\text{i.e., } x = 16 \text{ or } x = -\frac{33}{2}$$

Since x cannot be negative, being a dimension, the breadth of the plot is 16 metres and hence, the length of the plot is 33m.

You should verify that these values satisfy the conditions of the problem.

Example 11 : Find two consecutive odd positive integers, sum of whose squares is 290.

Solution : Let the smaller of the two consecutive odd positive integers be x . Then, the second integer will be $x + 2$. According to the question,

$$x^2 + (x + 2)^2 = 290$$

$$\text{i.e.,} \quad x^2 + x^2 + 4x + 4 = 290$$

$$\text{i.e.,} \quad 2x^2 + 4x - 286 = 0$$

$$\text{i.e.,} \quad x^2 + 2x - 143 = 0$$

which is a quadratic equation in x .

Using the quadratic formula, we get

$$x = \frac{-2 \pm \sqrt{4 + 572}}{2} = \frac{-2 \pm \sqrt{576}}{2} = \frac{-2 \pm 24}{2}$$

$$\text{i.e.,} \quad x = 11 \quad \text{or} \quad x = -13$$

But x is given to be an odd positive integer. Therefore, $x \neq -13$, $x = 11$.

Thus, the two consecutive odd integers are 11 and 13.

Check : $11^2 + 13^2 = 121 + 169 = 290$.

Example 12 : A rectangular park is to be designed whose breadth is 3 m less than its length. Its area is to be 4 square metres more than the area of a park that has already been made in the shape of an isosceles triangle with its base as the breadth of the rectangular park and of altitude 12 m (see Fig. 10.3). Find its length and breadth.

Solution : Let the breadth of the rectangular park be x m.

So, its length = $(x + 3)$ m.

Therefore, the area of the rectangular park = $x(x + 3) \text{ m}^2 = (x^2 + 3x) \text{ m}^2$.

Now, base of the isosceles triangle = x m.

Therefore, its area = $\frac{1}{2} \times x \times 12 = 6x \text{ m}^2$.

According to our requirements,

$$x^2 + 3x = 6x + 4$$

$$\text{i.e.,} \quad x^2 - 3x - 4 = 0$$

Using the quadratic formula, we get

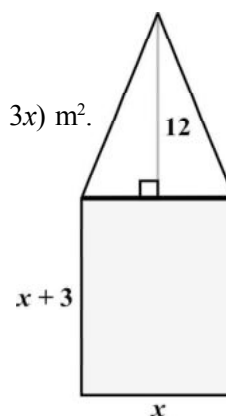


Fig. 10.3

$$x = \frac{3 \pm \sqrt{25}}{2} = \frac{3 \pm 5}{2} = 4 \text{ or } -1$$

But $x \neq -1$ (Why?). Therefore, $x = 4$.

So, the breadth of the park = 4m and its length will be 7m.

Verification : Area of rectangular park = 28 m²,

$$\text{area of triangular park} = 24 \text{ m}^2 = (28 - 4) \text{ m}^2$$

Example 13 : Find the roots of the following quadratic equations, if they exist, using the quadratic formula:

(i) $3x^2 - 5x + 2 = 0$

(ii) $x^2 + 4x + 5 = 0$

(iii) $2x^2 - 2\sqrt{2}x + 1 = 0$

Solution :

(i) $3x^2 - 5x + 2 = 0$. Here, $a = 3$, $b = -5$, $c = 2$. So, $b^2 - 4ac = 25 - 24 = 1 > 0$.

$$\text{Therefore, } x = \frac{5 \pm \sqrt{1}}{6} = \frac{5 \pm 1}{6}, \text{ i.e., } x = 1 \text{ or } x = \frac{2}{3}$$

So, the roots are $\frac{2}{3}$ and 1.

(ii) $x^2 + 4x + 5 = 0$. Here, $a = 1$, $b = 4$, $c = 5$. So, $b^2 - 4ac = 16 - 20 = -4 < 0$.

Since the square of a real number cannot be negative, therefore $\sqrt{b^2 - 4ac}$ will not have any real value.

So, there are no real roots for the given equation.

(iii) $2x^2 - 2\sqrt{2}x + 1 = 0$. Here, $a = 2$, $b = -2\sqrt{2}$, $c = 1$.

$$\text{So, } b^2 - 4ac = 8 - 8 = 0$$

$$\text{Therefore, } x = \frac{2\sqrt{2} \pm \sqrt{0}}{4} = \frac{\sqrt{2}}{2} \pm 0, \text{ i.e., } x = \frac{1}{\sqrt{2}}$$

So, the roots are $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$.

Example 14 : Find the roots of the following equations:

(i) $x + \frac{1}{x} = 3, x \neq 0$

(ii) $\frac{1}{x} - \frac{1}{x-2} = 3, x \neq 0, 2$

Solution :

(i) $x + \frac{1}{x} = 3$. Multiplying throughout by x , we get

$$x^2 + 1 = 3x$$

i.e.,

$$x^2 - 3x + 1 = 0, \text{ which is a quadratic equation.}$$

Here,

$$a = 1, b = -3, c = 1$$

So,

$$b^2 - 4ac = 9 - 4 = 5 > 0$$

Therefore,

$$x = \frac{3 \pm \sqrt{5}}{2} \quad (\text{Why?})$$

So, the roots are $\frac{3 + \sqrt{5}}{2}$ and $\frac{3 - \sqrt{5}}{2}$.

(ii) $\frac{1}{x} - \frac{1}{x-2} = 3, x \neq 0, 2$.

As $x \neq 0, 2$, multiplying the equation by $x(x-2)$, we get

$$\begin{aligned} (x-2) - x &= 3x(x-2) \\ &= 3x^2 - 6x \end{aligned}$$

So, the given equation reduces to $3x^2 - 6x + 2 = 0$, which is a quadratic equation.

Here,

$$a = 3, b = -6, c = 2. \quad \text{So, } b^2 - 4ac = 36 - 24 = 12 > 0$$

Therefore,

$$x = \frac{6 \pm \sqrt{12}}{6} = \frac{6 \pm 2\sqrt{3}}{6} = \frac{3 \pm \sqrt{3}}{3}.$$

So, the roots are $\frac{3 + \sqrt{3}}{3}$ and $\frac{3 - \sqrt{3}}{3}$.

Example 15 : A motor boat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.

Solution : Let the speed of the stream be x km/h.

Therefore, the speed of the boat upstream = $(18 - x)$ km/h and the speed of the boat downstream = $(18 + x)$ km/h.

The time taken to go upstream = $\frac{\text{distance}}{\text{speed}} = \frac{24}{18 - x}$ hours.

Similarly, the time taken to go downstream = $\frac{24}{18 + x}$ hours.

According to the question,

$$\frac{24}{18 - x} - \frac{24}{18 + x} = 1$$

$$\text{i.e.,} \quad 24(18 + x) - 24(18 - x) = (18 - x)(18 + x)$$

$$\text{i.e.,} \quad x^2 + 48x - 324 = 0$$

Using the quadratic formula, we get

$$\begin{aligned} x &= \frac{-48 \pm \sqrt{48^2 + 1296}}{2} = \frac{-48 \pm \sqrt{3600}}{2} \\ &= \frac{-48 \pm 60}{2} = 6 \text{ or } -54 \end{aligned}$$

Since x is the speed of the stream, it cannot be negative. So, we ignore the root $x = -54$. Therefore, $x = 6$ gives the speed of the stream as 6 km/h.

EXERCISE 10.3

- Find the roots of the following quadratic equations, if they exist, by the method of completing the square:
 - $2x^2 - 7x + 3 = 0$
 - $2x^2 + x - 4 = 0$
 - $4x^2 + 4\sqrt{3}x + 3 = 0$
 - $2x^2 + x + 4 = 0$
- Find the roots of the quadratic equations given in Q.1 above by applying the quadratic formula.

3. Find the roots of the following equations:

(i) $x - \frac{1}{x} = 3, x \neq 0$

(ii) $\frac{1}{x+4} - \frac{1}{x-7} = \frac{11}{30}, x \neq -4, 7$

4. The sum of the reciprocals of Rehman's ages, (in years) 3 years ago and 5 years from now is $\frac{1}{3}$. Find his present age.

5. In a class test, the sum of Shefali's marks in Mathematics and English is 30. Had she got 2 marks more in Mathematics and 3 marks less in English, the product of their marks would have been 210. Find her marks in the two subjects.

6. The diagonal of a rectangular field is 60 metres more than the shorter side. If the longer side is 30 metres more than the shorter side, find the sides of the field.

7. The difference of squares of two numbers is 180. The square of the smaller number is 8 times the larger number. Find the two numbers.

8. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train.

9. Two water taps together can fill a tank in $9\frac{3}{8}$ hours. The tap of larger diameter takes 10 hours less than the smaller one to fill the tank separately. Find the time in which each tap can separately fill the tank.

10. An express train takes 1 hour less than a passenger train to travel 132 km between Mysore and Bangalore (without taking into consideration the time they stop at intermediate stations). If the average speed of the express train is 11 km/h more than that of the passenger train, find the average speed of the two trains.

11. Sum of the areas of two squares is 468 m². If the difference of their perimeters is 24 m, find the sides of the two squares.

10.5 Nature of Roots

In the previous section, you have seen that the roots of the equation $ax^2 + bx + c = 0$ are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

If $b^2 - 4ac > 0$, we get two distinct real roots $-\frac{b}{2a} + \frac{\sqrt{b^2 - 4ac}}{2a}$ and $-\frac{b}{2a} - \frac{\sqrt{b^2 - 4ac}}{2a}$.

If $b^2 - 4ac = 0$, then $x = -\frac{b}{2a} \pm 0$, i.e., $x = -\frac{b}{2a}$ or $-\frac{b}{2a}$.

So, the roots of the equation $ax^2 + bx + c = 0$ are both $-\frac{b}{2a}$.

Therefore, we say that the quadratic equation $ax^2 + bx + c = 0$ has two equal real roots in this case.

If $b^2 - 4ac < 0$, then there is no real number whose square is $b^2 - 4ac$. Therefore, there are no real roots for the given quadratic equation in this case.

Since $b^2 - 4ac$ determines whether the quadratic equation $ax^2 + bx + c = 0$ has real roots or not, $b^2 - 4ac$ is called the **discriminant** of this quadratic equation.

So, a quadratic equation $ax^2 + bx + c = 0$ has

- (i) two distinct real roots, if $b^2 - 4ac > 0$,
- (ii) two equal real roots, if $b^2 - 4ac = 0$,
- (iii) no real roots, if $b^2 - 4ac < 0$.

Let us consider some examples.

Example 16 : Find the discriminant of the quadratic equation $2x^2 - 4x + 3 = 0$, and hence find the nature of its roots.

Solution : The given equation is of the form $ax^2 + bx + c = 0$, where $a = 2$, $b = -4$ and $c = 3$. Therefore, the discriminant

$$b^2 - 4ac = (-4)^2 - (4 \times 2 \times 3) = 16 - 24 = -8 < 0$$

So, the given equation has no real roots.

Example 17 : A pole has to be erected at a point on the boundary of a circular park of diameter 13 metres in such a way that the differences of its distances from two diametrically opposite fixed gates A and B on the boundary is 7 metres. Is it possible to do so? If yes, at what distances from the two gates should the pole be erected?

Solution : Let us first draw the diagram (see Fig. 10.4).

Let P be the required location of the pole. Let the distance of the pole from the gate B be x m, i.e., $BP = x$ m. Now the difference of the distances of the pole from the two gates = $AP - BP$ (or, $BP - AP$) = 7 m. Therefore, $AP = (x + 7)$ m.

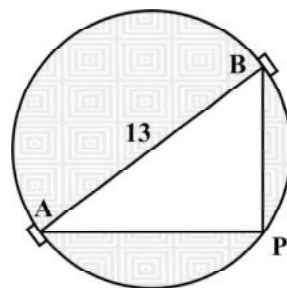


Fig. 10.4

Now, $AB = 13\text{m}$, and since AB is a diameter,

$$\angle APB = 90^\circ \quad (\text{Why?})$$

Therefore,

$$AP^2 + PB^2 = AB^2 \quad (\text{By Pythagoras theorem})$$

i.e.,

$$(x + 7)^2 + x^2 = 13^2$$

i.e.,

$$x^2 + 14x + 49 + x^2 = 169$$

i.e.,

$$2x^2 + 14x - 120 = 0$$

So, the distance ' x ' of the pole from gate B satisfies the equation

$$x^2 + 7x - 60 = 0$$

So, it would be possible to place the pole if this equation has real roots. To see if this is so or not, let us consider its discriminant. The discriminant is

$$b^2 - 4ac = 7^2 - 4 \times 1 \times (-60) = 289 > 0.$$

So, the given quadratic equation has two real roots, and it is possible to erect the pole on the boundary of the park.

Solving the quadratic equation $x^2 + 7x - 60 = 0$, by the quadratic formula, we get

$$x = \frac{-7 \pm \sqrt{289}}{2} = \frac{-7 \pm 17}{2}$$

Therefore, $x = 5$ or -12 .

Since x is the distance between the pole and the gate B, it must be positive. Therefore, $x = -12$ will have to be ignored. So, $x = 5$.

Thus, the pole has to be erected on the boundary of the park at a distance of 5m from the gate B and 12m from the gate A.

Example 18 : Find the discriminant of the equation $3x^2 - 2x + \frac{1}{3} = 0$ and hence find the nature of its roots. Find them, if they are real.

Solution : Here $a = 3$, $b = -2$ and $c = \frac{1}{3}$.

Therefore, discriminant $b^2 - 4ac = (-2)^2 - 4 \times 3 \times \frac{1}{3} = 4 - 4 = 0$.

Hence, the given quadratic equation has two equal real roots.

The roots are $\frac{-b}{2a}, \frac{-b}{2a}$, i.e., $\frac{2}{6}, \frac{2}{6}$, i.e., $\frac{1}{3}, \frac{1}{3}$.

EXERCISE 10.4

- Find the nature of the roots of the following quadratic equations. If the real roots exist, find them:
 - $2x^2 - 3x + 5 = 0$
 - $3x^2 - 4\sqrt{3}x + 4 = 0$
 - $2x^2 - 6x + 3 = 0$
- Find the values of k for each of the following quadratic equations, so that they have two equal roots.
 - $2x^2 + kx + 3 = 0$
 - $kx(x - 2) + 6 = 0$
- Is it possible to design a rectangular mango grove whose length is twice its breadth, and the area is 800 m^2 ? If so, find its length and breadth.
- Is the following situation possible? If so, determine their present ages.
The sum of the ages of two friends is 20 years. Four years ago, the product of their ages in years was 48.
- Is it possible to design a rectangular park of perimeter 80 m and area 400 m^2 ? If so, find its length and breadth.

10.6 Summary

In this chapter, you have studied the following points:

- A quadratic equation in the variable x is of the form $ax^2 + bx + c = 0$, where a, b, c are real numbers and $a \neq 0$.
- A real number α is said to be a root of the quadratic equation $ax^2 + bx + c = 0$, if $a\alpha^2 + b\alpha + c = 0$. The zeroes of the quadratic polynomial $ax^2 + bx + c$ and the roots of the quadratic equation $ax^2 + bx + c = 0$ are the same.
- If we can factorise $ax^2 + bx + c, a \neq 0$, into a product of two linear factors, then the roots of the quadratic equation $ax^2 + bx + c = 0$ can be found by equating each factor to zero.
- A quadratic equation can also be solved by the method of completing the square.
- Quadratic formula: The roots of a quadratic equation $ax^2 + bx + c = 0$ are given by

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \text{ provided } b^2 - 4ac \geq 0.$$

- A quadratic equation $ax^2 + bx + c = 0$ has
 - two distinct real roots, if $b^2 - 4ac > 0$,
 - two equal roots (i.e., coincident roots), if $b^2 - 4ac = 0$, and
 - no real roots, if $b^2 - 4ac < 0$.

A NOTE TO THE READER

In case of word problems, the obtained solutions should always be verified with the conditions of the original problem and not in the equations formed (see Examples 11, 13, 19 of Chapter 3 and Examples 10, 11, 12 of Chapter 10).

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INTRODUCTION TO

TRIGONOMETRY

11

There is perhaps nothing which so occupies the middle position of mathematics as trigonometry.

– J.F. Herbart (1890)

11.1 Introduction

You have already studied about triangles, and in particular, right triangles, in your earlier classes. Let us take some examples from our surroundings where right triangles can be imagined to be formed. For instance :

1. Suppose the students of a school are visiting Qutub Minar. Now, if a student is looking at the top of the Minar, a right triangle can be imagined to be made, as shown in Fig 11.1. Can the student find out the height of the Minar, without actually measuring it?
2. Suppose a girl is sitting on the balcony of her house located on the bank of a river. She is looking down at a flower pot placed on a stair of a temple situated nearby on the other bank of the river. A right triangle is imagined to be made in this situation as shown in Fig.11.2. If you know the height at which the person is sitting, can you find the width of the river?

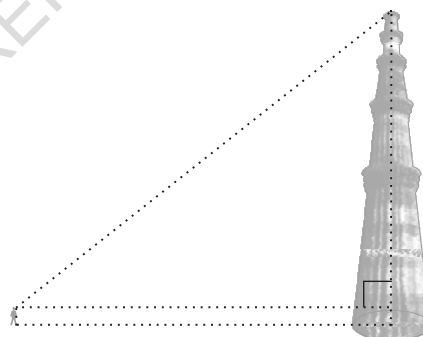


Fig. 11.1

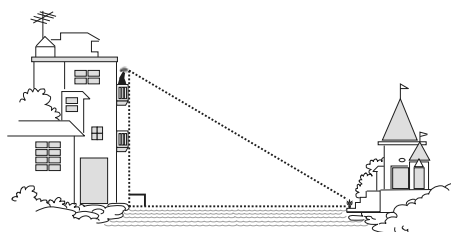


Fig. 11.2

3. Suppose a hot air balloon is flying in the air. A girl happens to spot the balloon in the sky and runs to her mother to tell her about it. Her mother rushes out of the house to look at the balloon. Now when the girl had spotted the balloon initially it was at point A. When both the mother and daughter came out to see it, it had already travelled to another point B. Can you find the altitude of B from the ground?

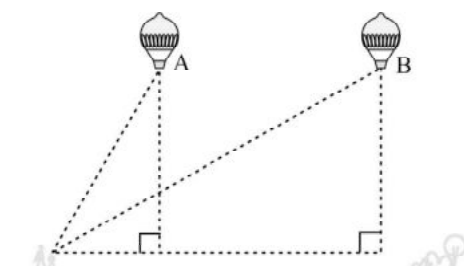


Fig. 11.3

In all the situations given above, the distances or heights can be found by using some mathematical techniques, which come under a branch of mathematics called 'trigonometry'. The word 'trigonometry' is derived from the Greek words 'tri' (meaning three), 'gon' (meaning sides) and 'metron' (meaning measure). In fact, **trigonometry** is the study of relationships between the sides and angles of a triangle. The earliest known work on trigonometry was recorded in Egypt and Babylon. Early astronomers used it to find out the distances of the stars and planets from the Earth. Even today, most of the technologically advanced methods used in Engineering and Physical Sciences are based on trigonometrical concepts.

In this chapter, we will study some ratios of the sides of a right triangle with respect to its acute angles, called **trigonometric ratios of the angle**. We will restrict our discussion to acute angles only. However, these ratios can be extended to other angles also. We will also define the trigonometric ratios for angles of measure 0° and 90° . We will calculate trigonometric ratios for some specific angles and establish some identities involving these ratios, called **trigonometric identities**.

11.2 Trigonometric Ratios

In Section 11.1, you have seen some right triangles imagined to be formed in different situations.

Let us take a right triangle ABC as shown in Fig. 11.4.

Here, $\angle CAB$ (or, in brief, angle A) is an acute angle. Note the position of the side BC with respect to angle A. It faces $\angle A$. We call it the *side opposite* to angle A. AC is the *hypotenuse* of the right triangle and the side AB is a part of $\angle A$. So, we call it the *side adjacent* to angle A.

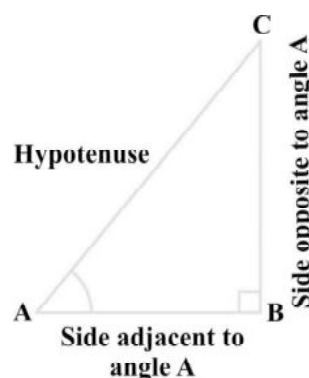


Fig. 11.4

Note that the position of sides change when you consider angle C in place of A (see Fig. 11.5).

You have studied the concept of 'ratio' in your earlier classes. We now define certain ratios involving the sides of a right triangle, and call them trigonometric ratios.

The trigonometric ratios of the angle A in right triangle ABC (see Fig. 11.4) are defined as follows :

$$\text{sine of } \angle A = \frac{\text{side opposite to angle A}}{\text{hypotenuse}} = \frac{BC}{AC}$$

$$\text{cosine of } \angle A = \frac{\text{side adjacent to angle A}}{\text{hypotenuse}} = \frac{AB}{AC}$$

$$\text{tangent of } \angle A = \frac{\text{side opposite to angle A}}{\text{side adjacent to angle A}} = \frac{BC}{AB}$$

$$\text{cosecant of } \angle A = \frac{1}{\text{sine of } \angle A} = \frac{\text{hypotenuse}}{\text{side opposite to angle A}} = \frac{AC}{BC}$$

$$\text{secant of } \angle A = \frac{1}{\text{cosine of } \angle A} = \frac{\text{hypotenuse}}{\text{side adjacent to angle A}} = \frac{AC}{AB}$$

$$\text{cotangent of } \angle A = \frac{1}{\text{tangent of } \angle A} = \frac{\text{side adjacent to angle A}}{\text{side opposite to angle A}} = \frac{AB}{BC}$$

The ratios defined above are abbreviated as $\sin A$, $\cos A$, $\tan A$, $\operatorname{cosec} A$, $\sec A$ and $\cot A$ respectively. Note that the ratios **cosec A**, **sec A** and **cot A** are respectively, the reciprocals of the ratios $\sin A$, $\cos A$ and $\tan A$.

$$\text{Also, observe that } \tan A = \frac{BC}{AB} = \frac{\frac{BC}{AC}}{\frac{AB}{AC}} = \frac{\sin A}{\cos A} \text{ and } \cot A = \frac{\cos A}{\sin A}.$$

So, the **trigonometric ratios** of an acute angle in a right triangle express the relationship between the angle and the length of its sides.

Why don't you try to define the trigonometric ratios for angle C in the right triangle? (See Fig. 11.5)

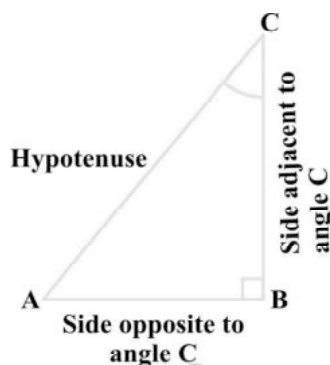


Fig. 11.5

The first use of the idea of ‘sine’ in the way we use it today was in the work *Aryabhatiyam* by Aryabhata, in A.D. 500. Aryabhata used the word *ardha-jya* for the half-chord, which was shortened to *jya* or *jiva* in due course. When the *Aryabhatiyam* was translated into Arabic, the word *jiva* was retained as it is. The word *jiva* was translated into *sinus*, which means curve, when the Arabic version was translated into Latin. Soon the word *sinus*, also used as *sine*, became common in mathematical texts throughout Europe. An English Professor of astronomy Edmund Gunter (1581–1626), first used the abbreviated notation ‘sin’.



Aryabhata
C.E. 476 – 550

The origin of the terms ‘cosine’ and ‘tangent’ was much later. The cosine function arose from the need to compute the sine of the complementary angle. Aryabhata called it *kotijya*. The name *cosinus* originated with Edmund Gunter. In 1674, the English Mathematician Sir Jonas Moore first used the abbreviated notation ‘cos’.

Remark : Note that the symbol $\sin A$ is used as an abbreviation for ‘the sine of the angle A ’. $\sin A$ is *not* the product of ‘sin’ and A . ‘sin’ separated from A has no meaning. Similarly, $\cos A$ is *not* the product of ‘cos’ and A . Similar interpretations follow for other trigonometric ratios also.

Now, if we take a point P on the hypotenuse AC or a point Q on AC extended, of the right triangle ABC and draw PM perpendicular to AB and QN perpendicular to AB extended (see Fig. 11.6), how will the trigonometric ratios of $\angle A$ in $\triangle PAM$ differ from those of $\angle A$ in $\triangle CAB$ or from those of $\angle A$ in $\triangle QAN$?

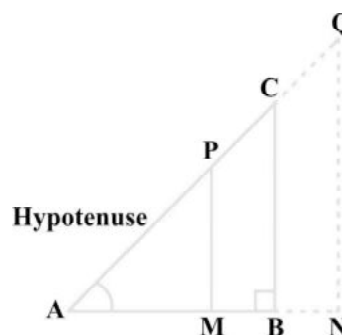


Fig. 11.6

To answer this, first look at these triangles. Is $\triangle PAM$ similar to $\triangle CAB$? From Chapter 6, recall the AA similarity criterion. Using the criterion, you will see that the triangles PAM and CAB are similar. Therefore, by the property of similar triangles, the corresponding sides of the triangles are proportional.

So, we have

$$\frac{AM}{AB} = \frac{AP}{AC} = \frac{MP}{BC}.$$

From this, we find $\frac{MP}{AP} = \frac{BC}{AC} = \sin A$.

Similarly, $\frac{AM}{AP} = \frac{AB}{AC} = \cos A$, $\frac{MP}{AM} = \frac{BC}{AB} = \tan A$ and so on.

This shows that the trigonometric ratios of angle A in $\triangle PAM$ not differ from those of angle A in $\triangle CAB$.

In the same way, you should check that the value of $\sin A$ (and also of other trigonometric ratios) remains the same in $\triangle QAN$ also.

From our observations, it is now clear that **the values of the trigonometric ratios of an angle do not vary with the lengths of the sides of the triangle, if the angle remains the same.**

Note : For the sake of convenience, we may write $\sin^2 A$, $\cos^2 A$, etc., in place of $(\sin A)^2$, $(\cos A)^2$, etc., respectively. But $\operatorname{cosec} A = (\sin A)^{-1} \neq \sin^{-1} A$ (it is called sine inverse A). $\sin^{-1} A$ has a different meaning, which will be discussed in higher classes. Similar conventions hold for the other trigonometric ratios as well. Sometimes, the Greek letter θ (theta) is also used to denote an angle.

We have defined six trigonometric ratios of an acute angle. If we know any one of the ratios, can we obtain the other ratios? Let us see.

If in a right triangle ABC , $\sin A = \frac{1}{3}$, then this means that $\frac{BC}{AC} = \frac{1}{3}$, i.e., the lengths of the sides BC and AC of the triangle ABC are in the ratio $1 : 3$ (see Fig. 11.7). So if BC is equal to k , then AC will be $3k$, where k is any positive number. To determine other trigonometric ratios for the angle A , we need to find the length of the third side AB . Do you remember the Pythagoras theorem? Let us use it to determine the required length AB .

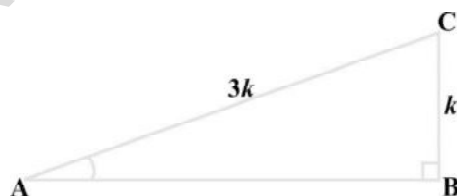


Fig. 11.7

$$AB^2 = AC^2 - BC^2 = (3k)^2 - (k)^2 = 8k^2 = (2\sqrt{2} k)^2$$

Therefore, $AB = \pm 2\sqrt{2} k$

So, we get $AB = 2\sqrt{2} k$ (Why is AB not $-2\sqrt{2} k$?)

$$\text{Now, } \cos A = \frac{AB}{AC} = \frac{2\sqrt{2} k}{3k} = \frac{2\sqrt{2}}{3}$$

Similarly, you can obtain the other trigonometric ratios of the angle A .

Remark : Since the hypotenuse is the longest side in a right triangle, the value of $\sin A$ or $\cos A$ is always less than 1 (or, in particular, equal to 1).

Let us consider some examples.

Example 1 : Given $\tan A = \frac{4}{3}$, find the other trigonometric ratios of the angle A .

Solution : Let us first draw a right ΔABC (see Fig 11.8).

Now, we know that $\tan A = \frac{BC}{AB} = \frac{4}{3}$.

Therefore, if $BC = 4k$, then $AB = 3k$, where k is a positive number.

Now, by using the Pythagoras Theorem, we have

$$AC^2 = AB^2 + BC^2 = (4k)^2 + (3k)^2 = 25k^2$$

So,

$$AC = 5k$$

Now, we can write all the trigonometric ratios using their definitions.

$$\sin A = \frac{BC}{AC} = \frac{4k}{5k} = \frac{4}{5}$$

$$\cos A = \frac{AB}{AC} = \frac{3k}{5k} = \frac{3}{5}$$

Therefore, $\cot A = \frac{1}{\tan A} = \frac{3}{4}$, $\operatorname{cosec} A = \frac{1}{\sin A} = \frac{5}{4}$ and $\sec A = \frac{1}{\cos A} = \frac{5}{3}$.

Example 2 : If $\angle B$ and $\angle Q$ are acute angles such that $\sin B = \sin Q$, then prove that $\angle B = \angle Q$.

Solution : Let us consider two right triangles ABC and PQR where $\sin B = \sin Q$ (see Fig. 11.9).

We have

$$\sin B = \frac{AC}{AB}$$

and

$$\sin Q = \frac{PR}{PQ}$$



Fig. 11.8

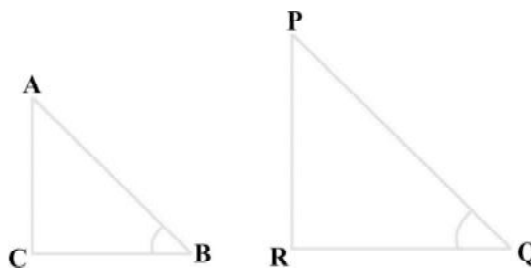


Fig. 11.9

Then
$$\frac{AC}{AB} = \frac{PR}{PQ}$$

Therefore,
$$\frac{AC}{PR} = \frac{AB}{PQ} = k, \text{ say} \quad (1)$$

Now, using Pythagoras theorem,

$$BC = \sqrt{AB^2 - AC^2}$$

and

$$QR = \sqrt{PQ^2 - PR^2}$$

So,
$$\frac{BC}{QR} = \frac{\sqrt{AB^2 - AC^2}}{\sqrt{PQ^2 - PR^2}} = \frac{\sqrt{k^2 PQ^2 - k^2 PR^2}}{\sqrt{PQ^2 - PR^2}} = \frac{k \sqrt{PQ^2 - PR^2}}{\sqrt{PQ^2 - PR^2}} = k \quad (2)$$

From (1) and (2), we have

$$\frac{AC}{PR} = \frac{AB}{PQ} = \frac{BC}{QR}$$

Then, by using Theorem 6.4, $\Delta ACB \sim \Delta PRQ$ and therefore, $\angle B = \angle Q$.

Example 3 : Consider ΔACB , right-angled at C, in which $AB = 29$ units, $BC = 21$ units and $\angle ABC = \theta$ (see Fig. 11.10). Determine the values of

(i) $\cos^2 \theta + \sin^2 \theta$,

(ii) $\cos^2 \theta - \sin^2 \theta$.

Solution : In ΔACB , we have

$$\begin{aligned} AC &= \sqrt{AB^2 - BC^2} = \sqrt{(29)^2 - (21)^2} \\ &= \sqrt{(29 - 21)(29 + 21)} = \sqrt{(8)(50)} = \sqrt{400} = 20 \text{ units} \end{aligned}$$

So, $\sin \theta = \frac{AC}{AB} = \frac{20}{29}, \cos \theta = \frac{BC}{AB} = \frac{21}{29}$.

Now, (i) $\cos^2 \theta + \sin^2 \theta = \left(\frac{20}{29}\right)^2 + \left(\frac{21}{29}\right)^2 = \frac{20^2 + 21^2}{29^2} = \frac{400 + 441}{841} = 1$,

and (ii) $\cos^2 \theta - \sin^2 \theta = \left(\frac{21}{29}\right)^2 - \left(\frac{20}{29}\right)^2 = \frac{(21 + 20)(21 - 20)}{29^2} = \frac{41}{841}$.

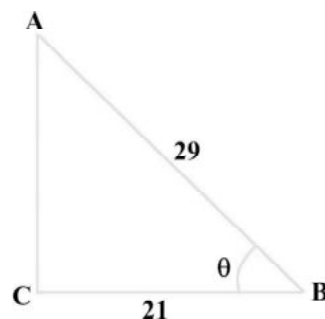


Fig. 11.10

Example 4 : In a right triangle ABC, right-angled at B, if $\tan A = 1$, then verify that

$$2 \sin A \cos A = 1.$$

Solution : In ΔABC , $\tan A = \frac{BC}{AB} = 1$ (see Fig 11.11)

i.e., $BC = AB$

Let $AB = BC = k$, where k is a positive number.

$$\begin{aligned} \text{Now,} \quad AC &= \sqrt{AB^2 + BC^2} \\ &= \sqrt{(k)^2 + (k)^2} = k\sqrt{2} \end{aligned}$$

Therefore, $\sin A = \frac{BC}{AC} = \frac{1}{\sqrt{2}}$ and $\cos A = \frac{AB}{AC} = \frac{1}{\sqrt{2}}$

So, $2 \sin A \cos A = 2 \left(\frac{1}{\sqrt{2}} \right) \left(\frac{1}{\sqrt{2}} \right) = 1$, which is the required value.

Example 5 : In ΔOPQ , right-angled at P, $OP = 7$ cm and $OQ - PQ = 1$ cm (see Fig. 11.12). Determine the values of $\sin Q$ and $\cos Q$.

Solution : In ΔOPQ , we have

$$OQ^2 = OP^2 + PQ^2$$

i.e., $(1 + PQ)^2 = OP^2 + PQ^2$ (Why?)

i.e., $1 + PQ^2 + 2PQ = OP^2 + PQ^2$

i.e., $1 + 2PQ = 7^2$ (Why?)

i.e., $PQ = 24$ cm and $OQ = 1 + PQ = 25$ cm

So, $\sin Q = \frac{7}{25}$ and $\cos Q = \frac{24}{25}$.

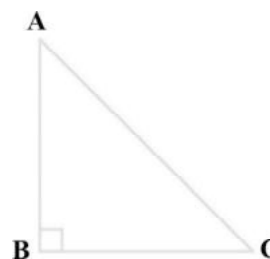


Fig. 11.11



Fig. 11.12

EXERCISE 11.1

- In $\triangle ABC$, right-angled at B, $AB = 24$ cm, $BC = 7$ cm. Determine :
 - $\sin A$, $\cos A$
 - $\sin C$, $\cos C$
- In Fig. 8.13, find $\tan P - \cot R$.
- If $\sin A = \frac{3}{4}$, calculate $\cos A$ and $\tan A$.
- Given $15 \cot A = 8$, find $\sin A$ and $\sec A$.
- Given $\sec \theta = \frac{13}{12}$, calculate all other trigonometric ratios.
- If $\angle A$ and $\angle B$ are acute angles such that $\cos A = \cos B$, then show that $\angle A = \angle B$.
- If $\cot \theta = \frac{7}{8}$, evaluate : (i) $\frac{(1 + \sin \theta)(1 - \sin \theta)}{(1 + \cos \theta)(1 - \cos \theta)}$, (ii) $\cot^2 \theta$
- If $3 \cot A = 4$, check whether $\frac{1 - \tan^2 A}{1 + \tan^2 A} = \cos^2 A - \sin^2 A$ or not.
- In triangle ABC, right-angled at B, if $\tan A = \frac{1}{\sqrt{3}}$, find the value of:
 - $\sin A \cos C + \cos A \sin C$
 - $\cos A \cos C - \sin A \sin C$
- In $\triangle PQR$, right-angled at Q, $PR + QR = 25$ cm and $PQ = 5$ cm. Determine the values of $\sin P$, $\cos P$ and $\tan P$.
- State whether the following are true or false. Justify your answer.
 - The value of $\tan A$ is always less than 1.
 - $\sec A = \frac{12}{5}$ for some value of angle A.
 - $\cos A$ is the abbreviation used for the cosecant of angle A.
 - $\cot A$ is the product of $\cot$ and A.
 - $\sin \theta = \frac{4}{3}$ for some angle θ .

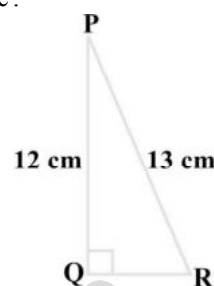


Fig. 11.13

11.3 Trigonometric Ratios of Some Specific Angles

From geometry, you are already familiar with the construction of angles of 30° , 45° , 60° and 90° . In this section, we will find the values of the trigonometric ratios for these angles and, of course, for 0° .

Trigonometric Ratios of 45°

In $\triangle ABC$, right-angled at B, if one angle is 45° , then the other angle is also 45° , i.e., $\angle A = \angle C = 45^\circ$ (see Fig. 11.14).

So, $BC = AB$ (Why?)

Now, Suppose $BC = AB = a$.

Then by Pythagoras Theorem, $AC^2 = AB^2 + BC^2 = a^2 + a^2 = 2a^2$,

and, therefore, $AC = a\sqrt{2}$.

Using the definitions of the trigonometric ratios, we have :

$$\sin 45^\circ = \frac{\text{side opposite to angle } 45^\circ}{\text{hypotenuse}} = \frac{BC}{AC} = \frac{a}{a\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\cos 45^\circ = \frac{\text{side adjacent to angle } 45^\circ}{\text{hypotenuse}} = \frac{AB}{AC} = \frac{a}{a\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$$\tan 45^\circ = \frac{\text{side opposite to angle } 45^\circ}{\text{side adjacent to angle } 45^\circ} = \frac{BC}{AB} = \frac{a}{a} = 1$$

$$\text{Also, } \operatorname{cosec} 45^\circ = \frac{1}{\sin 45^\circ} = \sqrt{2}, \sec 45^\circ = \frac{1}{\cos 45^\circ} = \sqrt{2}, \cot 45^\circ = \frac{1}{\tan 45^\circ} = 1.$$

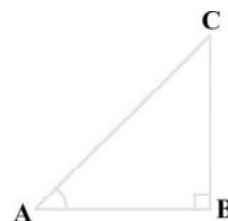


Fig. 11.14

Trigonometric Ratios of 30° and 60°

Let us now calculate the trigonometric ratios of 30° and 60° . Consider an equilateral triangle ABC. Since each angle in an equilateral triangle is 60° , therefore, $\angle A = \angle B = \angle C = 60^\circ$.

Draw the perpendicular AD from A to the side BC (see Fig. 11.15).

Now

$$\triangle ABD \cong \triangle ACD$$

(Why?) Fig. 11.15

Therefore,

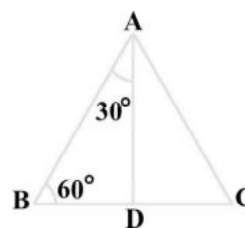
$$BD = DC$$

and

$$\angle BAD = \angle CAD \quad (\text{CPCT})$$

Now observe that:

$\triangle ABD$ is a right triangle, right-angled at D with $\angle BAD = 30^\circ$ and $\angle ABD = 60^\circ$ (see Fig. 11.15).



As you know, for finding the trigonometric ratios, we need to know the lengths of the sides of the triangle. So, let us suppose that $AB = 2a$.

Then,
$$BD = \frac{1}{2}BC = a$$

and
$$AD^2 = AB^2 - BD^2 = (2a)^2 - (a)^2 = 3a^2,$$

Therefore,
$$AD = a\sqrt{3}$$

Now, we have :

$$\sin 30^\circ = \frac{BD}{AB} = \frac{a}{2a} = \frac{1}{2}, \cos 30^\circ = \frac{AD}{AB} = \frac{a\sqrt{3}}{2a} = \frac{\sqrt{3}}{2}$$

$$\tan 30^\circ = \frac{BD}{AD} = \frac{a}{a\sqrt{3}} = \frac{1}{\sqrt{3}}.$$

Also,
$$\operatorname{cosec} 30^\circ = \frac{1}{\sin 30^\circ} = 2, \sec 30^\circ = \frac{1}{\cos 30^\circ} = \frac{2}{\sqrt{3}}$$

$$\cot 30^\circ = \frac{1}{\tan 30^\circ} = \sqrt{3}.$$

Similarly,

$$\sin 60^\circ = \frac{AD}{AB} = \frac{a\sqrt{3}}{2a} = \frac{\sqrt{3}}{2}, \cos 60^\circ = \frac{1}{2}, \tan 60^\circ = \sqrt{3},$$

$$\operatorname{cosec} 60^\circ = \frac{2}{\sqrt{3}}, \sec 60^\circ = 2 \text{ and } \cot 60^\circ = \frac{1}{\sqrt{3}}.$$

Trigonometric Ratios of 0° and 90°

Let us see what happens to the trigonometric ratios of angle A, if it is made smaller and smaller in the right triangle ABC (see Fig. 11.16), till it becomes zero. As $\angle A$ gets smaller and smaller, the length of the side BC decreases. The point C gets closer to point B, and finally when $\angle A$ becomes very close to 0° , AC becomes almost the same as AB (see Fig. 11.17).

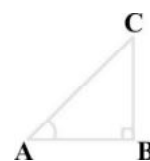


Fig. 11.16

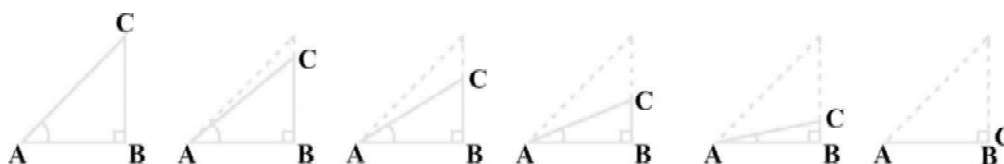


Fig. 11.17

When $\angle A$ is very close to 0° , BC gets very close to 0 and so the value of $\sin A = \frac{BC}{AC}$ is very close to 0. Also, when $\angle A$ is very close to 0° , AC is nearly the

same as AB and so the value of $\cos A = \frac{AB}{AC}$ is very close to 1.

This helps us to see how we can define the values of $\sin A$ and $\cos A$ when $A = 0^\circ$. We define : **$\sin 0^\circ = 0$ and $\cos 0^\circ = 1$.**

Using these, we have :

$$\tan 0^\circ = \frac{\sin 0^\circ}{\cos 0^\circ} = 0, \cot 0^\circ = \frac{1}{\tan 0^\circ}, \text{ which is not defined. (Why?)}$$

$$\sec 0^\circ = \frac{1}{\cos 0^\circ} = 1 \text{ and } \operatorname{cosec} 0^\circ = \frac{1}{\sin 0^\circ}, \text{ which is again not defined. (Why?)}$$

Now, let us see what happens to the trigonometric ratios of $\angle A$, when it is made larger and larger in $\triangle ABC$ till it becomes 90° . As $\angle A$ gets larger and larger, $\angle C$ gets smaller and smaller. Therefore, as in the case above, the length of the side AB goes on decreasing. The point A gets closer to point B . Finally when $\angle A$ is very close to 90° , $\angle C$ becomes very close to 0° and the side AC almost coincides with side BC (see Fig. 11.18).

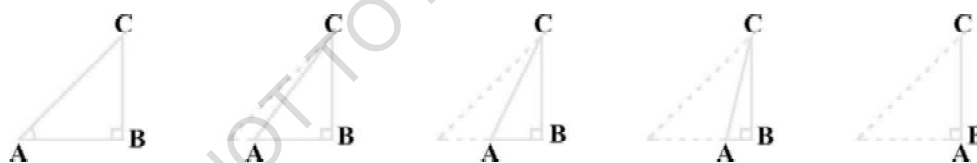


Fig. 11.18

When $\angle C$ is very close to 0° , $\angle A$ is very close to 90° , side AC is nearly the same as side BC , and so $\sin A$ is very close to 1. Also when $\angle A$ is very close to 90° , $\angle C$ is very close to 0° , and the side AB is nearly zero, so $\cos A$ is very close to 0.

So, we define : **$\sin 90^\circ = 1$ and $\cos 90^\circ = 0$.**

Now, why don't you find the other trigonometric ratios of 90° ?

We shall now give the values of all the trigonometric ratios of 0° , 30° , 45° , 60° and 90° in Table 11.1, for ready reference.

Table 11.1

$\angle A$	0°	30°	45°	60°	90°
$\sin A$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos A$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan A$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined
$\operatorname{cosec} A$	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec A$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined
$\cot A$	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

Remark : From the table above you can observe that as $\angle A$ increases from 0° to 90° , $\sin A$ increases from 0 to 1 and $\cos A$ decreases from 1 to 0.

Let us illustrate the use of the values in the table above through some examples.

Example 6 : In $\triangle ABC$, right-angled at B, $AB = 5$ cm and $\angle ACB = 30^\circ$ (see Fig. 11.19). Determine the lengths of the sides BC and AC.

Solution : To find the length of the side BC, we will choose the trigonometric ratio involving BC and the given side AB. Since BC is the side adjacent to angle C and AB is the side opposite to angle C, therefore

$$\frac{AB}{BC} = \tan C$$

$$\text{i.e.,} \quad \frac{5}{BC} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$

which gives

$$BC = 5\sqrt{3} \text{ cm}$$

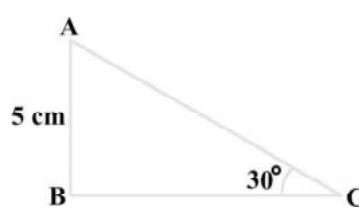


Fig. 11.19

To find the length of the side AC, we consider

$$\sin 30^\circ = \frac{AB}{AC} \quad (\text{Why?})$$

$$\text{i.e.,} \quad \frac{1}{2} = \frac{5}{AC}$$

$$\text{i.e.,} \quad AC = 10 \text{ cm}$$

Note that alternatively we could have used Pythagoras theorem to determine the third side in the example above,

$$\text{i.e.,} \quad AC = \sqrt{AB^2 + BC^2} = \sqrt{5^2 + (5\sqrt{3})^2} \text{ cm} = 10 \text{ cm}.$$

Example 7 : In ΔPQR , right-angled at Q (see Fig. 11.20), $PQ = 3 \text{ cm}$ and $PR = 6 \text{ cm}$. Determine $\angle QPR$ and $\angle PRQ$.

Solution : Given $PQ = 3 \text{ cm}$ and $PR = 6 \text{ cm}$.

$$\text{Therefore,} \quad \frac{PQ}{PR} = \sin R$$

$$\text{or} \quad \sin R = \frac{3}{6} = \frac{1}{2}$$

$$\text{So,} \quad \angle PRQ = 30^\circ$$

$$\text{and therefore,} \quad \angle QPR = 60^\circ. \quad (\text{Why?})$$

You may note that if one of the sides and any other part (either an acute angle or any side) of a right triangle is known, the remaining sides and angles of the triangle can be determined.

Example 8 : If $\sin (A - B) = \frac{1}{2}$, $\cos (A + B) = \frac{1}{2}$, $0^\circ < A + B \leq 90^\circ$, $A > B$, find A and B.

$$\text{Solution : Since, } \sin (A - B) = \frac{1}{2}, \text{ therefore, } A - B = 30^\circ \quad (\text{Why?}) \quad (1)$$

$$\text{Also, since } \cos (A + B) = \frac{1}{2}, \text{ therefore, } A + B = 60^\circ \quad (\text{Why?}) \quad (2)$$

Solving (1) and (2), we get : $A = 45^\circ$ and $B = 15^\circ$.

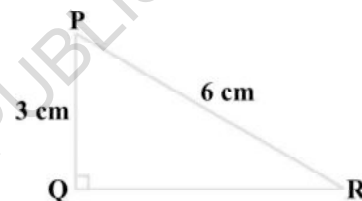


Fig. 11.20

EXERCISE 11.2

1. Evaluate the following :

- (i) $\sin 60^\circ \cos 30^\circ + \sin 30^\circ \cos 60^\circ$ (ii) $2 \tan^2 45^\circ + \cos^2 30^\circ - \sin^2 60^\circ$
- (iii) $\frac{\cos 45^\circ}{\sec 30^\circ + \operatorname{cosec} 30^\circ}$ (iv) $\frac{\sin 30^\circ + \tan 45^\circ - \operatorname{cosec} 60^\circ}{\sec 30^\circ + \cos 60^\circ + \cot 45^\circ}$
- (v) $\frac{5 \cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ}{\sin^2 30^\circ + \cos^2 30^\circ}$

2. Choose the correct option and justify your choice :

- (i) $\frac{2 \tan 30^\circ}{1 + \tan^2 30^\circ} =$
 (A) $\sin 60^\circ$ (B) $\cos 60^\circ$ (C) $\tan 60^\circ$ (D) $\sin 30^\circ$
- (ii) $\frac{1 - \tan^2 45^\circ}{1 + \tan^2 45^\circ} =$
 (A) $\tan 90^\circ$ (B) 1 (C) $\sin 45^\circ$ (D) 0
- (iii) $\sin 2A = 2 \sin A$ is true when $A =$
 (A) 0° (B) 30° (C) 45° (D) 60°
- (iv) $\frac{2 \tan 30^\circ}{1 - \tan^2 30^\circ} =$
 (A) $\cos 60^\circ$ (B) $\sin 60^\circ$ (C) $\tan 60^\circ$ (D) $\sin 30^\circ$

3. If $\tan (A + B) = \sqrt{3}$ and $\tan (A - B) = \frac{1}{\sqrt{3}}$; $0^\circ < A + B \leq 90^\circ$; $A > B$, find A and B.

4. State whether the following are true or false. Justify your answer.

- (i) $\sin (A + B) = \sin A + \sin B$.
 (ii) The value of $\sin \theta$ increases as θ increases.
 (iii) The value of $\cos \theta$ increases as θ increases.
 (iv) $\sin \theta = \cos \theta$ for all values of θ .
 (v) $\cot A$ is not defined for $A = 0^\circ$.

8.4 Trigonometric Ratios of Complementary Angles

Recall that two angles are said to be complementary if their sum equals 90° . In $\triangle ABC$, right-angled at B, do you see any pair of complementary angles? (See Fig. 11.21)

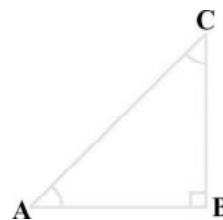


Fig. 11.21

Since $\angle A + \angle C = 90^\circ$, they form such a pair. We have:

$$\left. \begin{array}{lll} \sin A = \frac{BC}{AC} & \cos A = \frac{AB}{AC} & \tan A = \frac{BC}{AB} \\ \operatorname{cosec} A = \frac{AC}{BC} & \sec A = \frac{AC}{AB} & \cot A = \frac{AB}{BC} \end{array} \right\} \quad (1)$$

Now let us write the trigonometric ratios for $\angle C = 90^\circ - \angle A$.

For convenience, we shall write $90^\circ - A$ instead of $90^\circ - \angle A$.

What would be the side opposite and the side adjacent to the angle $90^\circ - A$?

You will find that AB is the side opposite and BC is the side adjacent to the angle $90^\circ - A$. Therefore,

$$\left. \begin{array}{lll} \sin (90^\circ - A) = \frac{AB}{AC}, & \cos (90^\circ - A) = \frac{BC}{AC}, & \tan (90^\circ - A) = \frac{AB}{BC} \\ \operatorname{cosec} (90^\circ - A) = \frac{AC}{AB}, & \sec (90^\circ - A) = \frac{AC}{BC}, & \cot (90^\circ - A) = \frac{BC}{AB} \end{array} \right\} \quad (2)$$

Now, compare the ratios in (1) and (2). Observe that :

$$\sin (90^\circ - A) = \frac{AB}{AC} = \cos A \text{ and } \cos (90^\circ - A) = \frac{BC}{AC} = \sin A$$

$$\text{Also, } \tan (90^\circ - A) = \frac{AB}{BC} = \cot A, \quad \cot (90^\circ - A) = \frac{BC}{AB} = \tan A$$

$$\sec (90^\circ - A) = \frac{AC}{BC} = \operatorname{cosec} A, \quad \operatorname{cosec} (90^\circ - A) = \frac{AC}{AB} = \sec A$$

$$\begin{array}{ll} \text{So, } \sin (90^\circ - A) = \cos A, & \cos (90^\circ - A) = \sin A, \\ \tan (90^\circ - A) = \cot A, & \cot (90^\circ - A) = \tan A, \\ \sec (90^\circ - A) = \operatorname{cosec} A, & \operatorname{cosec} (90^\circ - A) = \sec A, \end{array}$$

for all values of angle A lying between 0° and 90° . Check whether this holds for $A = 0^\circ$ or $A = 90^\circ$.

Note : $\tan 0^\circ = 0 = \cot 90^\circ$, $\sec 0^\circ = 1 = \operatorname{cosec} 90^\circ$ and $\sec 90^\circ$, $\operatorname{cosec} 0^\circ$, $\tan 90^\circ$ and $\cot 0^\circ$ are not defined.

Now, let us consider some examples.

Example 9 : Evaluate $\frac{\tan 65^\circ}{\cot 25^\circ}$.

Solution : We know : $\cot A = \tan (90^\circ - A)$
 So, $\cot 25^\circ = \tan (90^\circ - 25^\circ) = \tan 65^\circ$
 i.e., $\frac{\tan 65^\circ}{\cot 25^\circ} = \frac{\tan 65^\circ}{\tan 65^\circ} = 1$

Example 10 : If $\sin 3A = \cos (A - 26^\circ)$, where $3A$ is an acute angle, find the value of A .

Solution : We are given that $\sin 3A = \cos (A - 26^\circ)$. (1)

Since $\sin 3A = \cos (90^\circ - 3A)$, we can write (1) as

$$\cos (90^\circ - 3A) = \cos (A - 26^\circ)$$

Since $90^\circ - 3A$ and $A - 26^\circ$ are both acute angles, therefore,

$$90^\circ - 3A = A - 26^\circ$$

which gives

$$A = 29^\circ$$

Example 11 : Express $\cot 85^\circ + \cos 75^\circ$ in terms of trigonometric ratios of angles between 0° and 45° .

Solution : $\cot 85^\circ + \cos 75^\circ = \cot (90^\circ - 5^\circ) + \cos (90^\circ - 15^\circ)$
 $= \tan 5^\circ + \sin 15^\circ$

EXERCISE 11.3

1. Evaluate :

$$(i) \frac{\sin 18^\circ}{\cos 72^\circ} \quad (ii) \frac{\tan 26^\circ}{\cot 64^\circ} \quad (iii) \cos 48^\circ - \sin 42^\circ \quad (iv) \operatorname{cosec} 31^\circ - \sec 59^\circ$$

2. Show that :

$$(i) \tan 48^\circ \tan 23^\circ \tan 42^\circ \tan 67^\circ = 1$$

$$(ii) \cos 38^\circ \cos 52^\circ - \sin 38^\circ \sin 52^\circ = 0$$

3. If $\tan 2A = \cot (A - 18^\circ)$, where $2A$ is an acute angle, find the value of A .

4. If $\tan A = \cot B$, prove that $A + B = 90^\circ$.

5. If $\sec 4A = \operatorname{cosec} (A - 20^\circ)$, where $4A$ is an acute angle, find the value of A .

6. If A, B and C are interior angles of a triangle ABC, then show that

$$\sin\left(\frac{B+C}{2}\right) = \cos \frac{A}{2}.$$

7. Express $\sin 67^\circ + \cos 75^\circ$ in terms of trigonometric ratios of angles between 0° and 45° .

11.5 Trigonometric Identities

You may recall that an equation is called an identity when it is true for all values of the variables involved. Similarly, an equation involving trigonometric ratios of an angle is called a **trigonometric identity**, if it is true for all values of the angle(s) involved.

In this section, we will prove one trigonometric identity, and use it further to prove other useful trigonometric identities.

In $\triangle ABC$, right-angled at B (see Fig. 11.22), we have:

$$AB^2 + BC^2 = AC^2 \quad (1)$$

Dividing each term of (1) by AC^2 , we get

$$\frac{AB^2}{AC^2} + \frac{BC^2}{AC^2} = \frac{AC^2}{AC^2}$$

$$\text{i.e.,} \quad \left(\frac{AB}{AC}\right)^2 + \left(\frac{BC}{AC}\right)^2 = \left(\frac{AC}{AC}\right)^2$$

$$\text{i.e.,} \quad (\cos A)^2 + (\sin A)^2 = 1$$

$$\text{i.e.,} \quad \cos^2 A + \sin^2 A = 1 \quad (2)$$

This is true for all A such that $0^\circ \leq A \leq 90^\circ$. So, this is a trigonometric identity.

Let us now divide (1) by AB^2 . We get

$$\frac{AB^2}{AB^2} + \frac{BC^2}{AB^2} = \frac{AC^2}{AB^2}$$

$$\text{or,} \quad \left(\frac{AB}{AB}\right)^2 + \left(\frac{BC}{AB}\right)^2 = \left(\frac{AC}{AB}\right)^2$$

$$\text{i.e.,} \quad 1 + \tan^2 A = \sec^2 A \quad (3)$$

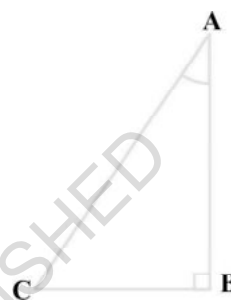


Fig. 11.22

Is this equation true for $A = 0^\circ$? Yes, it is. What about $A = 90^\circ$? Well, $\tan A$ and $\sec A$ are not defined for $A = 90^\circ$. So, (3) is true for all A such that $0^\circ \leq A < 90^\circ$.

Let us see what we get on dividing (1) by BC^2 . We get

$$\frac{AB^2}{BC^2} + \frac{BC^2}{BC^2} = \frac{AC^2}{BC^2}$$

$$\text{i.e.,} \quad \left(\frac{AB}{BC}\right)^2 + \left(\frac{BC}{BC}\right)^2 = \left(\frac{AC}{BC}\right)^2$$

$$\text{i.e.,} \quad \cot^2 A + 1 = \operatorname{cosec}^2 A \quad (4)$$

Note that $\operatorname{cosec} A$ and $\cot A$ are not defined for $A = 0^\circ$. Therefore (4) is true for all A such that $0^\circ < A \leq 90^\circ$.

Using these identities, we can express each trigonometric ratio in terms of other trigonometric ratios, i.e., if any one of the ratios is known, we can also determine the values of other trigonometric ratios.

Let us see how we can do this using these identities. Suppose we know that $\tan A = \frac{1}{\sqrt{3}}$. Then, $\cot A = \sqrt{3}$.

Since, $\sec^2 A = 1 + \tan^2 A = 1 + \frac{1}{3} = \frac{4}{3}$, $\sec A = \frac{2}{\sqrt{3}}$, and $\cos A = \frac{\sqrt{3}}{2}$.

Again, $\sin A = \sqrt{1 - \cos^2 A} = \sqrt{1 - \frac{3}{4}} = \frac{1}{2}$. Therefore, $\operatorname{cosec} A = 2$.

Example 12 : Express the ratios $\cos A$, $\tan A$ and $\sec A$ in terms of $\sin A$.

Solution : Since $\cos^2 A + \sin^2 A = 1$, therefore,

$$\cos^2 A = 1 - \sin^2 A, \text{ i.e., } \cos A = \pm \sqrt{1 - \sin^2 A}$$

This gives $\cos A = \sqrt{1 - \sin^2 A}$ (Why?)

Hence, $\tan A = \frac{\sin A}{\cos A} = \frac{\sin A}{\sqrt{1 - \sin^2 A}}$ and $\sec A = \frac{1}{\cos A} = \frac{1}{\sqrt{1 - \sin^2 A}}$

Example 13 : Prove that $\sec A (1 - \sin A)(\sec A + \tan A) = 1$.

Solution :

$$\begin{aligned} \text{LHS} &= \sec A (1 - \sin A)(\sec A + \tan A) = \left(\frac{1}{\cos A} \right) (1 - \sin A) \left(\frac{1}{\cos A} + \frac{\sin A}{\cos A} \right) \\ &= \frac{(1 - \sin A)(1 + \sin A)}{\cos^2 A} = \frac{1 - \sin^2 A}{\cos^2 A} \\ &= \frac{\cos^2 A}{\cos^2 A} = 1 = \text{RHS} \end{aligned}$$

Example 14 : Prove that $\frac{\cot A - \cos A}{\cot A + \cos A} = \frac{\operatorname{cosec} A - 1}{\operatorname{cosec} A + 1}$

$$\begin{aligned} \text{Solution : LHS} &= \frac{\cot A - \cos A}{\cot A + \cos A} = \frac{\frac{\cos A}{\sin A} - \cos A}{\frac{\cos A}{\sin A} + \cos A} \\ &= \frac{\cos A \left(\frac{1}{\sin A} - 1 \right)}{\cos A \left(\frac{1}{\sin A} + 1 \right)} = \frac{\left(\frac{1}{\sin A} - 1 \right)}{\left(\frac{1}{\sin A} + 1 \right)} = \frac{\operatorname{cosec} A - 1}{\operatorname{cosec} A + 1} = \text{RHS} \end{aligned}$$

Example 15 : Prove that $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = \frac{1}{\sec \theta - \tan \theta}$, using the identity $\sec^2 \theta = 1 + \tan^2 \theta$.

Solution : Since we will apply the identity involving $\sec \theta$ and $\tan \theta$, let us first convert the LHS (of the identity we need to prove) in terms of $\sec \theta$ and $\tan \theta$ by dividing numerator and denominator by $\cos \theta$.

$$\begin{aligned} \text{LHS} &= \frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = \frac{\tan \theta - 1 + \sec \theta}{\tan \theta + 1 - \sec \theta} \\ &= \frac{(\tan \theta + \sec \theta) - 1}{(\tan \theta - \sec \theta) + 1} = \frac{\{(\tan \theta + \sec \theta) - 1\} (\tan \theta - \sec \theta)}{\{(\tan \theta - \sec \theta) + 1\} (\tan \theta - \sec \theta)} \end{aligned}$$

$$\begin{aligned}
 &= \frac{(\tan^2 \theta - \sec^2 \theta) - (\tan \theta - \sec \theta)}{\{\tan \theta - \sec \theta + 1\} (\tan \theta - \sec \theta)} \\
 &= \frac{-1 - \tan \theta + \sec \theta}{(\tan \theta - \sec \theta + 1) (\tan \theta - \sec \theta)} \\
 &= \frac{-1}{\tan \theta - \sec \theta} = \frac{1}{\sec \theta - \tan \theta},
 \end{aligned}$$

which is the RHS of the identity, we are required to prove.

EXERCISE 11.4

- Express the trigonometric ratios $\sin A$, $\sec A$ and $\tan A$ in terms of $\cot A$.
- Write all the other trigonometric ratios of $\angle A$ in terms of $\sec A$.
- Evaluate :

(i) $\frac{\sin^2 63^\circ + \sin^2 27^\circ}{\cos^2 17^\circ + \cos^2 73^\circ}$

(ii) $\sin 25^\circ \cos 65^\circ + \cos 25^\circ \sin 65^\circ$

- Choose the correct option. Justify your choice.

(i) $9 \sec^2 A - 9 \tan^2 A =$

- (A) 1 (B) 9 (C) 8 (D) 0

(ii) $(1 + \tan \theta + \sec \theta)(1 + \cot \theta - \operatorname{cosec} \theta) =$

- (A) 0 (B) 1 (C) 2 (D) -1

(iii) $(\sec A + \tan A)(1 - \sin A) =$

- (A) $\sec A$ (B) $\sin A$ (C) $\operatorname{cosec} A$ (D) $\cos A$

(iv) $\frac{1 + \tan^2 A}{1 + \cot^2 A} =$

- (A) $\sec^2 A$ (B) -1 (C) $\cot^2 A$ (D) $\tan^2 A$

- Prove the following identities, where the angles involved are acute angles for which the expressions are defined.

(i) $(\operatorname{cosec} \theta - \cot \theta)^2 = \frac{1 - \cos \theta}{1 + \cos \theta}$

(ii) $\frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A} = 2 \sec A$

$$(iii) \frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 - \tan \theta} = 1 + \sec \theta \operatorname{cosec} \theta$$

[Hint : Write the expression in terms of $\sin \theta$ and $\cos \theta$]

$$(iv) \frac{1 + \sec A}{\sec A} = \frac{\sin^2 A}{1 - \cos A} \quad [\text{Hint : Simplify LHS and RHS separately}]$$

$$(v) \frac{\cos A - \sin A + 1}{\cos A + \sin A - 1} = \operatorname{cosec} A + \cot A, \text{ using the identity } \operatorname{cosec}^2 A = 1 + \cot^2 A.$$

$$(vi) \sqrt{\frac{1 + \sin A}{1 - \sin A}} = \sec A + \tan A \quad (vii) \frac{\sin \theta - 2 \sin^3 \theta}{2 \cos^3 \theta - \cos \theta} = \tan \theta$$

$$(viii) (\sin A + \operatorname{cosec} A)^2 + (\cos A + \sec A)^2 = 7 + \tan^2 A + \cot^2 A$$

$$(ix) (\operatorname{cosec} A - \sin A)(\sec A - \cos A) = \frac{1}{\tan A + \cot A}$$

[Hint : Simplify LHS and RHS separately]

$$(x) \left(\frac{1 + \tan^2 A}{1 + \cot^2 A} \right) = \left(\frac{1 - \tan A}{1 - \cot A} \right)^2 = \tan^2 A$$

11.6 Summary

In this chapter, you have studied the following points :

1. In a right triangle ABC, right-angled at B,

$$\sin A = \frac{\text{side opposite to angle } A}{\text{hypotenuse}}, \cos A = \frac{\text{side adjacent to angle } A}{\text{hypotenuse}}$$

$$\tan A = \frac{\text{side opposite to angle } A}{\text{side adjacent to angle } A}.$$

2. $\operatorname{cosec} A = \frac{1}{\sin A}$; $\sec A = \frac{1}{\cos A}$; $\tan A = \frac{1}{\cot A}$, $\tan A = \frac{\sin A}{\cos A}$.
3. If one of the trigonometric ratios of an acute angle is known, the remaining trigonometric ratios of the angle can be easily determined.
4. The values of trigonometric ratios for angles 0° , 30° , 45° , 60° and 90° .
5. The value of $\sin A$ or $\cos A$ never exceeds 1, whereas the value of $\sec A$ or $\operatorname{cosec} A$ is always greater than or equal to 1.
6. $\sin(90^\circ - A) = \cos A$, $\cos(90^\circ - A) = \sin A$;
 $\tan(90^\circ - A) = \cot A$, $\cot(90^\circ - A) = \tan A$;
 $\sec(90^\circ - A) = \operatorname{cosec} A$, $\operatorname{cosec}(90^\circ - A) = \sec A$.
7. $\sin^2 A + \cos^2 A = 1$,
 $\sec^2 A - \tan^2 A = 1$ for $0^\circ \leq A < 90^\circ$,
 $\operatorname{cosec}^2 A = 1 + \cot^2 A$ for $0^\circ < A \leq 90^\circ$.



SOME APPLICATIONS OF

12

TRIGONOMETRY

12.1 Introduction

In the previous chapter, you have studied about trigonometric ratios. In this chapter, you will be studying about some ways in which trigonometry is used in the life around you. Trigonometry is one of the most ancient subjects studied by scholars all over the world. As we have said in Chapter 11, trigonometry was invented because its need arose in astronomy. Since then the astronomers have used it, for instance, to calculate distances from the Earth to the planets and stars. Trigonometry is also used in geography and in navigation. The knowledge of trigonometry is used to construct maps, determine the position of an island in relation to the longitudes and latitudes.

Surveyors have used trigonometry for centuries. One such large surveying project of the nineteenth century was the '**Great Trigonometric Survey**' of British India for which the two largest-ever theodolites were built. During the survey in 1852, the highest mountain in the world was discovered. From a distance of over 160 km, the peak was observed from six different stations. In 1856, this peak was named after Sir George Everest, who had commissioned and first used the giant theodolites (see the figure alongside). The theodolites are now on display in the Museum of the Survey of India in Dehradun.



A Theodolite

(Surveying instrument, which is based on the Principles of trigonometry, is used for measuring angles with a rotating telescope)

In this chapter, we will see how trigonometry is used for finding the heights and distances of various objects, without actually measuring them.

12.2 Heights and Distances

Let us consider Fig. 11.1 of previous chapter, which is redrawn below in Fig. 12.1.

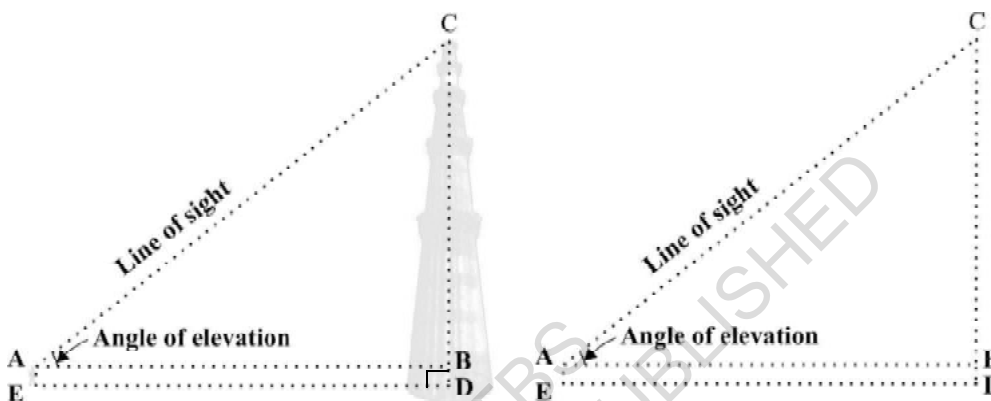


Fig. 12.1

In this figure, the line AC drawn from the eye of the student to the top of the minar is called the *line of sight*. The student is looking at the top of the minar. The angle BAC, so formed by the line of sight with the horizontal, is called the *angle of elevation* of the top of the minar from the eye of the student.

Thus, the **line of sight** is the line drawn from the eye of an observer to the point in the object viewed by the observer. The **angle of elevation** of the point viewed is the angle formed by the line of sight with the horizontal when the point being viewed is above the horizontal level, i.e., the case when we raise our head to look at the object (see Fig. 12.2).

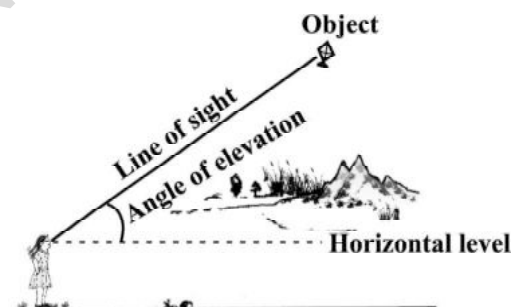


Fig. 12.2

Now, consider the situation given in Fig. 11.2. The girl sitting on the balcony is *looking down* at a flower pot placed on a stair of the temple. In this case, the line of sight is *below* the horizontal level. The angle so formed by the line of sight with the horizontal is called the *angle of depression*.

Thus, the **angle of depression** of a point on the object being viewed is the angle formed by the line of sight with the horizontal when the point is below the horizontal level, i.e., the case when we lower our head to look at the point being viewed (see Fig. 12.3).

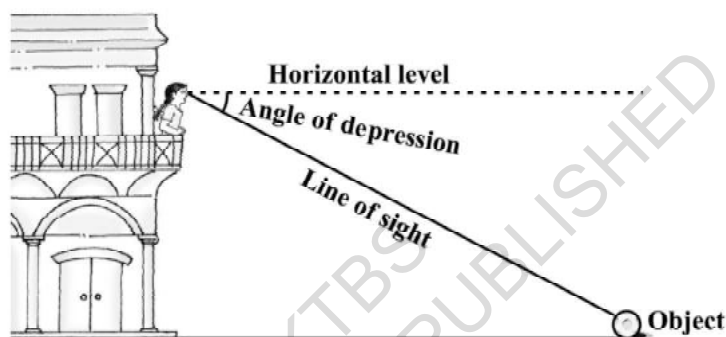


Fig. 12.3

Now, you may identify the lines of sight, and the angles so formed in Fig. 11.3. Are they angles of elevation or angles of depression?

Let us refer to Fig. 12.1 again. If you want to find the height CD of the minar without actually measuring it, what information do you need? You would need to know the following:

- (i) the distance DE at which the student is standing from the foot of the minar
- (ii) the angle of elevation, $\angle BAC$, of the top of the minar
- (iii) the height AE of the student.

Assuming that the above three conditions are known, how can we determine the height of the minar?

In the figure, $CD = CB + BD$. Here, $BD = AE$, which is the height of the student.

To find BC , we will use trigonometric ratios of $\angle BAC$ or $\angle A$.

In $\triangle ABC$, the side BC is the opposite side in relation to the known $\angle A$. Now, which of the trigonometric ratios can we use? Which one of them has the two values that we have and the one we need to determine? Our search narrows down to using either $\tan A$ or $\cot A$, as these ratios involve AB and BC .

Therefore, $\tan A = \frac{BC}{AB}$ or $\cot A = \frac{AB}{BC}$, which on solving would give us BC.

By adding AE to BC, you will get the height of the minar.

Now let us explain the process, we have just discussed, by solving some problems.

Example 1 : A tower stands vertically on the ground. From a point on the ground, which is 15 m away from the foot of the tower, the angle of elevation of the top of the tower is found to be 60° . Find the height of the tower.

Solution : First let us draw a simple diagram to represent the problem (see Fig. 12.4). Here AB represents the tower, CB is the distance of the point from the tower and $\angle ACB$ is the angle of elevation. We need to determine the height of the tower, i.e., AB. Also, ACB is a triangle, right-angled at B.

To solve the problem, we choose the trigonometric ratio $\tan 60^\circ$ (or $\cot 60^\circ$), as the ratio involves AB and BC.

$$\text{Now,} \quad \tan 60^\circ = \frac{AB}{BC}$$

$$\text{i.e.,} \quad \sqrt{3} = \frac{AB}{15}$$

$$\text{i.e.,} \quad AB = 15\sqrt{3}$$

Hence, the height of the tower is $15\sqrt{3}$ m.

Example 2 : An electrician has to repair an electric fault on a pole of height 5 m. She needs to reach a point 1.3m below the top of the pole to undertake the repair work (see Fig. 12.5). What should be the length of the ladder that she should use which, when inclined at an angle of 60° to the horizontal, would enable her to reach the required position? Also, how far from the foot of the pole should she place the foot of the ladder? (You may take $\sqrt{3} = 1.73$)



Fig. 12.4

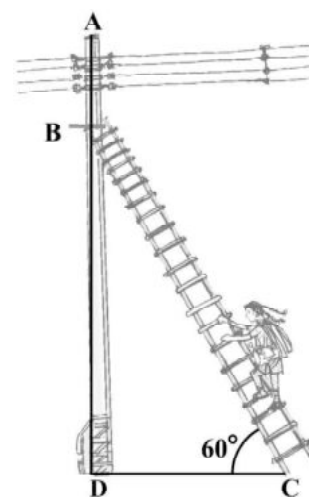


Fig. 12.5

Solution : In Fig. 9.5, the electrician is required to reach the point B on the pole AD.

So, $BD = AD - AB = (5 - 1.3)\text{m} = 3.7\text{ m}.$

Here, BC represents the ladder. We need to find its length, i.e., the hypotenuse of the right triangle BDC.

Now, can you think which trigonometric ratio should we consider?

It should be $\sin 60^\circ$.

$$\text{So, } \frac{BD}{BC} = \sin 60^\circ \text{ or } \frac{3.7}{BC} = \frac{\sqrt{3}}{2}$$

$$\text{Therefore, } BC = \frac{3.7 \times 2}{\sqrt{3}} = 4.28\text{ m (approx.)}$$

i.e., the length of the ladder should be 4.28 m.

$$\text{Now, } \frac{DC}{BD} = \cot 60^\circ = \frac{1}{\sqrt{3}}$$

$$\text{i.e., } DC = \frac{3.7}{\sqrt{3}} = 2.14\text{ m (approx.)}$$

Therefore, she should place the foot of the ladder at a distance of 2.14 m from the pole.

Example 3 : An observer 1.5 m tall is 28.5 m away from a chimney. The angle of elevation of the top of the chimney from her eyes is 45° . What is the height of the chimney?

Solution : Here, AB is the chimney, CD the observer and $\angle ADE$ the angle of elevation (see Fig. 12.6). In this case, ADE is a triangle, right-angled at E and we are required to find the height of the chimney.

We have $AB = AE + BE = AE + 1.5$

and $DE = CB = 28.5\text{ m}$

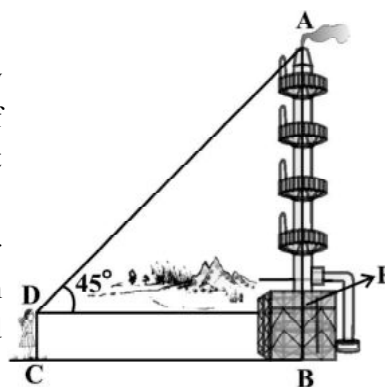


Fig. 12.6

To determine AE, we choose a trigonometric ratio, which involves both AE and DE. Let us choose the tangent of the angle of elevation.

Now, $\tan 45^\circ = \frac{AE}{DE}$

i.e., $1 = \frac{AE}{28.5}$

Therefore, $AE = 28.5$

So the height of the chimney (AB) = (28.5 + 1.5) m = 30 m.

Example 4 : From a point P on the ground the angle of elevation of the top of a 10 m tall building is 30° . A flag is hoisted at the top of the building and the angle of elevation of the top of the flagstaff from P is 45° . Find the length of the flagstaff and the distance of the building from the point P. (You may take $\sqrt{3} = 1.732$)

Solution : In Fig. 12.7, AB denotes the height of the building, BD the flagstaff and P the given point. Note that there are two right triangles PAB and PAD. We are required to find the length of the flagstaff, i.e., DB and the distance of the building from the point P, i.e., PA.

Since, we know the height of the building AB, we will first consider the right Δ PAB.

We have $\tan 30^\circ = \frac{AB}{AP}$

i.e., $\frac{1}{\sqrt{3}} = \frac{10}{AP}$

Therefore, $AP = 10\sqrt{3}$

i.e., the distance of the building from P is $10\sqrt{3}$ m = 17.32 m.

Next, let us suppose DB = x m. Then AD = (10 + x) m.

Now, in right Δ PAD, $\tan 45^\circ = \frac{AD}{AP} = \frac{10 + x}{10\sqrt{3}}$

Therefore, $1 = \frac{10 + x}{10\sqrt{3}}$

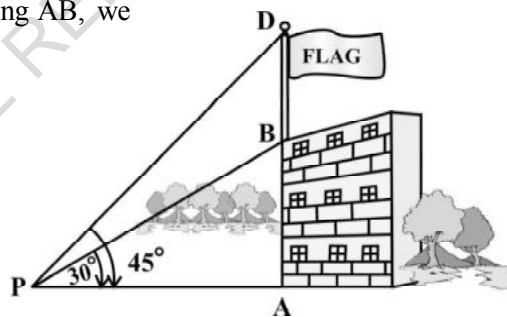


Fig. 12.7

i.e.,
$$x = 10 (\sqrt{3} - 1) = 7.32$$

So, the length of the flagstaff is 7.32 m.

Example 5 : The shadow of a tower standing on a level ground is found to be 40 m longer when the Sun's altitude is 30° than when it is 60° . Find the height of the tower.

Solution : In Fig. 12.8, AB is the tower and BC is the length of the shadow when the Sun's altitude is 60° , i.e., the angle of elevation of the top of the tower from the tip of the shadow is 60° and DB is the length of the shadow, when the angle of elevation is 30° .

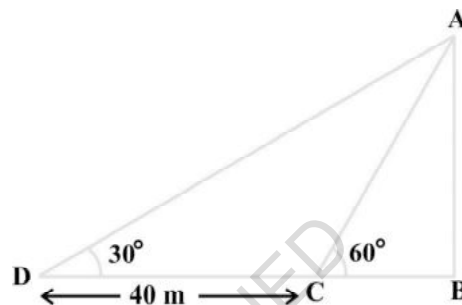


Fig. 12.8

Now, let AB be h m and BC be x m. According to the question, DB is 40 m longer than BC.

So,
$$DB = (40 + x) \text{ m}$$

Now, we have two right triangles ABC and ABD.

In $\triangle ABC$,
$$\tan 60^\circ = \frac{AB}{BC}$$

or,
$$\sqrt{3} = \frac{h}{x} \quad (1)$$

In $\triangle ABD$,
$$\tan 30^\circ = \frac{AB}{BD}$$

i.e.,
$$\frac{1}{\sqrt{3}} = \frac{h}{x + 40} \quad (2)$$

From (1), we have
$$h = x\sqrt{3}$$

Putting this value in (2), we get $(x\sqrt{3})\sqrt{3} = x + 40$, i.e., $3x = x + 40$

i.e.,
$$x = 20$$

So,
$$h = 20\sqrt{3} \quad [\text{From (1)}]$$

Therefore, the height of the tower is $20\sqrt{3}$ m.

Example 6 : The angles of depression of the top and the bottom of an 8 m tall building from the top of a multi-storeyed building are 30° and 45° , respectively. Find the height of the multi-storeyed building and the distance between the two buildings.

Solution : In Fig. 12.9, PC denotes the multi-storeyed building and AB denotes the 8 m tall building. We are interested to determine the height of the multi-storeyed building, i.e., PC and the distance between the two buildings, i.e., AC.

Look at the figure carefully. Observe that PB is a transversal to the parallel lines PQ and BD. Therefore, $\angle QPB$ and $\angle PBD$ are alternate angles, and so are equal. So $\angle PBD = 30^\circ$. Similarly, $\angle PAC = 45^\circ$.

In right $\triangle PBD$, we have

$$\frac{PD}{BD} = \tan 30^\circ = \frac{1}{\sqrt{3}} \text{ or } BD = PD\sqrt{3}$$

In right $\triangle PAC$, we have

$$\frac{PC}{AC} = \tan 45^\circ = 1$$

i.e.,

$$PC = AC$$

Also,

$$PC = PD + DC, \text{ therefore, } PD + DC = AC.$$

Since, $AC = BD$ and $DC = AB = 8$ m, we get $PD + 8 = BD = PD\sqrt{3}$ (Why?)

This gives
$$PD = \frac{8}{\sqrt{3} - 1} = \frac{8(\sqrt{3} + 1)}{(\sqrt{3} + 1)(\sqrt{3} - 1)} = 4(\sqrt{3} + 1)\text{m}.$$

So, the height of the multi-storeyed building is $\{4(\sqrt{3} + 1) + 8\}$ m $= 4(3 + \sqrt{3})$ m and the distance between the two buildings is also $4(3 + \sqrt{3})$ m.

Example 7 : From a point on a bridge across a river, the angles of depression of the banks on opposite sides of the river are 30° and 45° , respectively. If the bridge is at a height of 3 m from the banks, find the width of the river.

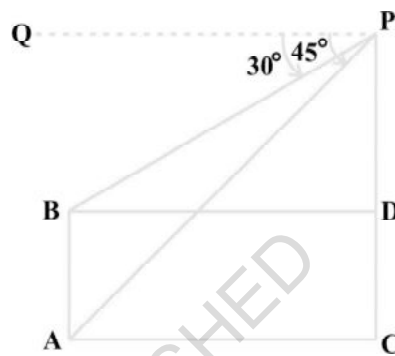


Fig. 12.9

Solution : In Fig 12.10, A and B represent points on the bank on opposite sides of the river, so that AB is the width of the river. P is a point on the bridge at a height of 3 m, i.e., $DP = 3$ m. We are interested to determine the width of the river, which is the length of the side AB of the $\triangle APB$.

Now, $AB = AD + DB$

In right $\triangle APD$, $\angle A = 30^\circ$.

$$\text{So, } \tan 30^\circ = \frac{PD}{AD}$$

$$\text{i.e., } \frac{1}{\sqrt{3}} = \frac{3}{AD} \text{ or } AD = 3\sqrt{3} \text{ m}$$

Also, in right $\triangle PBD$, $\angle B = 45^\circ$. So, $BD = PD = 3$ m.

$$\text{Now, } AB = BD + AD = 3 + 3\sqrt{3} = 3(1 + \sqrt{3}) \text{ m.}$$

Therefore, the width of the river is $3(\sqrt{3} + 1)$ m.

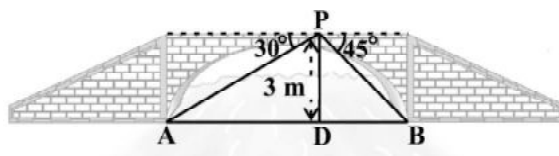


Fig. 12.10

EXERCISE 12.1

1. A circus artist is climbing a 20 m long rope, which is tightly stretched and tied from the top of a vertical pole to the ground. Find the height of the pole, if the angle made by the rope with the ground level is 30° (see Fig. 9.11).
2. A tree breaks due to storm and the broken part bends so that the top of the tree touches the ground making an angle 30° with it. The distance between the foot of the tree to the point where the top touches the ground is 8 m. Find the height of the tree.
3. A contractor plans to install two slides for the children to play in a park. For the children below the age of 5 years, she prefers to have a slide whose top is at a height of 1.5 m, and

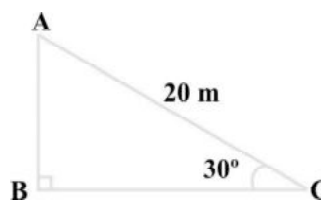


Fig. 12.11

is inclined at an angle of 30° to the ground, whereas for elder children, she wants to have a steep slide at a height of 3 m, and inclined at an angle of 60° to the ground. What should be the length of the slide in each case?

4. The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower, is 30° . Find the height of the tower.
5. A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 60° . Find the length of the string, assuming that there is no slack in the string.
6. A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked towards the building.
7. From a point on the ground, the angles of elevation of the bottom and the top of a transmission tower fixed at the top of a 20 m high building are 45° and 60° respectively. Find the height of the tower.
8. A statue, 1.6 m tall, stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and from the same point the angle of elevation of the top of the pedestal is 45° . Find the height of the pedestal.
9. The angle of elevation of the top of a building from the foot of the tower is 30° and the angle of elevation of the top of the tower from the foot of the building is 60° . If the tower is 50 m high, find the height of the building.
10. Two poles of equal heights are standing opposite each other on either side of the road, which is 80 m wide. From a point between them on the road, the angles of elevation of the top of the poles are 60° and 30° , respectively. Find the height of the poles and the distances of the point from the poles.
11. A TV tower stands vertically on a bank of a canal. From a point on the other bank directly opposite the tower, the angle of elevation of the top of the tower is 60° . From another point 20 m away from this point on the line joining this point to the foot of the tower, the angle of elevation of the top of the tower is 30° (see Fig. 12.12). Find the height of the tower and the width of the canal.

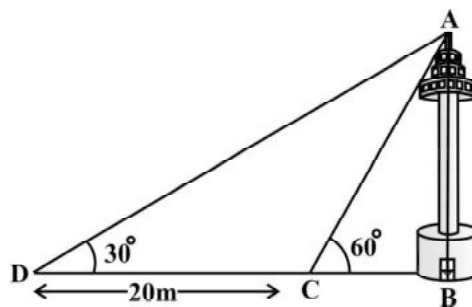


Fig. 12.12

12. From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45° . Determine the height of the tower.
13. As observed from the top of a 75 m high lighthouse from the sea-level, the angles of depression of two ships are 30° and 45° . If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.

14. A 1.2 m tall girl spots a balloon moving with the wind in a horizontal line at a height of 88.2 m from the ground. The angle of elevation of the balloon from the eyes of the girl at any instant is 60° . After some time, the angle of elevation reduces to 30° (see Fig. 12.13). Find the distance travelled by the balloon during the interval.

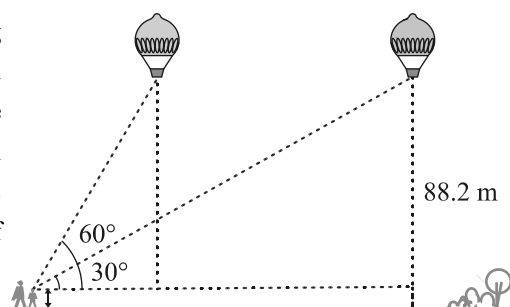


Fig. 12.13

15. A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of 30° , which is approaching the foot of the tower with a uniform speed. Six seconds later, the angle of depression of the car is found to be 60° . Find the time taken by the car to reach the foot of the tower from this point.
16. The angles of elevation of the top of a tower from two points at a distance of 4 m and 9 m from the base of the tower and in the same straight line with it are complementary. Prove that the height of the tower is 6 m.

12.3 Summary

In this chapter, you have studied the following points :

- The **line of sight** is the line drawn from the eye of an observer to the point in the object viewed by the observer.
 - The **angle of elevation** of an object viewed, is the angle formed by the line of sight with the horizontal when it is above the horizontal level, i.e., the case when we raise our head to look at the object.
 - The **angle of depression** of an object viewed, is the angle formed by the line of sight with the horizontal when it is below the horizontal level, i.e., the case when we lower our head to look at the object.
- The height or length of an object or the distance between two distant objects can be determined with the help of trigonometric ratios.



13.1 Introduction

In Class IX, you have studied the classification of given data into ungrouped as well as grouped frequency distributions. You have also learnt to represent the data pictorially in the form of various graphs such as bar graphs, histograms (including those of varying widths) and frequency polygons. In fact, you went a step further by studying certain numerical representatives of the ungrouped data, also called measures of central tendency, namely, *mean*, *median* and *mode*. In this chapter, we shall extend the study of these three measures, i.e., mean, median and mode from ungrouped data to that of *grouped data*. We shall also discuss the concept of cumulative frequency, the cumulative frequency distribution and how to draw cumulative frequency curves, called *ogives*.

13.2 Mean of Grouped Data

The mean (or average) of observations, as we know, is the sum of the values of all the observations divided by the total number of observations. From Class IX, recall that if $x_1, x_2, \dots, x_n$ are observations with respective frequencies $f_1, f_2, \dots, f_n$, then this means observation x_1 occurs f_1 times, x_2 occurs f_2 times, and so on.

Now, the sum of the values of all the observations $= f_1x_1 + f_2x_2 + \dots + f_nx_n$, and the number of observations $= f_1 + f_2 + \dots + f_n$.

So, the mean $\bar{x}$ of the data is given by

$$= \frac{f_1x_1 + f_2x_2 + \dots + f_nx_n}{f_1 + f_2 + \dots + f_n}$$

Recall that we can write this in short form by using the Greek letter Σ (capital sigma) which means summation. That is,

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$$

which, more briefly, is written as $\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i}$, if it is understood that i varies from 1 to n .

Let us apply this formula to find the mean in the following example.

Example 1 : The marks obtained by 30 students of Class X of a certain school in a Mathematics paper consisting of 100 marks are presented in table below. Find the mean of the marks obtained by the students.

Marks obtained (x_i)	10	20	36	40	50	56	60	70	72	80	88	92	95
Number of students (f_i)	1	1	3	4	3	2	4	4	1	1	2	3	1

Solution: Recall that to find the mean marks, we require the product of each x_i with the corresponding frequency f_i . So, let us put them in a column as shown in Table 13.1.

Table 13.1

Marks obtained (x_i)	Number of students (f_i)	$f_i x_i$
10	1	10
20	1	20
36	3	108
40	4	160
50	3	150
56	2	112
60	4	240
70	4	280
72	1	72
80	1	80
88	2	176
92	3	276
95	1	95
Total	$\Sigma f_i = 30$	$\Sigma f_i x_i = 1779$

Now,

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i} = \frac{1779}{30} = 59.3$$

Therefore, the mean marks obtained is 59.3.

In most of our real life situations, data is usually so large that to make a meaningful study it needs to be condensed as grouped data. So, we need to convert given ungrouped data into grouped data and devise some method to find its mean.

Let us convert the ungrouped data of Example 1 into grouped data by forming class-intervals of width, say 15. Remember that, while allocating frequencies to each class-interval, students falling in any upper class-limit would be considered in the next class, e.g., 4 students who have obtained 40 marks would be considered in the class-interval 40-55 and not in 25-40. With this convention in our mind, let us form a grouped frequency distribution table (see Table 13.2).

Table 13.2

Class interval	10 - 25	25 - 40	40 - 55	55 - 70	70 - 85	85 - 100
Number of students	2	3	7	6	6	6

Now, for each class-interval, we require a point which would serve as the representative of the whole class. *It is assumed that the frequency of each class-interval is centred around its mid-point.* So the *mid-point* (or *class mark*) of each class can be chosen to represent the observations falling in the class. Recall that we find the mid-point of a class (or its class mark) by finding the average of its upper and lower limits. That is,

$$\text{Class mark} = \frac{\text{Upper class limit} + \text{Lower class limit}}{2}$$

With reference to Table 13.2, for the class 10-25, the class mark is $\frac{10+25}{2}$, i.e.,

17.5. Similarly, we can find the class marks of the remaining class intervals. We put them in Table 13.3. These class marks serve as our x_i 's. Now, in general, for the i th class interval, we have the frequency f_i corresponding to the class mark x_i . We can now proceed to compute the mean in the same manner as in Example 1.

Table 13.3

Class interval	Number of students (f_i)	Class mark (x_i)	$f_i x_i$
10 - 25	2	17.5	35.0
25 - 40	3	32.5	97.5
40 - 55	7	47.5	332.5
55 - 70	6	62.5	375.0
70 - 85	6	77.5	465.0
85 - 100	6	92.5	555.0
Total	$\Sigma f_i = 30$		$\Sigma f_i x_i = 1860.0$

The sum of the values in the last column gives us $\Sigma f_i x_i$. So, the mean $\bar{x}$ of the given data is given by

$$\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{1860.0}{30} = 62$$

This new method of finding the mean is known as the **Direct Method**.

We observe that Tables 13.1 and 13.3 are using the same data and employing the same formula for the calculation of the mean but the results obtained are different. Can you think why this is so, and which one is more accurate? The difference in the two values is because of the mid-point assumption in Table 13.3, 59.3 being the exact mean, while 62 an approximate mean.

Sometimes when the numerical values of x_i and f_i are large, finding the product of x_i and f_i becomes tedious and time consuming. So, for such situations, let us think of a method of reducing these calculations.

We can do nothing with the f_i 's, but we can change each x_i to a smaller number so that our calculations become easy. How do we do this? What about subtracting a fixed number from each of these x_i 's? Let us try this method.

The first step is to choose one among the x_i 's as the *assumed mean*, and denote it by ' a '. Also, to further reduce our calculation work, we may take ' a ' to be that x_i which lies in the centre of $x_1, x_2, \dots, x_n$. So, we can choose $a = 47.5$ or $a = 62.5$. Let us choose $a = 47.5$.

The next step is to find the difference d_i between a and each of the x_i 's, that is, the **deviation** of ' a ' from each of the x_i 's.

i.e.,
$$d_i = x_i - a = x_i - 47.5$$

The third step is to find the product of d_i with the corresponding f_i , and take the sum of all the $f_i d_i$'s. The calculations are shown in Table 13.4.

Table 13.4

Class interval	Number of students (f_i)	Class mark (x_i)	$d_i = x_i - 47.5$	$f_i d_i$
10 - 25	2	17.5	-30	-60
25 - 40	3	32.5	-15	-45
40 - 55	7	47.5	0	0
55 - 70	6	62.5	15	90
70 - 85	6	77.5	30	180
85 - 100	6	92.5	45	270
Total	$\Sigma f_i = 30$			$\Sigma f_i d_i = 435$

So, from Table 13.4, the mean of the deviations, $\bar{d} = \frac{\Sigma f_i d_i}{\Sigma f_i}$.

Now, let us find the relation between $\bar{d}$ and $\bar{x}$.

Since in obtaining d_i , we subtracted 'a' from each x_i , so, in order to get the mean $\bar{x}$, we need to add 'a' to $\bar{d}$. This can be explained mathematically as:

$$\text{Mean of deviations, } \bar{d} = \frac{\Sigma f_i d_i}{\Sigma f_i}$$

$$\begin{aligned} \text{So, } \bar{d} &= \frac{\Sigma f_i (x_i - a)}{\Sigma f_i} \\ &= \frac{\Sigma f_i x_i}{\Sigma f_i} - \frac{\Sigma f_i a}{\Sigma f_i} \\ &= \bar{x} - a \frac{\Sigma f_i}{\Sigma f_i} \\ &= \bar{x} - a \end{aligned}$$

$$\text{So, } \bar{x} = a + \bar{d}$$

$$\text{i.e., } \bar{x} = a + \frac{\Sigma f_i d_i}{\Sigma f_i}$$

Substituting the values of a , $\Sigma f_i d_i$ and Σf_i from Table 13.4, we get

$$\bar{x} = 47.5 + \frac{435}{30} = 47.5 + 14.5 = 62.$$

Therefore, the mean of the marks obtained by the students is 62.

The method discussed above is called the **Assumed Mean Method**.

Activity 1 : From the Table 13.3 find the mean by taking each of x_i (i.e., 17.5, 32.5, and so on) as ' a '. What do you observe? You will find that the mean determined in each case is the same, i.e., 62. (Why ?)

So, we can say that the value of the mean obtained does not depend on the choice of ' a '.

Observe that in Table 13.4, the values in Column 4 are all multiples of 15. So, if we divide the values in the entire Column 4 by 15, we would get smaller numbers to multiply with f_i . (Here, 15 is the class size of each class interval.)

So, let $u_i = \frac{x_i - a}{h}$, where a is the assumed mean and h is the class size.

Now, we calculate u_i in this way and continue as before (i.e., find $f_i u_i$ and then $\Sigma f_i u_i$). Taking $h = 15$, let us form Table 13.5.

Table 13.5

Class interval	f_i	x_i	$d_i = x_i - a$	$u_i = \frac{x_i - a}{h}$	$f_i u_i$
10 - 25	2	17.5	-30	-2	-4
25 - 40	3	32.5	-15	-1	-3
40 - 55	7	47.5	0	0	0
55 - 70	6	62.5	15	1	6
70 - 85	6	77.5	30	2	12
85 - 100	6	92.5	45	3	18
Total	$\Sigma f_i = 30$				$\Sigma f_i u_i = 29$

Let
$$\bar{u} = \frac{\Sigma f_i u_i}{\Sigma f_i}$$

Here, again let us find the relation between $\bar{u}$ and $\bar{x}$.

We have, $u_i = \frac{x_i - a}{h}$

Therefore,

$$\begin{aligned}\bar{u} &= \frac{\sum f_i \frac{(x_i - a)}{h}}{\sum f_i} = \frac{1}{h} \left[\frac{\sum f_i x_i - a \sum f_i}{\sum f_i} \right] \\ &= \frac{1}{h} \left[\frac{\sum f_i x_i}{\sum f_i} - a \frac{\sum f_i}{\sum f_i} \right] \\ &= \frac{1}{h} [\bar{x} - a]\end{aligned}$$

So, $h\bar{u} = \bar{x} - a$

i.e., $\bar{x} = a + h\bar{u}$

So, $\bar{x} = a + h \left(\frac{\sum f_i u_i}{\sum f_i} \right)$

Now, substituting the values of a , h , $\sum f_i u_i$ and $\sum f_i$ from Table 13.5, we get

$$\begin{aligned}\bar{x} &= 47.5 + 15 \times \left(\frac{29}{30} \right) \\ &= 47.5 + 14.5 = 62\end{aligned}$$

So, the mean marks obtained by a student is 62.

The method discussed above is called the **Step-deviation** method.

We note that :

- the step-deviation method will be convenient to apply if all the d_i 's have a common factor.
- The mean obtained by all the three methods is the same.
- The assumed mean method and step-deviation method are just simplified forms of the direct method.
- The formula $\bar{x} = a + h\bar{u}$ still holds if a and h are not as given above, but are any non-zero numbers such that $u_i = \frac{x_i - a}{h}$.

Let us apply these methods in another example.

Example 2 : The table below gives the percentage distribution of female teachers in the primary schools of rural areas of various states and union territories (U.T.) of India. Find the mean percentage of female teachers by all the three methods discussed in this section.

Percentage of female teachers	15 - 25	25 - 35	35 - 45	45 - 55	55 - 65	65 - 75	75 - 85
Number of States/U.T.	6	11	7	4	4	2	1

Source : *Seventh All India School Education Survey conducted by NCERT*

Solution : Let us find the class marks, x_i , of each class, and put them in a column (see Table 13.6):

Table 13.6

Percentage of female teachers	Number of States /U.T. (f_i)	x_i
15 - 25	6	20
25 - 35	11	30
35 - 45	7	40
45 - 55	4	50
55 - 65	4	60
65 - 75	2	70
75 - 85	1	80

Here we take $a = 50$, $h = 10$, then $d_i = x_i - 50$ and $u_i = \frac{x_i - 50}{10}$.

We now find d_i and u_i and put them in Table 13.7.

Table 13.7

Percentage of female teachers	Number of states/U.T. (f_i)	x_i	$d_i = x_i - 50$	$u_i = \frac{x_i - 50}{10}$	$f_i x_i$	$f_i d_i$	$f_i u_i$
15 - 25	6	20	-30	-3	120	-180	-18
25 - 35	11	30	-20	-2	330	-220	-22
35 - 45	7	40	-10	-1	280	-70	-7
45 - 55	4	50	0	0	200	0	0
55 - 65	4	60	10	1	240	40	4
65 - 75	2	70	20	2	140	40	4
75 - 85	1	80	30	3	80	30	3
Total	35				1390	-360	-36

From the table above, we obtain $\Sigma f_i = 35$, $\Sigma f_i x_i = 1390$,

$$\Sigma f_i d_i = -360, \quad \Sigma f_i u_i = -36.$$

Using the direct method, $\bar{x} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{1390}{35} = 39.71$

Using the assumed mean method,

$$\bar{x} = a + \frac{\Sigma f_i d_i}{\Sigma f_i} = 50 + \frac{(-360)}{35} = 39.71$$

Using the step-deviation method,

$$\bar{x} = a + \left(\frac{\Sigma f_i u_i}{\Sigma f_i} \right) \times h = 50 + \left(\frac{-36}{35} \right) \times 10 = 39.71$$

Therefore, the mean percentage of female teachers in the primary schools of rural areas is 39.71.

Remark : The result obtained by all the three methods is the same. So the choice of method to be used depends on the numerical values of x_i and f_i . If x_i and f_i are sufficiently small, then the direct method is an appropriate choice. If x_i and f_i are numerically large numbers, then we can go for the assumed mean method or step-deviation method. If the class sizes are unequal, and x_i are large numerically, we can still apply the step-deviation method by taking h to be a suitable divisor of all the d_i 's.

Example 3 : The distribution below shows the number of wickets taken by bowlers in one-day cricket matches. Find the mean number of wickets by choosing a suitable method. What does the mean signify?

Number of wickets	20 - 60	60 - 100	100 - 150	150 - 250	250 - 350	350 - 450
Number of bowlers	7	5	16	12	2	3

Solution : Here, the class size varies, and the x_i 's are large. Let us still apply the step-deviation method with $a = 200$ and $h = 20$. Then, we obtain the data as in Table 13.8.

Table 13.8

Number of wickets taken	Number of bowlers (f_i)	x_i	$d_i = x_i - 200$	$u_i = \frac{d_i}{20}$	$u_i f_i$
20 - 60	7	40	-160	-8	-56
60 - 100	5	80	-120	-6	-30
100 - 150	16	125	-75	-3.75	-60
150 - 250	12	200	0	0	0
250 - 350	2	300	100	5	10
350 - 450	3	400	200	10	30
Total	45				-106

$$\text{So, } \bar{u} = \frac{-106}{45}. \text{ Therefore, } \bar{x} = 200 + 20 \left(\frac{-106}{45} \right) = 200 - 47.11 = 152.89.$$

This tells us that, on an average, the number of wickets taken by these 45 bowlers in one-day cricket is 152.89.

Now, let us see how well you can apply the concepts discussed in this section!

Activity 2 :

Divide the students of your class into three groups and ask each group to do one of the following activities.

1. Collect the marks obtained by all the students of your class in Mathematics in the latest examination conducted by your school. Form a grouped frequency distribution of the data obtained.
2. Collect the daily maximum temperatures recorded for a period of 30 days in your city. Present this data as a grouped frequency table.
3. Measure the heights of all the students of your class (in cm) and form a grouped frequency distribution table of this data.

After all the groups have collected the data and formed grouped frequency distribution tables, the groups should find the mean in each case by the method which they find appropriate.

EXERCISE 13.1

1. A survey was conducted by a group of students as a part of their environment awareness programme, in which they collected the following data regarding the number of plants in 20 houses in a locality. Find the mean number of plants per house.

Number of plants	0 - 2	2 - 4	4 - 6	6 - 8	8 - 10	10 - 12	12 - 14
Number of houses	1	2	1	5	6	2	3

Which method did you use for finding the mean, and why?

2. Consider the following distribution of daily wages of 50 workers of a factory.

Daily wages (in ₹)	500 - 520	520 - 540	540 - 560	560 - 580	580 - 600
Number of workers	12	14	8	6	10

Find the mean daily wages of the workers of the factory by using an appropriate method.

3. The following distribution shows the daily pocket allowance of children of a locality. The mean pocket allowance is ₹ 18. Find the missing frequency f .

Daily pocket allowance (in ₹)	11 - 13	13 - 15	15 - 17	17 - 19	19 - 21	21 - 23	23 - 25
Number of children	7	6	9	13	f	5	4

4. Thirty women were examined in a hospital by a doctor and the number of heartbeats per minute were recorded and summarised as follows. Find the mean heartbeats per minute for these women, choosing a suitable method.

Number of heartbeats per minute	65 - 68	68 - 71	71 - 74	74 - 77	77 - 80	80 - 83	83 - 86
Number of women	2	4	3	8	7	4	2

5. In a retail market, fruit vendors were selling mangoes kept in packing boxes. These boxes contained varying number of mangoes. The following was the distribution of mangoes according to the number of boxes.

Number of mangoes	50 - 52	53 - 55	56 - 58	59 - 61	62 - 64
Number of boxes	15	110	135	115	25

Find the mean number of mangoes kept in a packing box. Which method of finding the mean did you choose?

6. The table below shows the daily expenditure on food of 25 households in a locality.

Daily expenditure (in ₹)	100 - 150	150 - 200	200 - 250	250 - 300	300 - 350
Number of households	4	5	12	2	2

Find the mean daily expenditure on food by a suitable method.

7. To find out the concentration of SO_2 in the air (in parts per million, i.e., ppm), the data was collected for 30 localities in a certain city and is presented below:

Concentration of SO_2 (in ppm)	Frequency
0.00 - 0.04	4
0.04 - 0.08	9
0.08 - 0.12	9
0.12 - 0.16	2
0.16 - 0.20	4
0.20 - 0.24	2

Find the mean concentration of SO_2 in the air.

8. A class teacher has the following absentee record of 40 students of a class for the whole term. Find the mean number of days a student was absent.

Number of days	0 - 6	6 - 10	10 - 14	14 - 20	20 - 28	28 - 38	38 - 40
Number of students	11	10	7	4	4	3	1

9. The following table gives the literacy rate (in percentage) of 35 cities. Find the mean literacy rate.

Literacy rate (in %)	45 - 55	55 - 65	65 - 75	75 - 85	85 - 95
Number of cities	3	10	11	8	3

13.3 Mode of Grouped Data

Recall from Class IX, a mode is that value among the observations which occurs most often, that is, the value of the observation having the maximum frequency. Further, we discussed finding the mode of ungrouped data. Here, we shall discuss ways of obtaining a mode of grouped data. It is possible that more than one value may have the same maximum frequency. In such situations, the data is said to be multimodal. Though grouped data can also be multimodal, we shall restrict ourselves to problems having a single mode only.

Let us first recall how we found the mode for ungrouped data through the following example.

Example 4 : The wickets taken by a bowler in 10 cricket matches are as follows:

2 6 4 5 0 2 1 3 2 3

Find the mode of the data.

Solution : Let us form the frequency distribution table of the given data as follows:

Number of wickets	0	1	2	3	4	5	6
Number of matches	1	1	3	2	1	1	1

Clearly, 2 is the number of wickets taken by the bowler in the maximum number (i.e., 3) of matches. So, the mode of this data is 2.

In a grouped frequency distribution, it is not possible to determine the mode by looking at the frequencies. Here, we can only locate a class with the maximum frequency, called the **modal class**. The mode is a value inside the modal class, and is given by the formula:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

where l = lower limit of the modal class,

h = size of the class interval (assuming all class sizes to be equal),

f_1 = frequency of the modal class,

f_0 = frequency of the class preceding the modal class,

f_2 = frequency of the class succeeding the modal class.

Let us consider the following examples to illustrate the use of this formula.

Example 5 : A survey conducted on 20 households in a locality by a group of students resulted in the following frequency table for the number of family members in a household:

Family size	1 - 3	3 - 5	5 - 7	7 - 9	9 - 11
Number of families	7	8	2	2	1

Find the mode of this data.

Solution : Here the maximum class frequency is 8, and the class corresponding to this frequency is 3 – 5. So, the modal class is 3 – 5.

Now

modal class = 3 – 5, lower limit (l) of modal class = 3, class size (h) = 2

frequency (f_1) of the modal class = 8,

frequency (f_0) of class preceding the modal class = 7,

frequency (f_2) of class succeeding the modal class = 2.

Now, let us substitute these values in the formula :

$$\begin{aligned}\text{Mode} &= l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h \\ &= 3 + \left(\frac{8 - 7}{2 \times 8 - 7 - 2} \right) \times 2 = 3 + \frac{2}{7} = 3.286\end{aligned}$$

Therefore, the mode of the data above is 3.286.

Example 6 : The marks distribution of 30 students in a mathematics examination are given in Table 13.3 of Example 1. Find the mode of this data. Also compare and interpret the mode and the mean.

Solution : Refer to Table 13.3 of Example 1. Since the maximum number of students (i.e., 7) have got marks in the interval 40 - 55, the modal class is 40 - 55. Therefore,

the lower limit (l) of the modal class = 40,

the class size (h) = 15,

the frequency (f_1) of modal class = 7,

the frequency (f_0) of the class preceding the modal class = 3,

the frequency (f_2) of the class succeeding the modal class = 6.

Now, using the formula:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h,$$

we get

$$\text{Mode} = 40 + \left(\frac{7 - 3}{14 - 6 - 3} \right) \times 15 = 52$$

So, the mode marks is 52.

Now, from Example 1, you know that the mean marks is 62.

So, the maximum number of students obtained 52 marks, while on an average a student obtained 62 marks.

Remarks :

1. In Example 6, the mode is less than the mean. But for some other problems it may be equal or more than the mean also.
2. It depends upon the demand of the situation whether we are interested in finding the average marks obtained by the students or the average of the marks obtained by most

of the students. In the first situation, the mean is required and in the second situation, the mode is required.

Activity 3 : Continuing with the same groups as formed in Activity 2 and the situations assigned to the groups. Ask each group to find the mode of the data. They should also compare this with the mean, and interpret the meaning of both.

Remark : The mode can also be calculated for grouped data with unequal class sizes. However, we shall not be discussing it.

EXERCISE 13.2

1. The following table shows the ages of the patients admitted in a hospital during a year:

Age (in years)	5 - 15	15 - 25	25 - 35	35 - 45	45 - 55	55 - 65
Number of patients	6	11	21	23	14	5

Find the mode and the mean of the data given above. Compare and interpret the two measures of central tendency.

2. The following data gives the information on the observed lifetimes (in hours) of 225 electrical components :

Lifetimes (in hours)	0 - 20	20 - 40	40 - 60	60 - 80	80 - 100	100 - 120
Frequency	10	35	52	61	38	29

Determine the modal lifetimes of the components.

3. The following data gives the distribution of total monthly household expenditure of 200 families of a village. Find the modal monthly expenditure of the families. Also, find the mean monthly expenditure :

Expenditure (in ₹)	Number of families
1000 - 1500	24
1500 - 2000	40
2000 - 2500	33
2500 - 3000	28
3000 - 3500	30
3500 - 4000	22
4000 - 4500	16
4500 - 5000	7

4. The following distribution gives the state-wise teacher-student ratio in higher secondary schools of India. Find the mode and mean of this data. Interpret the two measures.

Number of students per teacher	Number of states / U.T.
15 - 20	3
20 - 25	8
25 - 30	9
30 - 35	10
35 - 40	3
40 - 45	0
45 - 50	0
50 - 55	2

5. The given distribution shows the number of runs scored by some top batsmen of the world in one-day international cricket matches.

Runs scored	Number of batsmen
3000 - 4000	4
4000 - 5000	18
5000 - 6000	9
6000 - 7000	7
7000 - 8000	6
8000 - 9000	3
9000 - 10000	1
10000 - 11000	1

Find the mode of the data.

6. A student noted the number of cars passing through a spot on a road for 100 periods each of 3 minutes and summarised it in the table given below. Find the mode of the data :

Number of cars	0 - 10	10 - 20	20 - 30	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80
Frequency	7	14	13	12	20	11	15	8

13.4 Median of Grouped Data

As you have studied in Class IX, the median is a measure of central tendency which gives the value of the middle-most observation in the data. Recall that for finding the median of ungrouped data, we first arrange the data values of the observations in

ascending order. Then, if n is odd, the median is the $\left(\frac{n+1}{2}\right)$ th observation. And, if n

is even, then the median will be the average of the $\frac{n}{2}$ th and the $\left(\frac{n}{2} + 1\right)$ th observations.

Suppose, we have to find the median of the following data, which gives the marks, out of 50, obtained by 100 students in a test :

Marks obtained	20	29	28	33	42	38	43	25
Number of students	6	28	24	15	2	4	1	20

First, we arrange the marks in ascending order and prepare a frequency table as follows :

Table 13.9

Marks obtained	Number of students (Frequency)
20	6
25	20
28	24
29	28
33	15
38	4
42	2
43	1
Total	100

Here $n = 100$, which is even. The median will be the average of the $\frac{n}{2}$ th and the

$\left(\frac{n}{2} + 1\right)$ th observations, i.e., the 50th and 51st observations. To find these observations, we proceed as follows:

Table 13.10

Marks obtained	Number of students
20	6
upto 25	$6 + 20 = 26$
upto 28	$26 + 24 = 50$
upto 29	$50 + 28 = 78$
upto 33	$78 + 15 = 93$
upto 38	$93 + 4 = 97$
upto 42	$97 + 2 = 99$
upto 43	$99 + 1 = 100$

Now we add another column depicting this information to the frequency table above and name it as *cumulative frequency column*.

Table 13.11

Marks obtained	Number of students	Cumulative frequency
20	6	6
25	20	26
28	24	50
29	28	78
33	15	93
38	4	97
42	2	99
43	1	100

From the table above, we see that:

50th observation is 28 (Why?)

51st observation is 29

So,
$$\text{Median} = \frac{28 + 29}{2} = 28.5$$

Remark : The part of Table 13.11 consisting Column 1 and Column 3 is known as *Cumulative Frequency Table*. The median marks 28.5 conveys the information that about 50% students obtained marks less than 28.5 and another 50% students obtained marks more than 28.5.

Now, let us see how to obtain the median of grouped data, through the following situation.

Consider a grouped frequency distribution of marks obtained, out of 100, by 53 students, in a certain examination, as follows:

Table 13.12

Marks	Number of students
0 - 10	5
10 - 20	3
20 - 30	4
30 - 40	3
40 - 50	3
50 - 60	4
60 - 70	7
70 - 80	9
80 - 90	7
90 - 100	8

From the table above, try to answer the following questions:

How many students have scored marks less than 10? The answer is clearly 5.

How many students have scored less than 20 marks? Observe that the number of students who have scored less than 20 include the number of students who have scored marks from 0 - 10 as well as the number of students who have scored marks from 10 - 20. So, the total number of students with marks less than 20 is $5 + 3$, i.e., 8. We say that the cumulative frequency of the class 10-20 is 8.

Similarly, we can compute the cumulative frequencies of the other classes, i.e., the number of students with marks less than 30, less than 40, . . . , less than 100. We give them in Table 13.13 given below:

Table 13.13

Marks obtained	Number of students (Cumulative frequency)
Less than 10	5
Less than 20	$5 + 3 = 8$
Less than 30	$8 + 4 = 12$
Less than 40	$12 + 3 = 15$
Less than 50	$15 + 3 = 18$
Less than 60	$18 + 4 = 22$
Less than 70	$22 + 7 = 29$
Less than 80	$29 + 9 = 38$
Less than 90	$38 + 7 = 45$
Less than 100	$45 + 8 = 53$

The distribution given above is called the *cumulative frequency distribution of the less than type*. Here 10, 20, 30, . . . 100, are the upper limits of the respective class intervals.

We can similarly make the table for the number of students with scores, more than or equal to 0, more than or equal to 10, more than or equal to 20, and so on. From Table 13.12, we observe that all 53 students have scored marks more than or equal to 0. Since there are 5 students scoring marks in the interval 0 - 10, this means that there are $53 - 5 = 48$ students getting more than or equal to 10 marks. Continuing in the same manner, we get the number of students scoring 20 or above as $48 - 3 = 45$, 30 or above as $45 - 4 = 41$, and so on, as shown in Table 13.14.

Table 13.14

Marks obtained	Number of students (Cumulative frequency)
More than or equal to 0	53
More than or equal to 10	$53 - 5 = 48$
More than or equal to 20	$48 - 3 = 45$
More than or equal to 30	$45 - 4 = 41$
More than or equal to 40	$41 - 3 = 38$
More than or equal to 50	$38 - 3 = 35$
More than or equal to 60	$35 - 4 = 31$
More than or equal to 70	$31 - 7 = 24$
More than or equal to 80	$24 - 9 = 15$
More than or equal to 90	$15 - 7 = 8$

The table above is called a *cumulative frequency distribution of the more than type*. Here 0, 10, 20, . . . , 90 give the lower limits of the respective class intervals.

Now, to find the median of grouped data, we can make use of any of these cumulative frequency distributions.

Let us combine Tables 13.12 and 13.13 to get Table 13.15 given below:

Table 13.15

Marks	Number of students (f)	Cumulative frequency (cf)
0 - 10	5	5
10 - 20	3	8
20 - 30	4	12
30 - 40	3	15
40 - 50	3	18
50 - 60	4	22
60 - 70	7	29
70 - 80	9	38
80 - 90	7	45
90 - 100	8	53

Now in a grouped data, we may not be able to find the middle observation by looking at the cumulative frequencies as the middle observation will be some value in

a class interval. It is, therefore, necessary to find the value inside a class that divides the whole distribution into two halves. But which class should this be?

To find this class, we find the cumulative frequencies of all the classes and $\frac{n}{2}$. We now locate the class whose cumulative frequency is greater than (and nearest to) $\frac{n}{2}$. This is called the *median class*. In the distribution above, $n = 53$. So, $\frac{n}{2} = 26.5$. Now 60 – 70 is the class whose cumulative frequency 29 is greater than (and nearest to) $\frac{n}{2}$, i.e., 26.5.

Therefore, 60 – 70 is the **median class**.

After finding the median class, we use the following formula for calculating the median.

$$\text{Median} = l + \left(\frac{\frac{n}{2} - cf}{f} \right) \times h,$$

where l = lower limit of median class,
 n = number of observations,
 cf = cumulative frequency of class preceding the median class,
 f = frequency of median class,
 h = class size (assuming class size to be equal).

Substituting the values $\frac{n}{2} = 26.5$, $l = 60$, $cf = 22$, $f = 7$, $h = 10$

in the formula above, we get

$$\begin{aligned} \text{Median} &= 60 + \left(\frac{26.5 - 22}{7} \right) \times 10 \\ &= 60 + \frac{45}{7} \\ &= 66.4 \end{aligned}$$

So, about half the students have scored marks less than 66.4, and the other half have scored marks more than 66.4.

Example 7 : A survey regarding the heights (in cm) of 51 girls of Class X of a school was conducted and the following data was obtained:

Height (in cm)	Number of girls
Less than 140	4
Less than 145	11
Less than 150	29
Less than 155	40
Less than 160	46
Less than 165	51

Find the median height.

Solution : To calculate the median height, we need to find the class intervals and their corresponding frequencies.

The given distribution being of the *less than type*, 140, 145, 150, ..., 165 give the upper limits of the corresponding class intervals. So, the classes should be below 140, 140 - 145, 145 - 150, ..., 160 - 165. Observe that from the given distribution, we find that there are 4 girls with height less than 140, i.e., the frequency of class interval below 140 is 4. Now, there are 11 girls with heights less than 145 and 4 girls with height less than 140. Therefore, the number of girls with height in the interval 140 - 145 is $11 - 4 = 7$. Similarly, the frequency of 145 - 150 is $29 - 11 = 18$, for 150 - 155, it is $40 - 29 = 11$, and so on. So, our frequency distribution table with the given cumulative frequencies becomes:

Table 13.16

Class intervals	Frequency	Cumulative frequency
Below 140	4	4
140 - 145	7	11
145 - 150	18	29
150 - 155	11	40
155 - 160	6	46
160 - 165	5	51

Now $n = 51$. So, $\frac{n}{2} = \frac{51}{2} = 25.5$. This observation lies in the class 145 - 150. Then,

l (the lower limit) = 145,

cf (the cumulative frequency of the class preceding 145 - 150) = 11,

f (the frequency of the median class 145 - 150) = 18,

h (the class size) = 5.

Using the formula, Median = $l + \left(\frac{\frac{n}{2} - cf}{f} \right) \times h$, we have

$$\begin{aligned} \text{Median} &= 145 + \left(\frac{25.5 - 11}{18} \right) \times 5 \\ &= 145 + \frac{72.5}{18} = 149.03. \end{aligned}$$

So, the median height of the girls is 149.03 cm.

This means that the height of about 50% of the girls is less than this height, and 50% are taller than this height.

Example 8 : The median of the following data is 525. Find the values of x and y , if the total frequency is 100.

Class interval	Frequency
0 - 100	2
100 - 200	5
200 - 300	x
300 - 400	12
400 - 500	17
500 - 600	20
600 - 700	y
700 - 800	9
800 - 900	7
900 - 1000	4

Solution :

Class intervals	Frequency	Cumulative frequency
0 - 100	2	2
100 - 200	5	7
200 - 300	x	$7 + x$
300 - 400	12	$19 + x$
400 - 500	17	$36 + x$
500 - 600	20	$56 + x$
600 - 700	y	$56 + x + y$
700 - 800	9	$65 + x + y$
800 - 900	7	$72 + x + y$
900 - 1000	4	$76 + x + y$

It is given that $n = 100$

So, $76 + x + y = 100$, i.e., $x + y = 24$ (1)

The median is 525, which lies in the class 500 – 600

So, $l = 500$, $f = 20$, $cf = 36 + x$, $h = 100$

Using the formula : $\text{Median} = l + \left(\frac{\frac{n}{2} - cf}{f} \right) h$, we get

$$525 = 500 + \left(\frac{50 - 36 - x}{20} \right) \times 100$$

i.e., $525 - 500 = (14 - x) \times 5$

i.e., $25 = 70 - 5x$

i.e., $5x = 70 - 25 = 45$

So, $x = 9$

Therefore, from (1), we get $9 + y = 24$

i.e., $y = 15$

Now, that you have studied about all the three measures of central tendency, let us discuss **which measure would be best suited for a particular requirement**.

The mean is the most frequently used measure of central tendency because it takes into account all the observations, and lies between the extremes, i.e., the largest and the smallest observations of the entire data. It also enables us to compare two or more distributions. For example, by comparing the average (mean) results of students of different schools of a particular examination, we can conclude which school has a better performance.

However, extreme values in the data affect the mean. For example, the mean of classes having frequencies more or less the same is a good representative of the data. But, if one class has frequency, say 2, and the five others have frequency 20, 25, 20, 21, 18, then the mean will certainly not reflect the way the data behaves. So, in such cases, the mean is not a good representative of the data.

In problems where individual observations are not important, and we wish to find out a 'typical' observation, the median is more appropriate, e.g., finding the typical productivity rate of workers, average wage in a country, etc. These are situations where extreme values may be there. So, rather than the mean, we take the median as a better measure of central tendency.

In situations which require establishing the most frequent value or most popular item, the mode is the best choice, e.g., to find the most popular T.V. programme being watched, the consumer item in greatest demand, the colour of the vehicle used by most of the people, etc.

Remarks :

1. There is an empirical relationship between the three measures of central tendency :

$$3 \text{ Median} = \text{Mode} + 2 \text{ Mean}$$

2. The median of grouped data with unequal class sizes can also be calculated. However, we shall not discuss it here.

EXERCISE 13.3

1. The following frequency distribution gives the monthly consumption of electricity of 68 consumers of a locality. Find the median, mean and mode of the data and compare them.

Monthly consumption (in units)	Number of consumers
65 - 85	4
85 - 105	5
105 - 125	13
125 - 145	20
145 - 165	14
165 - 185	8
185 - 205	4

2. If the median of the distribution given below is 28.5, find the values of x and y .

Class interval	Frequency
0 - 10	5
10 - 20	x
20 - 30	20
30 - 40	15
40 - 50	y
50 - 60	5
Total	60

3. A life insurance agent found the following data for distribution of ages of 100 policy holders. Calculate the median age, if policies are given only to persons having age 18 years onwards but less than 60 year.

Age (in years)	Number of policy holders
Below 20	2
Below 25	6
Below 30	24
Below 35	45
Below 40	78
Below 45	89
Below 50	92
Below 55	98
Below 60	100

4. The lengths of 40 leaves of a plant are measured correct to the nearest millimetre, and the data obtained is represented in the following table :

Length (in mm)	Number of leaves
118 - 126	3
127 - 135	5
136 - 144	9
145 - 153	12
154 - 162	5
163 - 171	4
172 - 180	2

Find the median length of the leaves.

(**Hint :** The data needs to be converted to continuous classes for finding the median, since the formula assumes continuous classes. The classes then change to 117.5 - 126.5, 126.5 - 135.5, ..., 171.5 - 180.5.)

5. The following table gives the distribution of the life time of 400 neon lamps :

Life time (in hours)	Number of lamps
1500 - 2000	14
2000 - 2500	56
2500 - 3000	60
3000 - 3500	86
3500 - 4000	74
4000 - 4500	62
4500 - 5000	48

Find the median life time of a lamp.

6. 100 surnames were randomly picked up from a local telephone directory and the frequency distribution of the number of letters in the English alphabets in the surnames was obtained as follows:

Number of letters	1 - 4	4 - 7	7 - 10	10 - 13	13 - 16	16 - 19
Number of surnames	6	30	40	16	4	4

Determine the median number of letters in the surnames. Find the mean number of letters in the surnames? Also, find the modal size of the surnames.

7. The distribution below gives the weights of 30 students of a class. Find the median weight of the students.

Weight (in kg)	40 - 45	45 - 50	50 - 55	55 - 60	60 - 65	65 - 70	70 - 75
Number of students	2	3	8	6	6	3	2

13.5 Graphical Representation of Cumulative Frequency Distribution

As we all know, pictures speak better than words. A graphical representation helps us in understanding given data at a glance. In Class IX, we have represented the data through bar graphs, histograms and frequency polygons. Let us now represent a cumulative frequency distribution graphically.

For example, let us consider the cumulative frequency distribution given in Table 13.13.

Recall that the values 10, 20, 30, . . . , 100 are the upper limits of the respective class intervals. To represent the data in the table graphically, we mark the upper limits of the class intervals on the horizontal axis (x -axis) and their corresponding cumulative frequencies on the vertical axis (y -axis), choosing a convenient scale. The scale may not be the same on both the axis. Let us now plot the points corresponding to the ordered pairs given by (upper limit, corresponding cumulative frequency), i.e., (10, 5), (20, 8), (30, 12), (40, 15), (50, 18), (60, 22), (70, 29), (80, 38), (90, 45), (100, 53) on a graph paper and join them by a free hand smooth curve. The curve we get is called a **cumulative frequency curve**, or an **ogive** (of the less than type). (See Fig. 13.1)

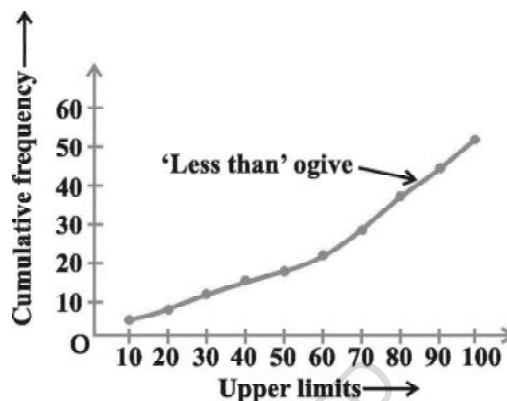


Fig. 13.1

The term 'ogive' is pronounced as 'ojeev' and is derived from the word **ogee**. An ogee is a shape consisting of a concave arc flowing into a convex arc, so forming an S-shaped curve with vertical ends. In architecture, the ogee shape is one of the characteristics of the 14th and 15th century Gothic styles.

Next, again we consider the cumulative frequency distribution given in Table 13.14 and draw its ogive (of the more than type).

Recall that, here 0, 10, 20, . . . , 90 are the lower limits of the respective class intervals 0 - 10, 10 - 20, . . . , 90 - 100. To represent 'the more than type' graphically, we plot the lower limits on the x -axis and the corresponding cumulative frequencies on the y -axis. Then we plot the points (lower limit, corresponding cumulative frequency), i.e., (0, 53), (10, 48), (20, 45), (30, 41), (40, 38), (50, 35), (60, 31), (70, 24), (80, 15), (90, 8), on a graph paper, and join them by a free hand smooth curve.

The curve we get is a *cumulative frequency curve*, or an *ogive* (of the more than type). (See Fig. 13.2)

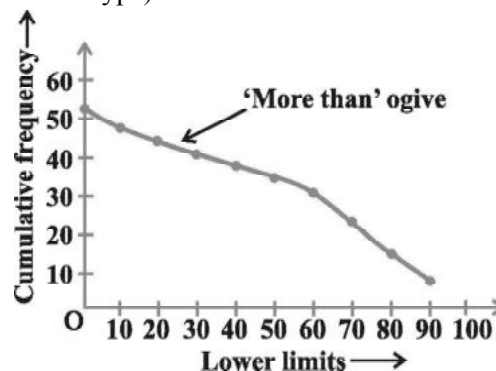


Fig. 13.2

Remark : Note that both the ogives (in Fig. 13.1 and Fig. 13.2) correspond to the same data, which is given in Table 13.12.

Now, are the ogives related to the median in any way? Is it possible to obtain the median from these two cumulative frequency curves corresponding to the data in Table 13.12? Let us see.

One obvious way is to locate

$$\frac{n}{2} = \frac{53}{2} = 26.5 \text{ on the } y\text{-axis (see Fig. 13.3).}$$

From this point, draw a line parallel to the x -axis cutting the curve at a point. From this point, draw a perpendicular to the x -axis. The point of intersection of this perpendicular with the x -axis determines the median of the data (see Fig. 13.3).

Another way of obtaining the median is the following :

Draw both ogives (i.e., of the less than type and of the more than type) on the same axis. The two ogives will intersect each other at a point. From this point, if we draw a perpendicular on the x -axis, the point at which it cuts the x -axis gives us the median (see Fig. 13.4).

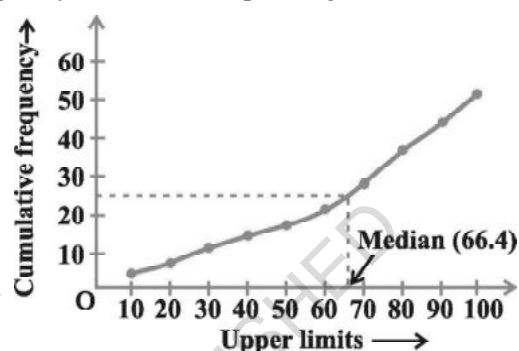


Fig 13.3

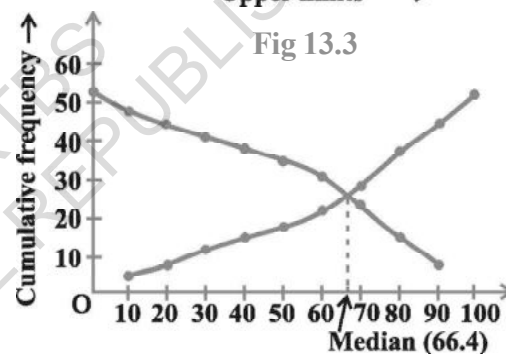


Fig 13.4

Example 9 : The annual profits earned by 30 shops of a shopping complex in a locality give rise to the following distribution :

Profit (₹ in lakhs)	Number of shops (frequency)
More than or equal to 5	30
More than or equal to 10	28
More than or equal to 15	16
More than or equal to 20	14
More than or equal to 25	10
More than or equal to 30	7
More than or equal to 35	3

Hence obtain the median profit.

Solution : We first draw the coordinate axes, with lower limits of the profit along the horizontal axis, and the cumulative frequency along the vertical axes. Then, we plot the points (5, 30), (10, 28), (15, 16), (20, 14), (25, 10), (30, 7) and (35, 3). We join these points with a smooth curve to get the 'more than' ogive, as shown in Fig. 13.5.

Now, let us obtain the classes, their frequencies and the cumulative frequency from the table above.

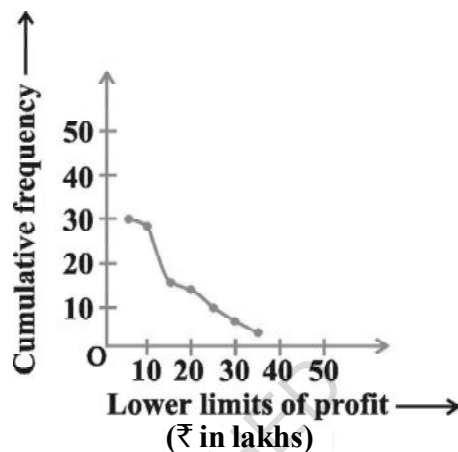


Fig. 13.5

Table 13.17

Classes	5 - 10	10 - 15	15 - 20	20 - 25	25 - 30	30 - 35	35 - 40
No. of shops	2	12	2	4	3	4	3
Cumulative frequency	2	14	16	20	23	27	30

Using these values, we plot the points (10, 2), (15, 14), (20, 16), (25, 20), (30, 23), (35, 27), (40, 30) on the same axes as in Fig. 13.5 to get the 'less than' ogive, as shown in Fig. 13.6..

The abscissa of their point of intersection is nearly 17.5, which is the median. This can also be verified by using the formula. Hence, the median profit (in lakhs) is ₹ 17.5.

Remark : In the above examples, it may be noted that the class intervals were continuous. For drawing ogives, it should be ensured that the class intervals are continuous. (Also see constructions of histograms in Class IX)

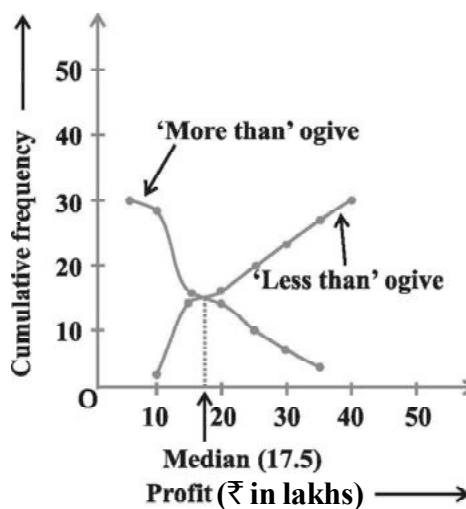


Fig. 13.6

EXERCISE 13.4

1. The following distribution gives the daily income of 50 workers of a factory.

Daily income (in ₹)	100 - 120	120 - 140	140 - 160	160 - 180	180 - 200
Number of workers	12	14	8	6	10

Convert the distribution above to a less than type cumulative frequency distribution, and draw its ogive.

2. During the medical check-up of 35 students of a class, their weights were recorded as follows:

Weight (in kg)	Number of students
Less than 38	0
Less than 40	3
Less than 42	5
Less than 44	9
Less than 46	14
Less than 48	28
Less than 50	32
Less than 52	35

Draw a less than type ogive for the given data. Hence obtain the median weight from the graph and verify the result by using the formula.

3. The following table gives production yield per hectare of wheat of 100 farms of a village.

Production yield (in kg/ha)	50 - 55	55 - 60	60 - 65	65 - 70	70 - 75	75 - 80
Number of farms	2	8	12	24	38	16

Change the distribution to a more than type distribution, and draw its ogive.

13.6 Summary

In this chapter, you have studied the following points:

1. The mean for grouped data can be found by :

(i) the direct method : $\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$

- (ii) the assumed mean method : $\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i}$
- (iii) the step deviation method : $\bar{x} = a + \left(\frac{\sum f_i u_i}{\sum f_i} \right) \times h$,

with the assumption that the frequency of a class is centred at its mid-point, called its class mark.

2. The mode for grouped data can be found by using the formula:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

where symbols have their usual meanings.

3. The cumulative frequency of a class is the frequency obtained by adding the frequencies of all the classes preceding the given class.
4. The median for grouped data is formed by using the formula:

$$\text{Median} = l + \left(\frac{\frac{n}{2} - \text{cf}}{f} \right) \times h,$$

where symbols have their usual meanings.

5. Representing a cumulative frequency distribution graphically as a cumulative frequency curve, or an ogive of the less than type and of the more than type.
6. The median of grouped data can be obtained graphically as the x -coordinate of the point of intersection of the two ogives for this data.

A NOTE TO THE READER

For calculating mode and median for grouped data, it should be ensured that the class intervals are continuous before applying the formulae. Same condition also apply for construction of an ogive. Further, in case of ogives, the scale may not be the same on both the axes.



PROBABILITY

14

The theory of probabilities and the theory of errors now constitute a formidable body of great mathematical interest and of great practical importance.

— R.S. Woodward

14.1 Introduction

In Class IX, you have studied about experimental (or empirical) probabilities of events which were based on the results of actual experiments. We discussed an experiment of tossing a coin 1000 times in which the frequencies of the outcomes were as follows:

Head : 455 Tail : 545

Based on this experiment, the empirical probability of a head is $\frac{455}{1000}$, i.e., 0.455 and that of getting a tail is 0.545. (Also see Example 1, Chapter 15 of Class IX Mathematics Textbook.) Note that these probabilities are based on the results of an actual experiment of tossing a coin 1000 times. For this reason, they are called *experimental* or *empirical probabilities*. In fact, experimental probabilities are based on the results of actual experiments and adequate recordings of the happening of the events. Moreover, these probabilities are only ‘estimates’. If we perform the same experiment for another 1000 times, we may get different data giving different probability estimates.

In Class IX, you tossed a coin many times and noted the number of times it turned up heads (or tails) (refer to Activities 1 and 2 of Chapter 15). You also noted that as the number of tosses of the coin increased, the experimental probability of getting a head

(or tail) came closer and closer to the number $\frac{1}{2}$. Not only you, but many other persons

persons from different parts of the world have done this kind of experiment and recorded the number of heads that turned up.

For example, the eighteenth century French naturalist Comte de Buffon tossed a coin 4040 times and got 2048 heads. The experimental probability of getting a head, in this case, was $\frac{2048}{4040}$, i.e., 0.507. J.E. Kerrich, from Britain, recorded 5067 heads in 10000 tosses of a coin. The experimental probability of getting a head, in this case, was $\frac{5067}{10000} = 0.5067$. Statistician Karl Pearson spent some more time, making 24000 tosses of a coin. He got 12012 heads, and thus, the experimental probability of a head obtained by him was 0.5005.

Now, suppose we ask, ‘What will the experimental probability of a head be if the experiment is carried on upto, say, one million times? Or 10 million times? And so on?’ You would intuitively feel that as the number of tosses increases, the experimental probability of a head (or a tail) seems to be settling down around the number 0.5, i.e., $\frac{1}{2}$, which is what we call the *theoretical probability* of getting a head (or getting a tail), as you will see in the next section. In this chapter, we provide an introduction to the theoretical (also called classical) probability of an event, and discuss simple problems based on this concept.

14.2 Probability — A Theoretical Approach

Let us consider the following situation :

Suppose a coin is tossed *at random*.

When we speak of a coin, we assume it to be ‘fair’, that is, it is symmetrical so that there is no reason for it to come down more often on one side than the other. We call this property of the coin as being ‘unbiased’. By the phrase ‘random toss’, we mean that the coin is allowed to fall freely without any *bias* or *interference*.

We know, in advance, that the coin can only land in one of two possible ways — either head up or tail up (we dismiss the possibility of its ‘landing’ on its edge, which may be possible, for example, if it falls on sand). We can reasonably assume that each outcome, head or tail, is *as likely to occur as the other*. We refer to this by saying that *the outcomes head and tail, are equally likely*.

For another example of equally likely outcomes, suppose we throw a die once. For us, a die will always mean a fair die. What are the possible outcomes? They are 1, 2, 3, 4, 5, 6. Each number has the same possibility of showing up. So the *equally likely outcomes* of throwing a die are 1, 2, 3, 4, 5 and 6.

Are the outcomes of every experiment equally likely? Let us see.

Suppose that a bag contains 4 red balls and 1 blue ball, and you draw a ball without looking into the bag. What are the outcomes? Are the outcomes — a red ball and a blue ball equally likely? Since there are 4 red balls and only one blue ball, you would agree that you are more likely to get a red ball than a blue ball. So, the outcomes (a red ball or a blue ball) are *not* equally likely. However, the outcome of drawing a ball of any colour from the bag is equally likely. So, all experiments do not necessarily have equally likely outcomes.

However, in this chapter, from now on, we will **assume that all the experiments have equally likely outcomes**.

In Class IX, we defined the experimental or empirical probability $P(E)$ of an event E as

$$P(E) = \frac{\text{Number of trials in which the event happened}}{\text{Total number of trials}}$$

The empirical interpretation of probability can be applied to every event associated with an experiment which can be repeated a large number of times. The requirement of repeating an experiment has some limitations, as it may be very expensive or unfeasible in many situations. Of course, it worked well in coin tossing or die throwing experiments. But how about repeating the experiment of launching a satellite in order to compute the empirical probability of its failure during launching, or the repetition of the phenomenon of an earthquake to compute the empirical probability of a multi-storeyed building getting destroyed in an earthquake?

In experiments where we are prepared to make certain assumptions, the repetition of an experiment can be avoided, as the assumptions help in directly calculating the exact (theoretical) probability. The assumption of equally likely outcomes (which is valid in many experiments, as in the two examples above, of a coin and of a die) is one such assumption that leads us to the following definition of probability of an event.

The **theoretical probability** (also called **classical probability**) of an event E , written as $P(E)$, is defined as

$$P(E) = \frac{\text{Number of outcomes favourable to } E}{\text{Number of all possible outcomes of the experiment}},$$

where we assume that the outcomes of the experiment are *equally likely*.

We will briefly refer to theoretical probability as probability.

This definition of probability was given by Pierre Simon Laplace in 1795.

Probability theory had its origin in the 16th century when an Italian physician and mathematician J. Cardan wrote the first book on the subject, *The Book on Games of Chance*. Since its inception, the study of probability has attracted the attention of great mathematicians. James Bernoulli (1654 – 1705), A. de Moivre (1667 – 1754), and Pierre Simon Laplace are among those who made significant contributions to this field. Laplace's *Theorie Analytique des Probabilités*, 1812, is considered to be the greatest contribution by a single person to the theory of probability. In recent years, probability has been used extensively in many areas such as biology, economics, genetics, physics, sociology etc.



Pierre Simon Laplace
(1749 – 1827)

Let us find the probability for some of the events associated with experiments where the equally likely assumption holds.

Example 1 : Find the probability of getting a head when a coin is tossed once. Also find the probability of getting a tail.

Solution : In the experiment of tossing a coin once, the number of possible outcomes is two — Head (H) and Tail (T). Let E be the event ‘getting a head’. The number of outcomes favourable to E, (i.e., of getting a head) is 1. Therefore,

$$P(E) = P(\text{head}) = \frac{\text{Number of outcomes favourable to E}}{\text{Number of all possible outcomes}} = \frac{1}{2}$$

Similarly, if F is the event ‘getting a tail’, then

$$P(F) = P(\text{tail}) = \frac{1}{2} \quad (\text{Why ?})$$

Example 2 : A bag contains a red ball, a blue ball and a yellow ball, all the balls being of the same size. Kritika takes out a ball from the bag without looking into it. What is the probability that she takes out the

- (i) yellow ball? (ii) red ball? (iii) blue ball?

Solution : Kritika takes out a ball from the bag without looking into it. So, it is equally likely that she takes out any one of them.

Let Y be the event ‘the ball taken out is yellow’, B be the event ‘the ball taken out is blue’, and R be the event ‘the ball taken out is red’.

Now, the number of possible outcomes = 3.

(i) The number of outcomes favourable to the event Y = 1.

So, $P(Y) = \frac{1}{3}$

Similarly, (ii) $P(R) = \frac{1}{3}$ and (iii) $P(B) = \frac{1}{3}$.

Remarks :

1. An event having only one outcome of the experiment is called an *elementary event*. In Example 1, both the events E and F are elementary events. Similarly, in Example 2, all the three events, Y, B and R are elementary events.

2. In Example 1, we note that : $P(E) + P(F) = 1$

In Example 2, we note that : $P(Y) + P(R) + P(B) = 1$

Observe that **the sum of the probabilities of all the elementary events of an experiment** is 1. This is true in general also.

Example 3 : Suppose we throw a die once. (i) What is the probability of getting a number greater than 4 ? (ii) What is the probability of getting a number less than or equal to 4 ?

Solution : (i) Here, let E be the event ‘getting a number greater than 4’. The number of possible outcomes is six : 1, 2, 3, 4, 5 and 6, and the outcomes favourable to E are 5 and 6. Therefore, the number of outcomes favourable to E is 2. So,

$$P(E) = P(\text{number greater than 4}) = \frac{2}{6} = \frac{1}{3}$$

(ii) Let F be the event ‘getting a number less than or equal to 4’.

Number of possible outcomes = 6

Outcomes favourable to the event F are 1, 2, 3, 4.

So, the number of outcomes favourable to F is 4.

Therefore, $P(F) = \frac{4}{6} = \frac{2}{3}$

Are the events E and F in the example above elementary events? No, they are **not** because the event E has 2 outcomes and the event F has 4 outcomes.

Remarks : From Example 1, we note that

$$P(E) + P(F) = \frac{1}{2} + \frac{1}{2} = 1 \quad (1)$$

where E is the event ‘getting a head’ and F is the event ‘getting a tail’.

From (i) and (ii) of Example 3, we also get

$$P(E) + P(F) = \frac{1}{3} + \frac{2}{3} = 1 \quad (2)$$

where E is the event ‘getting a number >4 ’ and F is the event ‘getting a number ≤ 4 ’.

Note that getting a number *not* greater than 4 is same as getting a number less than or equal to 4, and vice versa.

In (1) and (2) above, is F not the same as ‘not E’? Yes, it is. We denote the event ‘not E’ by $\bar{E}$.

So, $P(E) + P(\text{not } E) = 1$

i.e., $P(E) + P(\bar{E}) = 1$, which gives us $P(\bar{E}) = 1 - P(E)$.

In general, it is true that for an event E,

$$P(\bar{E}) = 1 - P(E)$$

The event $\bar{E}$, representing ‘not E’, is called the **complement** of the event E. We also say that E and $\bar{E}$ are **complementary** events.

Before proceeding further, let us try to find the answers to the following questions:

- (i) What is the probability of getting a number 8 in a single throw of a die?
- (ii) What is the probability of getting a number less than 7 in a single throw of a die?

Let us answer (i) :

We know that there are only six possible outcomes in a single throw of a die. These outcomes are 1, 2, 3, 4, 5 and 6. Since no face of the die is marked 8, so there is no outcome favourable to 8, i.e., the number of such outcomes is zero. In other words, getting 8 in a single throw of a die, is *impossible*.

So, $P(\text{getting } 8) = \frac{0}{6} = 0$

That is, the probability of an event which is *impossible* to occur is 0. Such an event is called an **impossible event**.

Let us answer (ii) :

Since every face of a die is marked with a number less than 7, it is *sure* that we will always get a number less than 7 when it is thrown once. So, the number of favourable outcomes is the same as the number of all possible outcomes, which is 6.

Therefore, $P(E) = P(\text{getting a number less than 7}) = \frac{6}{6} = 1$

So, the probability of an event which is *sure* (or *certain*) to occur is 1. Such an event is called a **sure event** or a **certain event**.

Note : From the definition of the probability $P(E)$, we see that the numerator (number of outcomes favourable to the event E) is always less than or equal to the denominator (the number of all possible outcomes). Therefore,

$$0 \leq P(E) \leq 1$$

Now, let us take an example related to playing cards. Have you seen a deck of playing cards? It consists of 52 cards which are divided into 4 suits of 13 cards each—spades (♠), hearts (♥), diamonds (♦) and clubs (♣). Clubs and spades are of black colour, while hearts and diamonds are of red colour. The cards in each suit are ace, king, queen, jack, 10, 9, 8, 7, 6, 5, 4, 3 and 2. Kings, queens and jacks are called *face cards*.

Example 4 : One card is drawn from a well-shuffled deck of 52 cards. Calculate the probability that the card will

- (i) be an ace,
- (ii) not be an ace.

Solution : Well-shuffling ensures *equally likely* outcomes.

- (i) There are 4 aces in a deck. Let E be the event ‘the card is an ace’.

The number of outcomes favourable to $E = 4$

The number of possible outcomes = 52 (Why ?)

Therefore, $P(E) = \frac{4}{52} = \frac{1}{13}$

- (ii) Let F be the event ‘card drawn is not an ace’.

The number of outcomes favourable to the event $F = 52 - 4 = 48$ (Why?)

The number of possible outcomes = 52

$$\text{Therefore, } P(F) = \frac{48}{52} = \frac{12}{13}$$

Remark : Note that F is nothing but $\bar{E}$. Therefore, we can also calculate $P(F)$ as

$$\text{follows: } P(F) = P(\bar{E}) = 1 - P(E) = 1 - \frac{1}{13} = \frac{12}{13}.$$

Example 5 : Two players, Sangeeta and Reshma, play a tennis match. It is known that the probability of Sangeeta winning the match is 0.62. What is the probability of Reshma winning the match?

Solution : Let S and R denote the events that Sangeeta wins the match and Reshma wins the match, respectively.

The probability of Sangeeta's winning = $P(S) = 0.62$ (given)

The probability of Reshma's winning = $P(R) = 1 - P(S)$

$$\begin{aligned} &[\text{As the events } R \text{ and } S \text{ are complementary}] \\ &= 1 - 0.62 = 0.38 \end{aligned}$$

Example 6 : Savita and Hamida are friends. What is the probability that both will have (i) different birthdays? (ii) the same birthday? (ignoring a leap year).

Solution : Out of the two friends, one girl, say, Savita's birthday can be any day of the year. Now, Hamida's birthday can also be any day of 365 days in the year.

We assume that these 365 outcomes are equally likely.

- (i) If Hamida's birthday is different from Savita's, the number of favourable outcomes for her birthday is $365 - 1 = 364$

$$\text{So, } P(\text{Hamida's birthday is different from Savita's birthday}) = \frac{364}{365}$$

- (ii) $P(\text{Savita and Hamida have the same birthday})$

$$\begin{aligned} &= 1 - P(\text{both have different birthdays}) \\ &= 1 - \frac{364}{365} \quad [\text{Using } P(\bar{E}) = 1 - P(E)] \\ &= \frac{1}{365} \end{aligned}$$

Example 7 : There are 40 students in Class X of a school of whom 25 are girls and 15 are boys. The class teacher has to select one student as a class representative. She writes the name of each student on a separate card, the cards being identical. Then she puts cards in a bag and stirs them thoroughly. She then draws one card from the bag. What is the probability that the name written on the card is the name of (i) a girl? (ii) a boy?

Solution : There are 40 students, and only one name card has to be chosen.

(i) The number of all possible outcomes is 40

The number of outcomes favourable for a card with the name of a girl = 25 (Why?)

$$\text{Therefore, } P(\text{card with name of a girl}) = P(\text{Girl}) = \frac{25}{40} = \frac{5}{8}$$

(ii) The number of outcomes favourable for a card with the name of a boy = 15 (Why?)

$$\text{Therefore, } P(\text{card with name of a boy}) = P(\text{Boy}) = \frac{15}{40} = \frac{3}{8}$$

Note : We can also determine $P(\text{Boy})$, by taking

$$P(\text{Boy}) = 1 - P(\text{not Boy}) = 1 - P(\text{Girl}) = 1 - \frac{5}{8} = \frac{3}{8}$$

Example 8 : A box contains 3 blue, 2 white, and 4 red marbles. If a marble is drawn at random from the box, what is the probability that it will be

(i) white? (ii) blue? (iii) red?

Solution : Saying that a marble is drawn at random is a short way of saying that all the marbles are equally likely to be drawn. Therefore, the

$$\text{number of possible outcomes} = 3 + 2 + 4 = 9 \quad (\text{Why?})$$

Let W denote the event 'the marble is white', B denote the event 'the marble is blue' and R denote the event 'marble is red'.

(i) The number of outcomes favourable to the event $W = 2$

$$\text{So, } P(W) = \frac{2}{9}$$

$$\text{Similarly, (ii) } P(B) = \frac{3}{9} = \frac{1}{3} \quad \text{and} \quad \text{(iii) } P(R) = \frac{4}{9}$$

Note that $P(W) + P(B) + P(R) = 1$.

Example 9 : Harpreet tosses two different coins simultaneously (say, one is of ₹ 1 and other of ₹ 2). What is the probability that she gets *at least* one head?

Solution : We write H for ‘head’ and T for ‘tail’. When two coins are tossed simultaneously, the possible outcomes are (H, H), (H, T), (T, H), (T, T), which are all *equally likely*. Here (H, H) means head up on the first coin (say on ₹ 1) and head up on the second coin (₹ 2). Similarly (H, T) means head up on the first coin and tail up on the second coin and so on.

The outcomes favourable to the event E, ‘at least one head’ are (H, H), (H, T) and (T, H). (Why?)

So, the number of outcomes favourable to E is 3.

Therefore, $P(E) = \frac{3}{4}$

i.e., the probability that Harpreet gets at least one head is $\frac{3}{4}$.

Note : You can also find $P(E)$ as follows:

$$P(E) = 1 - P(\bar{E}) = 1 - \frac{1}{4} = \frac{3}{4} \quad \left(\text{Since } P(\bar{E}) = P(\text{no head}) = \frac{1}{4} \right)$$

Did you observe that in all the examples discussed so far, the number of possible outcomes in each experiment was finite? If not, check it now.

There are many experiments in which the outcome is any number between two given numbers, or in which the outcome is every point within a circle or rectangle, etc. Can you now count the number of all possible outcomes? As you know, this is not possible since there are infinitely many numbers between two given numbers, or there are infinitely many points within a circle. So, the definition of (theoretical) probability which you have learnt so far cannot be applied in the present form. What is the way out? To answer this, let us consider the following example :

Example 10* : In a musical chair game, the person playing the music has been advised to stop playing the music at any time within 2 minutes after she starts playing. What is the probability that the music will stop within the first half-minute after starting?

Solution : Here the possible outcomes are all the numbers between 0 and 2. This is the portion of the number line from 0 to 2 (see Fig. 14.1).



Fig. 14.1

* Not from the examination point of view.

Let E be the event that ‘the music is stopped within the first half-minute’.

The outcomes favourable to E are points on the number line from 0 to $\frac{1}{2}$.

The distance from 0 to 2 is 2, while the distance from 0 to $\frac{1}{2}$ is $\frac{1}{2}$.

Since all the outcomes are equally likely, we can argue that, of the total distance of 2, the distance favourable to the event E is $\frac{1}{2}$.

$$\text{So, } P(E) = \frac{\text{Distance favourable to the event } E}{\text{Total distance in which outcomes can lie}} = \frac{\frac{1}{2}}{2} = \frac{1}{4}$$

Can we now extend the idea of Example 10 for finding the probability as the ratio of the favourable area to the total area?

Example 11* : A missing helicopter is reported to have crashed somewhere in the rectangular region shown in Fig. 14.2. What is the probability that it crashed inside the lake shown in the figure?

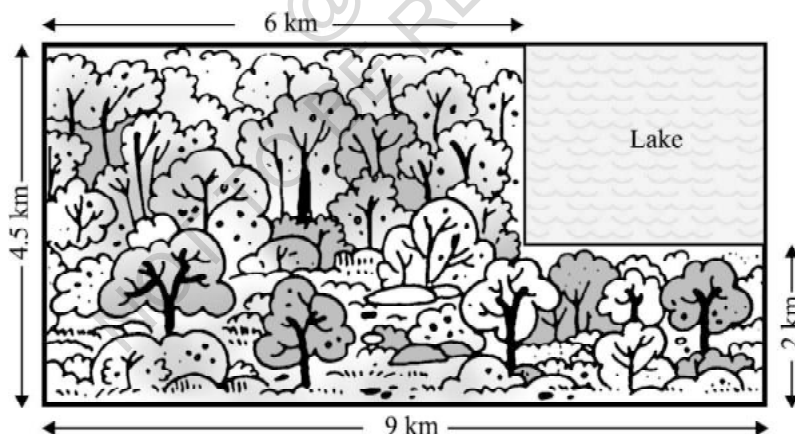


Fig. 14.2

Solution : The helicopter is equally likely to crash anywhere in the region.

Area of the entire region where the helicopter can crash

$$= (4.5 \times 9) \text{ km}^2 = 40.5 \text{ km}^2$$

* Not from the examination point of view.

Area of the lake = $(2.5 \times 3) \text{ km}^2 = 7.5 \text{ km}^2$

Therefore, $P(\text{helicopter crashed in the lake}) = \frac{7.5}{40.5} = \frac{75}{405} = \frac{5}{27}$

Example 12 : A carton consists of 100 shirts of which 88 are good, 8 have minor defects and 4 have major defects. Jimmy, a trader, will only accept the shirts which are good, but Sujatha, another trader, will only reject the shirts which have major defects. One shirt is drawn at random from the carton. What is the probability that

(i) it is acceptable to Jimmy?

(ii) it is acceptable to Sujatha?

Solution : One shirt is drawn at random from the carton of 100 shirts. Therefore, there are 100 equally likely outcomes.

(i) The number of outcomes favourable (i.e., acceptable) to Jimmy = 88 (Why?)

Therefore, $P(\text{shirt is acceptable to Jimmy}) = \frac{88}{100} = 0.88$

(ii) The number of outcomes favourable to Sujatha = $88 + 8 = 96$ (Why?)

So, $P(\text{shirt is acceptable to Sujatha}) = \frac{96}{100} = 0.96$

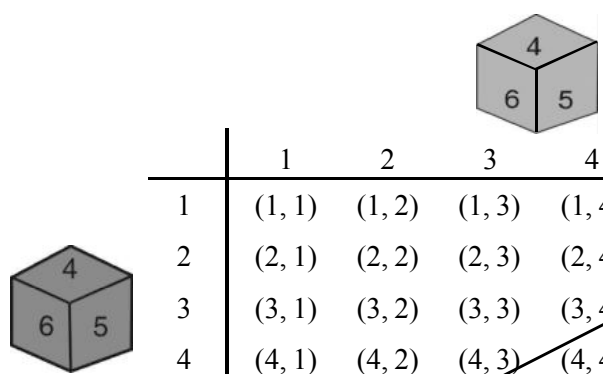
Example 13 : Two dice, one blue and one grey, are thrown at the same time. Write down all the possible outcomes. What is the probability that the sum of the two numbers appearing on the top of the dice is

(i) 8?

(ii) 13?

(iii) less than or equal to 12?

Solution : When the blue die shows '1', the grey die could show any one of the numbers 1, 2, 3, 4, 5, 6. The same is true when the blue die shows '2', '3', '4', '5' or '6'. The possible outcomes of the experiment are listed in the table below; the first number in each ordered pair is the number appearing on the blue die and the second number is that on the grey die.



	1	2	3	4	5	6
1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

Fig. 14.3

Note that the pair (1, 4) is different from (4, 1). (Why?)

So, the number of possible outcomes = $6 \times 6 = 36$.

- (i) The outcomes favourable to the event 'the sum of the two numbers is 8' denoted by E, are: (2, 6), (3, 5), (4, 4), (5, 3), (6, 2) (see Fig. 15.3)

i.e., the number of outcomes favourable to E = 5.

Hence,
$$P(E) = \frac{5}{36}$$

- (ii) As you can see from Fig. 15.3, there is no outcome favourable to the event F, 'the sum of two numbers is 13'.

So,
$$P(F) = \frac{0}{36} = 0$$

- (iii) As you can see from Fig. 15.3, all the outcomes are favourable to the event G, 'sum of two numbers ≤ 12 '.

So,
$$P(G) = \frac{36}{36} = 1$$

EXERCISE 14.1

1. Complete the following statements:
 - (i) Probability of an event E + Probability of the event 'not E ' = _____.
 - (ii) The probability of an event that cannot happen is _____. Such an event is called _____.
 - (iii) The probability of an event that is certain to happen is _____. Such an event is called _____.
 - (iv) The sum of the probabilities of all the elementary events of an experiment is _____.
 - (v) The probability of an event is greater than or equal to _____ and less than or equal to _____.
2. Which of the following experiments have equally likely outcomes? Explain.
 - (i) A driver attempts to start a car. The car starts or does not start.
 - (ii) A player attempts to shoot a basketball. She/he shoots or misses the shot.
 - (iii) A trial is made to answer a true-false question. The answer is right or wrong.
 - (iv) A baby is born. It is a boy or a girl.
3. Why is tossing a coin considered to be a fair way of deciding which team should get the ball at the beginning of a football game?
4. Which of the following cannot be the probability of an event?
(A) $\frac{2}{3}$ (B) -1.5 (C) 15% (D) 0.7
5. If $P(E) = 0.05$, what is the probability of 'not E '?
6. A bag contains lemon flavoured candies only. Malini takes out one candy without looking into the bag. What is the probability that she takes out
 - (i) an orange flavoured candy?
 - (ii) a lemon flavoured candy?
7. It is given that in a group of 3 students, the probability of 2 students not having the same birthday is 0.992 . What is the probability that the 2 students have the same birthday?
8. A bag contains 3 red balls and 5 black balls. A ball is drawn at random from the bag. What is the probability that the ball drawn is (i) red ? (ii) not red?
9. A box contains 5 red marbles, 8 white marbles and 4 green marbles. One marble is taken out of the box at random. What is the probability that the marble taken out will be (i) red ? (ii) white ? (iii) not green?

10. A piggy bank contains hundred 50p coins, fifty ₹ 1 coins, twenty ₹ 2 coins and ten ₹ 5 coins. If it is equally likely that one of the coins will fall out when the bank is turned upside down, what is the probability that the coin (i) will be a 50 p coin ? (ii) will not be a ₹ 5 coin?

11. Gopi buys a fish from a shop for his aquarium. The shopkeeper takes out one fish at random from a tank containing 5 male fish and 8 female fish (see Fig. 14.4). What is the probability that the fish taken out is a male fish?



Fig. 14.4

12. A game of chance consists of spinning an arrow which comes to rest pointing at one of the numbers 1, 2, 3, 4, 5, 6, 7, 8 (see Fig. 14.5), and these are equally likely outcomes. What is the probability that it will point at



Fig. 14.5

- (i) 8 ?
 - (ii) an odd number?
 - (iii) a number greater than 2?
 - (iv) a number less than 9?
13. A die is thrown once. Find the probability of getting
- (i) a prime number; (ii) a number lying between 2 and 6; (iii) an odd number.
14. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting
- (i) a king of red colour (ii) a face card (iii) a red face card
 - (iv) the jack of hearts (v) a spade (vi) the queen of diamonds
15. Five cards—the ten, jack, queen, king and ace of diamonds, are well-shuffled with their face downwards. One card is then picked up at random.
- (i) What is the probability that the card is the queen?
 - (ii) If the queen is drawn and put aside, what is the probability that the second card picked up is (a) an ace? (b) a queen?
16. 12 defective pens are accidentally mixed with 132 good ones. It is not possible to just look at a pen and tell whether or not it is defective. One pen is taken out at random from this lot. Determine the probability that the pen taken out is a good one.
17. (i) A lot of 20 bulbs contain 4 defective ones. One bulb is drawn at random from the lot. What is the probability that this bulb is defective?
- (ii) Suppose the bulb drawn in (i) is not defective and is not replaced. Now one bulb is drawn at random from the rest. What is the probability that this bulb is not defective ?
18. A box contains 90 discs which are numbered from 1 to 90. If one disc is drawn at random from the box, find the probability that it bears (i) a two-digit number (ii) a perfect square number (iii) a number divisible by 5.

19. A child has a die whose six faces show the letters as given below:



The die is thrown once. What is the probability of getting (i) A? (ii) D?

- 20*. Suppose you drop a die at random on the rectangular region shown in Fig. 14.6. What is the probability that it will land inside the circle with diameter 1m?

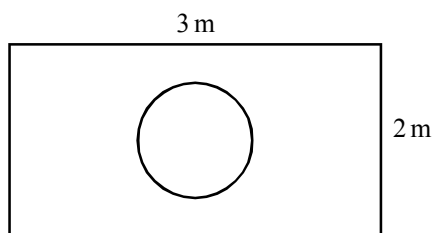


Fig. 14.6

21. A lot consists of 144 ball pens of which 20 are defective and the others are good. Nuri will buy a pen if it is good, but will not buy if it is defective. The shopkeeper draws one pen at random and gives it to her. What is the probability that
- She will buy it ?
 - She will not buy it ?
22. Refer to Example 13. (i) Complete the following table:

Event: 'Sum on 2 dice'	2	3	4	5	6	7	8	9	10	11	12
Probability	$\frac{1}{36}$						$\frac{5}{36}$				$\frac{1}{36}$

- A student argues that 'there are 11 possible outcomes 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 and 12. Therefore, each of them has a probability $\frac{1}{11}$. Do you agree with this argument? Justify your answer.
23. A game consists of tossing a one rupee coin 3 times and noting its outcome each time. Hanif wins if all the tosses give the same result i.e., three heads or three tails, and loses otherwise. Calculate the probability that Hanif will lose the game.
24. A die is thrown twice. What is the probability that
- 5 will not come up either time?
 - 5 will come up at least once?
- [Hint : Throwing a die twice and throwing two dice simultaneously are treated as the same experiment]

* Not from the examination point of view.

25. Which of the following arguments are correct and which are not correct? Give reasons for your answer.
- (i) If two coins are tossed simultaneously there are three possible outcomes—two heads, two tails or one of each. Therefore, for each of these outcomes, the probability is $\frac{1}{3}$.
 - (ii) If a die is thrown, there are two possible outcomes—an odd number or an even number. Therefore, the probability of getting an odd number is $\frac{1}{2}$.

EXERCISE 14.2 (Optional)*

- Two customers Shyam and Ekta are visiting a particular shop in the same week (Tuesday to Saturday). Each is equally likely to visit the shop on any day as on another day. What is the probability that both will visit the shop on (i) the same day? (ii) consecutive days? (iii) different days?
- A die is numbered in such a way that its faces show the numbers 1, 2, 2, 3, 3, 6. It is thrown two times and the total score in two throws is noted. Complete the following table which gives a few values of the total score on the two throws:

		Number in first throw					
Number in second throw	+	1	2	2	3	3	6
	1	2	3	3	4	4	7
	2	3	4	4	5	5	8
	2					5	
	3						
	3			5			9
	6	7	8	8	9	9	12

What is the probability that the total score is

- (i) even? (ii) 6? (iii) at least 6?
- A bag contains 5 red balls and some blue balls. If the probability of drawing a blue ball is *double* that of a red ball, determine the number of blue balls in the bag.
 - A box contains 12 balls out of which x are black. If one ball is drawn at random from the box, what is the probability that it will be a black ball?
If 6 more black balls are put in the box, the probability of drawing a black ball is now double of what it was before. Find x .

* These exercises are not from the examination point of view.

5. A jar contains 24 marbles, some are green and others are blue. If a marble is drawn at random from the jar, the probability that it is green is $\frac{2}{3}$. Find the number of blue balls in the jar.

14.3 Summary

In this chapter, you have studied the following points :

1. The difference between experimental probability and theoretical probability.
2. The theoretical (classical) probability of an event E, written as P(E), is defined as

$$P(E) = \frac{\text{Number of outcomes favourable to E}}{\text{Number of all possible outcomes of the experiment}}$$

where we assume that the outcomes of the experiment are equally likely.

3. The probability of a sure event (or certain event) is 1.
4. The probability of an impossible event is 0.
5. The probability of an event E is a number P(E) such that

$$0 \leq P(E) \leq 1$$
6. An event having only one outcome is called an elementary event. The sum of the probabilities of all the elementary events of an experiment is 1.
7. For any event E, $P(E) + P(\bar{E}) = 1$, where $\bar{E}$ stands for 'not E'. E and $\bar{E}$ are called complementary events.

A NOTE TO THE READER

The experimental or empirical probability of an event is based on what has actually happened while the theoretical probability of the event attempts to predict what will happen on the basis of certain assumptions. As the number of trials in an experiment, go on increasing we may expect the experimental and theoretical probabilities to be nearly the same.

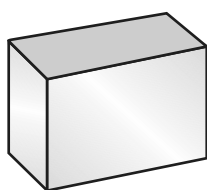


SURFACE AREAS AND VOLUMES

15

15.1 Introduction

From Class IX, you are familiar with some of the solids like cuboid, cone, cylinder, and sphere (see Fig. 13.1). You have also learnt how to find their surface areas and volumes.



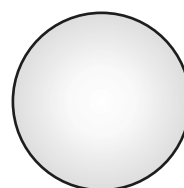
(i)



(ii)



(iii)



(iv)

Fig. 15.1

In our day-to-day life, we come across a number of solids made up of combinations of two or more of the basic solids as shown above.

You must have seen a truck with a container fitted on its back (see Fig. 15.2), carrying oil or water from one place to another. Is it in the shape of any of the four basic solids mentioned above? You may guess that it is made of a cylinder with two hemispheres as its ends.

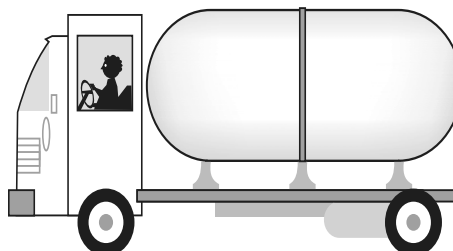


Fig. 15.2

Again, you may have seen an object like the one in Fig. 15.3. Can you name it? A test tube, right! You would have used one in your science laboratory. This tube is also a combination of a cylinder and a hemisphere. Similarly, while travelling, you may have seen some big and beautiful buildings or monuments made up of a combination of solids mentioned above.

If for some reason you wanted to find the surface areas, or volumes, or capacities of such objects, how would you do it? We cannot classify these under any of the solids you have already studied.

In this chapter, you will see how to find surface areas and volumes of such objects.



Fig. 15.3

15.2 Surface Area of a Combination of Solids

Let us consider the container seen in Fig. 13.2. How do we find the surface area of such a solid? Now, whenever we come across a new problem, we first try to see, if we can break it down into smaller problems, we have earlier solved. We can see that this solid is made up of a cylinder with two hemispheres stuck at either end. It would look like what we have in Fig. 13.4, after we put the pieces all together.



Fig. 15.4

If we consider the surface of the newly formed object, we would be able to see only the curved surfaces of the two hemispheres and the curved surface of the cylinder.

So, the *total* surface area of the new solid is the sum of the *curved* surface areas of each of the individual parts. This gives,

$$\begin{aligned} \text{TSA of new solid} &= \text{CSA of one hemisphere} + \text{CSA of cylinder} \\ &\quad + \text{CSA of other hemisphere} \end{aligned}$$

where TSA, CSA stand for 'Total Surface Area' and 'Curved Surface Area' respectively.

Let us now consider another situation. Suppose we are making a toy by putting together a hemisphere and a cone. Let us see the steps that we would be going through.

First, we would take a cone and a hemisphere and bring their flat faces together. Here, of course, we would take the base radius of the cone equal to the radius of the hemisphere, for the toy is to have a smooth surface. So, the steps would be as shown in Fig. 15.5.

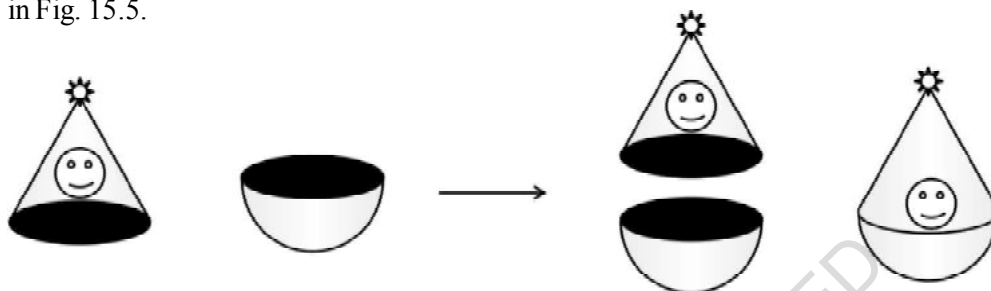


Fig. 15.5

At the end of our trial, we have got ourselves a nice round-bottomed toy. Now if we want to find how much paint we would require to colour the surface of this toy, what would we need to know? We would need to know the surface area of the toy, which consists of the CSA of the hemisphere and the CSA of the cone.

So, we can say:

$$\text{Total surface area of the toy} = \text{CSA of hemisphere} + \text{CSA of cone}$$

Now, let us consider some examples.

Example 1 : Rasheed got a playing top (*lattu*) as his birthday present, which surprisingly had no colour on it. He wanted to colour it with his crayons. The top is shaped like a cone surmounted by a hemisphere (see Fig 13.6). The entire top is 5 cm in height and the diameter of the top is 3.5 cm. Find the area he

has to colour. (Take $\pi = \frac{22}{7}$)

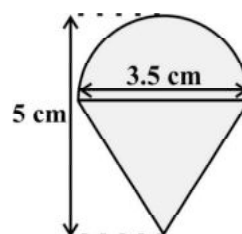


Fig. 15.6

Solution : This top is exactly like the object we have discussed in Fig. 13.5. So, we can conveniently use the result we have arrived at there. That is :

$$\text{TSA of the toy} = \text{CSA of hemisphere} + \text{CSA of cone}$$

$$\text{Now, the curved surface area of the hemisphere} = \frac{1}{2}(4\pi r^2) = 2\pi r^2$$

$$= \left(2 \times \frac{22}{7} \times \frac{3.5}{2} \times \frac{3.5}{2} \right) \text{cm}^2$$

Also, the height of the cone = height of the top – height (radius) of the hemispherical part

$$= \left(5 - \frac{3.5}{2} \right) \text{ cm} = 3.25 \text{ cm}$$

So, the slant height of the cone (l) = $\sqrt{r^2 + h^2} = \sqrt{\left(\frac{3.5}{2}\right)^2 + (3.25)^2} \text{ cm} = 3.7 \text{ cm (approx.)}$

Therefore, CSA of cone = $\pi rl = \left(\frac{22}{7} \times \frac{3.5}{2} \times 3.7 \right) \text{ cm}^2$

This gives the surface area of the top as

$$\begin{aligned} &= \left(2 \times \frac{22}{7} \times \frac{3.5}{2} \times \frac{3.5}{2} \right) \text{ cm}^2 + \left(\frac{22}{7} \times \frac{3.5}{2} \times 3.7 \right) \text{ cm}^2 \\ &= \frac{22}{7} \times \frac{3.5}{2} (3.5 + 3.7) \text{ cm}^2 = \frac{11}{2} \times (3.5 + 3.7) \text{ cm}^2 = 39.6 \text{ cm}^2 \text{ (approx.)} \end{aligned}$$

You may note that ‘total surface area of the top’ is *not* the sum of the total surface areas of the cone and hemisphere.

Example 2 : The decorative block shown in Fig. 13.7 is made of two solids — a cube and a hemisphere. The base of the block is a cube with edge 5 cm, and the hemisphere fixed on the top has a diameter of 4.2 cm. Find the total surface area of the block.

(Take $\pi = \frac{22}{7}$)

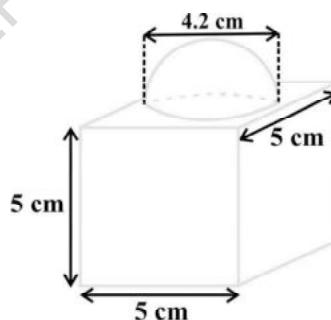


Fig. 15.7

Solution : The total surface area of the cube = $6 \times (\text{edge})^2 = 6 \times 5 \times 5 \text{ cm}^2 = 150 \text{ cm}^2$.

Note that the part of the cube where the hemisphere is attached is not included in the surface area.

So, the surface area of the block = TSA of cube – base area of hemisphere + CSA of hemisphere

$$= 150 - \pi r^2 + 2 \pi r^2 = (150 + \pi r^2) \text{ cm}^2$$

$$= 150 \text{ cm}^2 + \left(\frac{22}{7} \times \frac{4.2}{2} \times \frac{4.2}{2} \right) \text{ cm}^2$$

$$= (150 + 13.86) \text{ cm}^2 = 163.86 \text{ cm}^2$$

Example 3 : A wooden toy rocket is in the shape of a cone mounted on a cylinder, as shown in Fig. 15.8. The height of the entire rocket is 26 cm, while the height of the conical part is 6 cm. The base of the conical portion has a diameter of 5 cm, while the base diameter of the cylindrical portion is 3 cm. If the conical portion is to be painted orange and the cylindrical portion yellow, find the area of the rocket painted with each of these colours. (Take $\pi = 3.14$)

Solution : Denote radius of cone by r , slant height of cone by l , height of cone by h , radius of cylinder by r' and height of cylinder by h' . Then $r = 2.5$ cm, $h = 6$ cm, $r' = 1.5$ cm, $h' = 26 - 6 = 20$ cm and

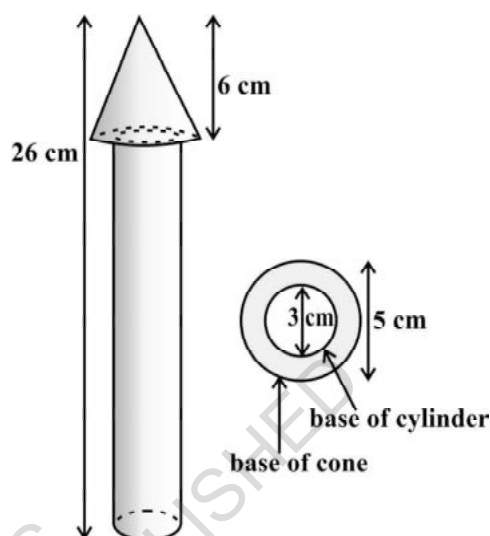


Fig. 15.8

$$l = \sqrt{r^2 + h^2} = \sqrt{2.5^2 + 6^2} \text{ cm} = 6.5 \text{ cm}$$

Here, the conical portion has its circular base resting on the base of the cylinder, but the base of the cone is larger than the base of the cylinder. So, a part of the base of the cone (a ring) is to be painted.

So, the area to be painted orange = CSA of the cone + base area of the cone
 – base area of the cylinder

$$\begin{aligned}
 &= \pi r l + \pi r^2 - \pi (r')^2 \\
 &= \pi [(2.5 \times 6.5) + (2.5)^2 - (1.5)^2] \text{ cm}^2 \\
 &= \pi [20.25] \text{ cm}^2 = 3.14 \times 20.25 \text{ cm}^2 \\
 &= 63.585 \text{ cm}^2
 \end{aligned}$$

Now, the area to be painted yellow = CSA of the cylinder
 + area of one base of the cylinder

$$\begin{aligned}
 &= 2\pi r' h' + \pi (r')^2 \\
 &= \pi r' (2h' + r') \\
 &= (3.14 \times 1.5) (2 \times 20 + 1.5) \text{ cm}^2 \\
 &= 4.71 \times 41.5 \text{ cm}^2 \\
 &= 195.465 \text{ cm}^2
 \end{aligned}$$

Example 4 : Mayank made a bird-bath for his garden in the shape of a cylinder with a hemispherical depression at one end (see Fig. 15.9). The height of the cylinder is 1.45 m and its radius is 30 cm. Find the total surface area of the bird-bath. (Take $\pi = \frac{22}{7}$)

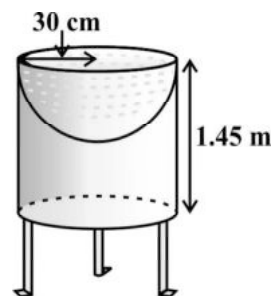


Fig. 15.9

Solution : Let h be height of the cylinder, and r the common radius of the cylinder and hemisphere. Then,

$$\begin{aligned}
 \text{the total surface area of the bird-bath} &= \text{CSA of cylinder} + \text{CSA of hemisphere} \\
 &= 2\pi rh + 2\pi r^2 = 2\pi r(h + r) \\
 &= 2 \times \frac{22}{7} \times 30(145 + 30) \text{ cm}^2 \\
 &= 33000 \text{ cm}^2 = 3.3 \text{ m}^2
 \end{aligned}$$

EXERCISE 15.1

Unless stated otherwise, take $\pi = \frac{22}{7}$.

- 2 cubes each of volume 64 cm^3 are joined end to end. Find the surface area of the resulting cuboid.
- A vessel is in the form of a hollow hemisphere mounted by a hollow cylinder. The diameter of the hemisphere is 14 cm and the total height of the vessel is 13 cm. Find the inner surface area of the vessel.
- A toy is in the form of a cone of radius 3.5 cm mounted on a hemisphere of same radius. The total height of the toy is 15.5 cm. Find the total surface area of the toy.
- A cubical block of side 7 cm is surmounted by a hemisphere. What is the greatest diameter the hemisphere can have? Find the surface area of the solid.
- A hemispherical depression is cut out from one face of a cubical wooden block such that the diameter l of the hemisphere is equal to the edge of the cube. Determine the surface area of the remaining solid.
- A medicine capsule is in the shape of a cylinder with two hemispheres stuck to each of its ends (see Fig. 13.10). The length of the entire capsule is 14 mm and the diameter of the capsule is 5 mm. Find its surface area.

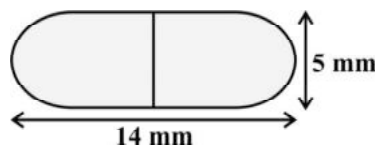


Fig. 13.10

7. A tent is in the shape of a cylinder surmounted by a conical top. If the height and diameter of the cylindrical part are 2.1 m and 4 m respectively, and the slant height of the top is 2.8 m, find the area of the canvas used for making the tent. Also, find the cost of the canvas of the tent at the rate of ₹500 per m^2 . (Note that the base of the tent will not be covered with canvas.)
8. From a solid cylinder whose height is 2.4 cm and diameter 1.4 cm, a conical cavity of the same height and same diameter is hollowed out. Find the total surface area of the remaining solid to the nearest cm^2 .
9. A wooden article was made by scooping out a hemisphere from each end of a solid cylinder, as shown in Fig. 15.11. If the height of the cylinder is 10 cm, and its base is of radius 3.5 cm, find the total surface area of the article.

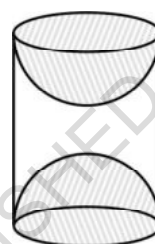


Fig. 15.11

15.3 Volume of a Combination of Solids

In the previous section, we have discussed how to find the surface area of solids made up of a combination of two basic solids. Here, we shall see how to calculate their volumes. It may be noted that in calculating the surface area, we have not added the surface areas of the two constituents, because some part of the surface area disappeared in the process of joining them. However, this will not be the case when we calculate the volume. The volume of the solid formed by joining two basic solids will actually be the sum of the volumes of the constituents, as we see in the examples below.

Example 5 : Shanta runs an industry in a shed which is in the shape of a cuboid surmounted by a half cylinder (see Fig. 15.12). If the base of the shed is of dimension $7\text{ m} \times 15\text{ m}$, and the height of the cuboidal portion is 8 m, find the volume of air that the shed can hold. Further, suppose the machinery in the shed occupies a total space of 300 m^3 , and there are 20 workers, each of whom occupy about 0.08 m^3 space on an average. Then, how much air is in the

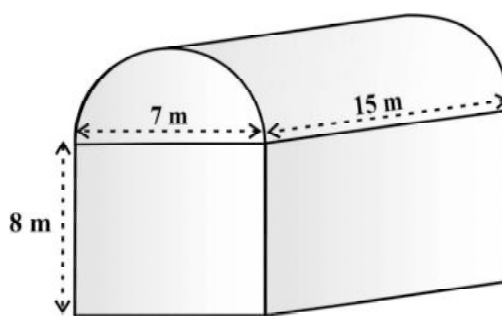


Fig. 15.12

shed? (Take $\pi = \frac{22}{7}$)

Solution : The volume of air inside the shed (when there are no people or machinery) is given by the volume of air inside the cuboid and inside the half cylinder, taken together.

Now, the length, breadth and height of the cuboid are 15 m, 7 m and 8 m, respectively. Also, the diameter of the half cylinder is 7 m and its height is 15 m.

So, the required volume = volume of the cuboid + $\frac{1}{2}$ volume of the cylinder

$$= \left[15 \times 7 \times 8 + \frac{1}{2} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 15 \right] \text{ m}^3 = 1128.75 \text{ m}^3$$

Next, the total space occupied by the machinery = 300 m^3

And the total space occupied by the workers = $20 \times 0.08 \text{ m}^3 = 1.6 \text{ m}^3$

Therefore, the volume of the air, when there are machinery and workers

$$= 1128.75 - (300.00 + 1.60) = 827.15 \text{ m}^3$$

Example 6 : A juice seller was serving his customers using glasses as shown in Fig. 15.13. The inner diameter of the cylindrical glass was 5 cm, but the bottom of the glass had a hemispherical raised portion which reduced the capacity of the glass. If the height of a glass was 10 cm, find the apparent capacity of the glass and its actual capacity. (Use $\pi = 3.14$.)



Fig. 15.13

Solution : Since the inner diameter of the glass = 5 cm and height = 10 cm,

the apparent capacity of the glass = $\pi r^2 h$

$$= 3.14 \times 2.5 \times 2.5 \times 10 \text{ cm}^3 = 196.25 \text{ cm}^3$$

But the actual capacity of the glass is less by the volume of the hemisphere at the base of the glass.

$$\text{i.e., it is less by } \frac{2}{3} \pi r^3 = \frac{2}{3} \times 3.14 \times 2.5 \times 2.5 \times 2.5 \text{ cm}^3 = 32.71 \text{ cm}^3$$

So, the actual capacity of the glass = apparent capacity of glass – volume of the hemisphere

$$= (196.25 - 32.71) \text{ cm}^3$$

$$= 163.54 \text{ cm}^3$$

Example 7 : A solid toy is in the form of a hemisphere surmounted by a right circular cone. The height of the cone is 2 cm and the diameter of the base is 4 cm. Determine the volume of the toy. If a right circular cylinder circumscribes the toy, find the difference of the volumes of the cylinder and the toy. (Take $\pi = 3.14$)

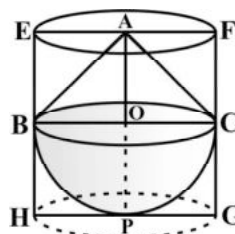


Fig. 15.14

Solution : Let BPC be the hemisphere and ABC be the cone standing on the base of the hemisphere (see Fig. 15.14). The radius BO of the hemisphere (as well as of the cone) = $\frac{1}{2} \times 4 \text{ cm} = 2 \text{ cm}$.

$$\begin{aligned} \text{So, volume of the toy} &= \frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 h \\ &= \left[\frac{2}{3} \times 3.14 \times (2)^3 + \frac{1}{3} \times 3.14 \times (2)^2 \times 2 \right] \text{ cm}^3 = 25.12 \text{ cm}^3 \end{aligned}$$

Now, let the right circular cylinder EFGH circumscribe the given solid. The radius of the base of the right circular cylinder = HP = BO = 2 cm, and its height is

$$EH = AO + OP = (2 + 2) \text{ cm} = 4 \text{ cm}$$

$$\begin{aligned} \text{So, the volume required} &= \text{volume of the right circular cylinder} - \text{volume of the toy} \\ &= (3.14 \times 2^2 \times 4 - 25.12) \text{ cm}^3 \\ &= 25.12 \text{ cm}^3 \end{aligned}$$

Hence, the required difference of the two volumes = 25.12 cm³.

EXERCISE 15.2

Unless stated otherwise, take $\pi = \frac{22}{7}$.

1. A solid is in the shape of a cone standing on a hemisphere with both their radii being equal to 1 cm and the height of the cone is equal to its radius. Find the volume of the solid in terms of π .
2. Rachel, an engineering student, was asked to make a model shaped like a cylinder with two cones attached at its two ends by using a thin aluminium sheet. The diameter of the model is 3 cm and its length is 12 cm. If each cone has a height of 2 cm, find the volume of air contained in the model that Rachel made. (Assume the outer and inner dimensions of the model to be nearly the same.)

3. A *gulab jamun*, contains sugar syrup up to about 30% of its volume. Find approximately how much syrup would be found in 45 *gulab jamuns*, each shaped like a cylinder with two hemispherical ends with length 5 cm and diameter 2.8 cm (see Fig. 15.15).

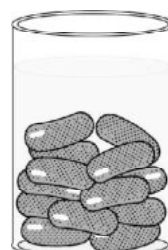


Fig. 15.15

4. A pen stand made of wood is in the shape of a cuboid with four conical depressions to hold pens. The dimensions of the cuboid are 15 cm by 10 cm by 3.5 cm. The radius of each of the depressions is 0.5 cm and the depth is 1.4 cm. Find the volume of wood in the entire stand (see Fig. 15.16).

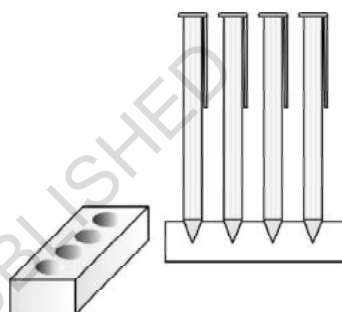


Fig. 15.16

5. A vessel is in the form of an inverted cone. Its height is 8 cm and the radius of its top, which is open, is 5 cm. It is filled with water up to the brim. When lead shots, each of which is a sphere of radius 0.5 cm are dropped into the vessel, one-fourth of the water flows out. Find the number of lead shots dropped in the vessel.
6. A solid iron pole consists of a cylinder of height 220 cm and base diameter 24 cm, which is surmounted by another cylinder of height 60 cm and radius 8 cm. Find the mass of the pole, given that 1 cm^3 of iron has approximately 8g mass. (Use $\pi = 3.14$)
7. A solid consisting of a right circular cone of height 120 cm and radius 60 cm standing on a hemisphere of radius 60 cm is placed upright in a right circular cylinder full of water such that it touches the bottom. Find the volume of water left in the cylinder, if the radius of the cylinder is 60 cm and its height is 180 cm.
8. A spherical glass vessel has a cylindrical neck 8 cm long, 2 cm in diameter; the diameter of the spherical part is 8.5 cm. By measuring the amount of water it holds, a child finds its volume to be 345 cm^3 . Check whether she is correct, taking the above as the inside measurements, and $\pi = 3.14$.

13.4 Conversion of Solid from One Shape to Another

We are sure you would have seen candles. Generally, they are in the shape of a cylinder. You may have also seen some candles shaped like an animal (see Fig. 15.17).

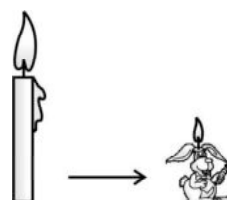


Fig. 15.17

How are they made? If you want a candle of any special shape, you will have to heat the wax in a metal container till it becomes completely liquid. Then you will have to pour it into another container which has the special shape that you want. For example, take a candle in the shape of a solid cylinder, melt it and pour whole of the molten wax into another container shaped like a rabbit. On cooling, you will obtain a candle in the shape of the rabbit. The volume of the new candle will be the same as the volume of the earlier candle. This is what we have to remember when we come across objects which are converted from one shape to another, or when a liquid which originally filled one container of a particular shape is poured into another container of a different shape or size, as you see in Fig 15.18.

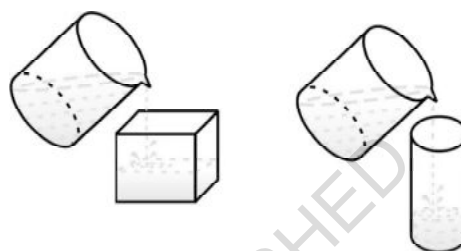


Fig. 15.18

To understand what has been discussed, let us consider some examples.

Example 8: A cone of height 24 cm and radius of base 6 cm is made up of modelling clay. A child reshapes it in the form of a sphere. Find the radius of the sphere.

Solution : Volume of cone = $\frac{1}{3} \times \pi \times 6 \times 6 \times 24 \text{ cm}^3$

If r is the radius of the sphere, then its volume is $\frac{4}{3} \pi r^3$.

Since, the volume of clay in the form of the cone and the sphere remains the same, we have

$$\frac{4}{3} \times \pi \times r^3 = \frac{1}{3} \times \pi \times 6 \times 6 \times 24$$

i.e.,

$$r^3 = 3 \times 3 \times 24 = 3^3 \times 2^3$$

So,

$$r = 3 \times 2 = 6 \text{ cm}$$

Therefore, the radius of the sphere is 6 cm.

Example 9 : Selvi's house has an overhead tank in the shape of a cylinder. This is filled by pumping water from a sump (an underground tank) which is in the shape of a cuboid. The sump has dimensions 1.57 m $\times$ 1.44 m $\times$ 95cm. The overhead tank has its radius 60 cm and height 95 cm. Find the height of the water left in the sump after the overhead tank has been completely filled with water from the sump which had been full. Compare the capacity of the tank with that of the sump. (Use $\pi = 3.14$)

Solution : The volume of water in the overhead tank equals the volume of the water removed from the sump.

Now, the volume of water in the overhead tank (cylinder) $= \pi r^2 h$

$$= 3.14 \times 0.6 \times 0.6 \times 0.95 \text{ m}^3$$

The volume of water in the sump when full $= l \times b \times h = 1.57 \times 1.44 \times 0.95 \text{ m}^3$

The volume of water left in the sump after filling the tank

$$= [(1.57 \times 1.44 \times 0.95) - (3.14 \times 0.6 \times 0.6 \times 0.95)] \text{ m}^3 = (1.57 \times 0.6 \times 0.6 \times 0.95 \times 2) \text{ m}^3$$

$$\begin{aligned} \text{So, the height of the water left in the sump} &= \frac{\text{volume of water left in the sump}}{l \times b} \\ &= \frac{1.57 \times 0.6 \times 0.6 \times 0.95 \times 2}{1.57 \times 1.44} \text{ m} \\ &= 0.475 \text{ m} = 47.5 \text{ cm} \end{aligned}$$

$$\text{Also, } \frac{\text{Capacity of tank}}{\text{Capacity of sump}} = \frac{3.14 \times 0.6 \times 0.6 \times 0.95}{1.57 \times 1.44 \times 0.95} = \frac{1}{2}$$

Therefore, the capacity of the tank is half the capacity of the sump.

Example 10 : A copper rod of diameter 1 cm and length 8 cm is drawn into a wire of length 18 m of uniform thickness. Find the thickness of the wire.

Solution : The volume of the rod $= \pi \times \left(\frac{1}{2}\right)^2 \times 8 \text{ cm}^3 = 2\pi \text{ cm}^3$.

The length of the new wire of the same volume $= 18 \text{ m} = 1800 \text{ cm}$

If r is the radius (in cm) of cross-section of the wire, its volume $= \pi \times r^2 \times 1800 \text{ cm}^3$

Therefore, $\pi \times r^2 \times 1800 = 2\pi$

$$\text{i.e., } r^2 = \frac{1}{900}$$

$$\text{i.e., } r = \frac{1}{30} \text{ cm}$$

So, the diameter of the cross section, i.e., the thickness of the wire is $\frac{1}{15} \text{ cm}$, i.e., 0.67mm (approx.).

Example 11 : A hemispherical tank full of water is emptied by a pipe at the rate of $3\frac{4}{7}$ litres per second. How much time will it take to empty half the tank, if it is 3m in diameter? (Take $\pi = \frac{22}{7}$)

Solution : Radius of the hemispherical tank = $\frac{3}{2}$ m

$$\text{Volume of the tank} = \frac{2}{3} \times \frac{22}{7} \times \left(\frac{3}{2}\right)^3 \text{ m}^3 = \frac{99}{14} \text{ m}^3$$

$$\begin{aligned} \text{So, the volume of the water to be emptied} &= \frac{1}{2} \times \frac{99}{14} \text{ m}^3 = \frac{99}{28} \times 1000 \text{ litres} \\ &= \frac{99000}{28} \text{ litres} \end{aligned}$$

Since, $\frac{25}{7}$ litres of water is emptied in 1 second, $\frac{99000}{28}$ litres of water will be emptied in $\frac{99000}{28} \times \frac{7}{25}$ seconds, i.e., in 16.5 minutes.

EXERCISE 15.3

Take $\pi = \frac{22}{7}$, unless stated otherwise.

1. A metallic sphere of radius 4.2 cm is melted and recast into the shape of a cylinder of radius 6 cm. Find the height of the cylinder.
2. Metallic spheres of radii 6 cm, 8 cm and 10 cm, respectively, are melted to form a single solid sphere. Find the radius of the resulting sphere.
3. A 20 m deep well with diameter 7 m is dug and the earth from digging is evenly spread out to form a platform 22 m by 14 m. Find the height of the platform.
4. A well of diameter 3 m is dug 14 m deep. The earth taken out of it has been spread evenly all around it in the shape of a circular ring of width 4 m to form an embankment. Find the height of the embankment.
5. A container shaped like a right circular cylinder having diameter 12 cm and height 15 cm is full of ice cream. The ice cream is to be filled into cones of height 12 cm and diameter 6 cm, having a hemispherical shape on the top. Find the number of such cones which can be filled with ice cream.
6. How many silver coins, 1.75 cm in diameter and of thickness 2 mm, must be melted to form a cuboid of dimensions 5.5 cm $\times$ 10 cm $\times$ 3.5 cm?

7. A cylindrical bucket, 32 cm high and with radius of base 18 cm, is filled with sand. This bucket is emptied on the ground and a conical heap of sand is formed. If the height of the conical heap is 24 cm, find the radius and slant height of the heap.
8. Water in a canal, 6 m wide and 1.5 m deep, is flowing with a speed of 10 km/h. How much area will it irrigate in 30 minutes, if 8 cm of standing water is needed?
9. A farmer connects a pipe of internal diameter 20 cm from a canal into a cylindrical tank in her field, which is 10 m in diameter and 2 m deep. If water flows through the pipe at the rate of 3 km/h, in how much time will the tank be filled?

15.5 Frustum of a Cone

In Section 15.2, we observed objects that are formed when two basic solids were joined together. Let us now do something different. We will take a right circular cone and *remove* a portion of it. There are so many ways in which we can do this. But one particular case that we are interested in is the removal of a smaller right circular cone by cutting the given cone by a plane parallel to its base. You must have observed that the glasses (tumblers), in general, used for drinking water, are of this shape. (See Fig. 15.19)



Fig. 15.19

Activity 1 : Take some clay, or any other such material (like plasticine, etc.) and form a cone. Cut it with a knife parallel to its base. Remove the smaller cone. What are you left with? You are left with a solid called a *frustum* of the cone. You can see that this has two circular ends with different radii.

So, given a cone, when we slice (or cut) through it with a plane parallel to its base (see Fig. 15.20) and remove the cone that is formed on one side of that plane, the part that is now left over on the other side of the plane is called a **frustum* of the cone**.

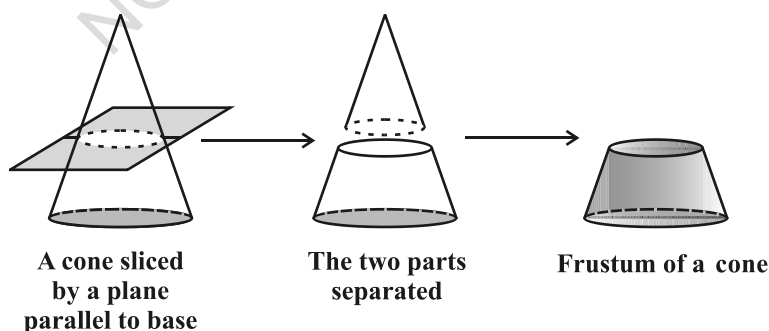


Fig. 15.20

*'Frustum' is a latin word meaning 'piece cut off', and its plural is 'frusta'.

How can we find the surface area and volume of a frustum of a cone? Let us explain it through an example.

Example 12 : The radii of the ends of a frustum of a cone 45 cm high are 28 cm and 7 cm (see Fig. 15.21). Find its volume, the curved surface area and the total surface area

(Take $\pi = \frac{22}{7}$).

Solution : The frustum can be viewed as a difference of two right circular cones OAB and OCD (see Fig. 13.21). Let the height (in cm) of the cone OAB be h_1 and its slant height l_1 , i.e., $OP = h_1$ and $OA = OB = l_1$. Let h_2 be the height of cone OCD and l_2 its slant height.

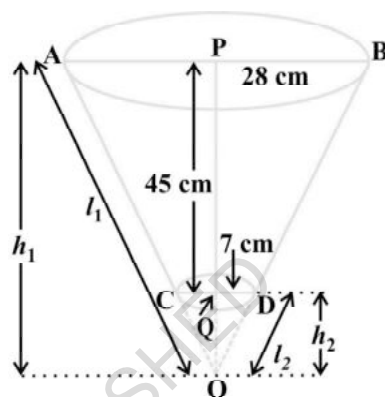


Fig. 15.21

We have : $r_1 = 28$ cm, $r_2 = 7$ cm

and the height of frustum (h) = 45 cm. Also,

$$h_1 = 45 + h_2 \quad (1)$$

We first need to determine the respective heights h_1 and h_2 of the cone OAB and OCD.

Since the triangles OPB and OQD are similar (Why?), we have

$$\frac{h_1}{h_2} = \frac{28}{7} = \frac{4}{1} \quad (2)$$

From (1) and (2), we get $h_2 = 15$ cm and $h_1 = 60$ cm.

Now, the volume of the frustum

= volume of the cone OAB – volume of the cone OCD

$$= \left[\frac{1}{3} \cdot \frac{22}{7} \cdot (28)^2 \cdot (60) - \frac{1}{3} \cdot \frac{22}{7} \cdot (7)^2 \cdot (15) \right] \text{cm}^3 = 48510 \text{ cm}^3$$

The respective slant height l_2 and l_1 of the cones OCD and OAB are given by

$$l_2 = \sqrt{(7)^2 + (15)^2} = 16.55 \text{ cm (approx.)}$$

$$l_1 = \sqrt{(28)^2 + (60)^2} = 4\sqrt{(7)^2 + (15)^2} = 4 \times 16.55 = 66.20 \text{ cm}$$

Thus, the curved surface area of the frustum $= \pi r_1 l_1 - \pi r_2 l_2$

$$= \frac{22}{7} (28)(66.20) - \frac{22}{7} (7)(16.55) = 5461.5 \text{ cm}^2$$

Now, the total surface area of the frustum

$$\begin{aligned} &= \text{the curved surface area} + \pi r_1^2 + \pi r_2^2 \\ &= 5461.5 \text{ cm}^2 + \frac{22}{7} (28)^2 \text{ cm}^2 + \frac{22}{7} (7)^2 \text{ cm}^2 \\ &= 5461.5 \text{ cm}^2 + 2464 \text{ cm}^2 + 154 \text{ cm}^2 = 8079.5 \text{ cm}^2. \end{aligned}$$

Let h be the height, l the slant height and r_1 and r_2 the radii of the ends ($r_1 > r_2$) of the frustum of a cone. Then we can directly find the volume, the curved surface area and the total surface area of frustum by using the formulae given below :

(i) *Volume of the frustum of the cone* $= \frac{1}{3} \pi h (r_1^2 + r_2^2 + r_1 r_2)$.

(ii) *the curved surface area of the frustum of the cone* $= \pi (r_1 + r_2) l$

where $l = \sqrt{h^2 + (r_1 - r_2)^2}$.

(iii) *Total surface area of the frustum of the cone* $= \pi l (r_1 + r_2) + \pi r_1^2 + \pi r_2^2$,

where $l = \sqrt{h^2 + (r_1 - r_2)^2}$.

These formulae can be derived using the idea of similarity of triangles but we shall not be doing derivations here.

Let us solve Example 12, using these formulae :

(i) Volume of the frustum $= \frac{1}{3} \pi h (r_1^2 + r_2^2 + r_1 r_2)$

$$\begin{aligned} &= \frac{1}{3} \cdot \frac{22}{7} \cdot 45 \cdot [(28)^2 + (7)^2 + (28)(7)] \text{ cm}^3 \\ &= 48510 \text{ cm}^3 \end{aligned}$$

(ii) We have $l = \sqrt{h^2 + (r_1 - r_2)^2} = \sqrt{(45)^2 + (28 - 7)^2} \text{ cm}$

$$= 3\sqrt{(15)^2 + (7)^2} = 49.65 \text{ cm}$$

So, the curved surface area of the frustum

$$= \pi(r_1 + r_2) l = \frac{22}{7} (28 + 7) (49.65) = 5461.5 \text{ cm}^2$$

(iii) Total curved surface area of the frustum

$$\begin{aligned} &= \pi(r_1 + r_2)l + \pi r_1^2 + \pi r_2^2 \\ &= \left[5461.5 + \frac{22}{7}(28)^2 + \frac{22}{7}(7)^2 \right] \text{ cm}^2 = 8079.5 \text{ cm}^2 \end{aligned}$$

Let us apply these formulae in some examples.

Example 13 : Hanumappa and his wife Gangamma are busy making jaggery out of sugarcane juice. They have processed the sugarcane juice to make the molasses, which is poured into moulds in the shape of a frustum of a cone having the diameters of its two circular faces as 30 cm and 35 cm and the vertical height of the mould is 14 cm (see Fig. 15.22). If each cm^3 of molasses has mass about 1.2 g, find the mass of the molasses that can



Fig. 15.22

be poured into each mould. (Take $\pi = \frac{22}{7}$)

Solution : Since the mould is in the shape of a frustum of a cone, the quantity (volume) of molasses that can be poured into it = $\frac{\pi}{3} h(r_1^2 + r_2^2 + r_1 r_2)$,

where r_1 is the radius of the larger base and r_2 is the radius of the smaller base.

$$= \frac{1}{3} \times \frac{22}{7} \times 14 \left[\left(\frac{35}{2} \right)^2 + \left(\frac{30}{2} \right)^2 + \left(\frac{35}{2} \times \frac{30}{2} \right) \right] \text{ cm}^3 = 11641.7 \text{ cm}^3.$$

It is given that 1 cm^3 of molasses has mass 1.2g. So, the mass of the molasses that can be poured into each mould = (11641.7×1.2) g

$$= 13970.04 \text{ g} = 13.97 \text{ kg} = 14 \text{ kg (approx.)}$$

Example 14 : An open metal bucket is in the shape of a frustum of a cone, mounted on a hollow cylindrical base made of the same metallic sheet (see Fig. 15.23). The diameters of the two circular ends of the bucket are 45 cm and 25 cm, the total vertical height of the bucket is 40 cm and that of the cylindrical base is 6 cm. Find the area of the metallic sheet used to make the bucket, where we do not take into account the handle of the bucket. Also, find the volume of water the bucket can hold.



Fig. 15.23

(Take $\pi = \frac{22}{7}$).

Solution : The total height of the bucket = 40 cm, which includes the height of the base. So, the height of the frustum of the cone = $h = (40 - 6)$ cm = 34 cm.

Therefore, the slant height of the frustum, $l = \sqrt{h^2 + (r_1 - r_2)^2}$,

where $r_1 = 22.5$ cm, $r_2 = 12.5$ cm and $h = 34$ cm.

$$\begin{aligned} \text{So,} \quad l &= \sqrt{34^2 + (22.5 - 12.5)^2} \text{ cm} \\ &= \sqrt{34^2 + 10^2} = 35.44 \text{ cm} \end{aligned}$$

The area of metallic sheet used = curved surface area of frustum of cone

+ area of circular base

+ curved surface area of cylinder

$$\begin{aligned} &= [\pi \times 35.44 (22.5 + 12.5) + \pi \times (12.5)^2 \\ &\quad + 2\pi \times 12.5 \times 6] \text{ cm}^2 \end{aligned}$$

$$= \frac{22}{7} (1240.4 + 156.25 + 150) \text{ cm}^2$$

$$= 4860.9 \text{ cm}^2$$

Now, the volume of water that the bucket can hold (also, known as the capacity of the bucket)

$$\begin{aligned}
 &= \frac{\pi \times h}{3} \times (r_1^2 + r_2^2 + r_1 r_2) \\
 &= \frac{22}{7} \times \frac{34}{3} \times [(22.5)^2 + (12.5)^2 + 22.5 \times 12.5] \text{ cm}^3 \\
 &= \frac{22}{7} \times \frac{34}{3} \times 943.75 = 33615.48 \text{ cm}^3 \\
 &= 33.62 \text{ litres (approx.)}
 \end{aligned}$$

EXERCISE 15.4

Use $\pi = \frac{22}{7}$ unless stated otherwise.

1. A drinking glass is in the shape of a frustum of a cone of height 14 cm. The diameters of its two circular ends are 4 cm and 2 cm. Find the capacity of the glass.
2. The slant height of a frustum of a cone is 4 cm and the perimeters (circumference) of its circular ends are 18 cm and 6 cm. Find the curved surface area of the frustum.
3. A *fez*, the cap used by the Turks, is shaped like the frustum of a cone (see Fig. 15.24). If its radius on the open side is 10 cm, radius at the upper base is 4 cm and its slant height is 15 cm, find the area of material used for making it.
4. A container, opened from the top and made up of a metal sheet, is in the form of a frustum of a cone of height 16 cm with radii of its lower and upper ends as 8 cm and 20 cm, respectively. Find the cost of the milk which can completely fill the container, at the rate of ₹ 20 per litre. Also find the cost of metal sheet used to make the container, if it costs ₹ 8 per 100 cm². (Take $\pi = 3.14$)
5. A metallic right circular cone 20 cm high and whose vertical angle is 60° is cut into two parts at the middle of its height by a plane parallel to its base. If the frustum so obtained be drawn into a wire of diameter $\frac{1}{16}$ cm, find the length of the wire.



Fig. 15.24

EXERCISE 15.5 (Optional)*

1. A copper wire, 3 mm in diameter, is wound about a cylinder whose length is 12 cm, and diameter 10 cm, so as to cover the curved surface of the cylinder. Find the length and mass of the wire, assuming the density of copper to be 8.88 g per cm^3 .
2. A right triangle, whose sides are 3 cm and 4 cm (other than hypotenuse) is made to revolve about its hypotenuse. Find the volume and surface area of the double cone so formed. (Choose value of π as found appropriate.)
3. A cistern, internally measuring $150 \text{ cm} \times 120 \text{ cm} \times 110 \text{ cm}$, has 129600 cm^3 of water in it. Porous bricks are placed in the water until the cistern is full to the brim. Each brick absorbs one-seventeenth of its own volume of water. How many bricks can be put in without overflowing the water, each brick being $22.5 \text{ cm} \times 7.5 \text{ cm} \times 6.5 \text{ cm}$?
4. In one fortnight of a given month, there was a rainfall of 10 cm in a river valley. If the area of the valley is 7280 km^2 , show that the total rainfall was approximately equivalent to the addition to the normal water of three rivers each 1072 km long, 75 m wide and 3 m deep.
5. An oil funnel made of tin sheet consists of a 10 cm long cylindrical portion attached to a frustum of a cone. If the total height is 22 cm, diameter of the cylindrical portion is 8 cm and the diameter of the top of the funnel is 18 cm, find the area of the tin sheet required to make the funnel (see Fig. 15.25).
6. Derive the formula for the curved surface area and total surface area of the frustum of a cone, given to you in Section 13.5, using the symbols as explained.
7. Derive the formula for the volume of the frustum of a cone, given to you in Section 13.5, using the symbols as explained.

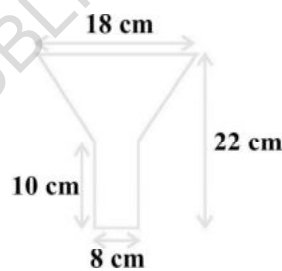


Fig. 15.25

13.6 Summary

In this chapter, you have studied the following points:

1. To determine the surface area of an object formed by combining any two of the basic solids, namely, cuboid, cone, cylinder, sphere and hemisphere.
2. To find the volume of objects formed by combining any two of a cuboid, cone, cylinder, sphere and hemisphere.

* These exercises are not for the examination point of view.

3. Given a right circular cone, which is sliced through by a plane parallel to its base, when the smaller conical portion is removed, the resulting solid is called a *Frustum of a Right Circular Cone*.
4. The formulae involving the frustum of a cone are:

(i) Volume of a frustum of a cone $= \frac{1}{3} \pi h (r_1^2 + r_2^2 + r_1 r_2)$.

(ii) Curved surface area of a frustum of a cone $= \pi l (r_1 + r_2)$ where $l = \sqrt{h^2 + (r_1 - r_2)^2}$.

(iii) Total surface area of frustum of a cone $= \pi l (r_1 + r_2) + \pi (r_1^2 + r_2^2)$ where
 h = vertical height of the frustum, l = slant height of the frustum
 r_1 and r_2 are radii of the two bases (ends) of the frustum.

PROOFS IN MATHEMATICS

A1

A1.1 Introduction

The ability to reason and think clearly is extremely useful in our daily life. For example, suppose a politician tells you, ‘If you are interested in a clean government, then you should vote for me.’ What he actually wants you to believe is that if you do not vote for him, then you may not get a clean government. Similarly, if an advertisement tells you, ‘The intelligent wear XYZ shoes’, what the company wants you to conclude is that if you do not wear XYZ shoes, then you are not intelligent enough. You can yourself observe that both the above statements may mislead the general public. So, if we understand the process of reasoning correctly, we do not fall into such traps unknowingly.

The correct use of reasoning is at the core of mathematics, especially in constructing proofs. In Class IX, you were introduced to the idea of proofs, and you actually proved many statements, especially in geometry. Recall that a proof is made up of several mathematical statements, each of which is logically deduced from a previous statement in the proof, or from a theorem proved earlier, or an axiom, or the hypotheses. The main tool, we use in constructing a proof, is the process of deductive reasoning.

We start the study of this chapter with a review of what a mathematical statement is. Then, we proceed to sharpen our skills in deductive reasoning using several examples. We shall also deal with the concept of negation and finding the negation of a given statement. Then, we discuss what it means to find the converse of a given statement. Finally, we review the ingredients of a proof learnt in Class IX by analysing the proofs of several theorems. Here, we also discuss the idea of proof by contradiction, which you have come across in Class IX and many other chapters of this book.

A1.2 Mathematical Statements Revisited

Recall, that a ‘statement’ is a meaningful sentence which is not an order, or an exclamation or a question. For example, ‘Which two teams are playing in the

Cricket World Cup Final?’ is a question, not a statement. ‘Go and finish your homework’ is an order, not a statement. ‘What a fantastic goal!’ is an exclamation, not a statement.

Remember, in general, statements can be one of the following:

- *always true*
- *always false*
- *ambiguous*

In Class IX, you have also studied that in mathematics, **a statement is acceptable only if it is either always true or always false**. So, ambiguous sentences are not considered as mathematical statements.

Let us review our understanding with a few examples.

Example 1 : State whether the following statements are always true, always false or ambiguous. Justify your answers.

- (i) The Sun orbits the Earth.
- (ii) Vehicles have four wheels.
- (iii) The speed of light is approximately 3×10^8 km/s.
- (iv) A road to Kolkata will be closed from November to March.
- (v) All humans are mortal.

Solution :

- (i) This statement is always false, since astronomers have established that the Earth orbits the Sun.
- (ii) This statement is ambiguous, because we cannot decide if it is always true or always false. This depends on what the vehicle is — vehicles can have 2, 3, 4, 6, 10, etc., wheels.
- (iii) This statement is always true, as verified by physicists.
- (iv) This statement is ambiguous, because it is not clear which road is being referred to.
- (v) This statement is always true, since every human being has to die some time.

Example 2 : State whether the following statements are true or false, and justify your answers.

- (i) All equilateral triangles are isosceles.
- (ii) Some isosceles triangles are equilateral.
- (iii) All isosceles triangles are equilateral.
- (iv) Some rational numbers are integers.

- (v) Some rational numbers are not integers.
- (vi) Not all integers are rational.
- (vii) Between any two rational numbers there is no rational number.

Solution :

- (i) This statement is true, because equilateral triangles have equal sides, and therefore are isosceles.
- (ii) This statement is true, because those isosceles triangles whose base angles are 60° are equilateral.
- (iii) This statement is false. Give a counter-example for it.
- (iv) This statement is true, since rational numbers of the form $\frac{p}{q}$, where p is an integer and $q = 1$, are integers (for example, $3 = \frac{3}{1}$).
- (v) This statement is true, because rational numbers of the form $\frac{p}{q}$, p, q are integers and q does not divide p , are not integers (for example, $\frac{3}{2}$).
- (vi) This statement is the same as saying ‘there is an integer which is not a rational number’. This is false, because all integers are rational numbers.
- (vii) This statement is false. As you know, between any two rational numbers r and s lies $\frac{r+s}{2}$, which is a rational number.

Example 3 : If $x < 4$, which of the following statements are true? Justify your answers.

- (i) $2x > 8$
- (ii) $2x < 6$
- (iii) $2x < 8$

Solution :

- (i) This statement is false, because, for example, $x = 3 < 4$ does not satisfy $2x > 8$.
- (ii) This statement is false, because, for example, $x = 3.5 < 4$ does not satisfy $2x < 6$.
- (iii) This statement is true, because it is the same as $x < 4$.

Example 4 : Restate the following statements with appropriate conditions, so that they become true statements:

- (i) If the diagonals of a quadrilateral are equal, then it is a rectangle.
- (ii) A line joining two points on two sides of a triangle is parallel to the third side.
- (iii) $\sqrt{p}$ is irrational for all positive integers p .
- (iv) All quadratic equations have two real roots.

Solution :

- (i) If the diagonals of a parallelogram are equal, then it is a rectangle.
- (ii) A line joining the mid-points of two sides of a triangle is parallel to the third side.
- (iii) $\sqrt{p}$ is irrational for all primes p .
- (iv) All quadratic equations have at most two real roots.

Remark : There can be other ways of restating the statements above. For instance, (iii) can also be restated as ‘ $\sqrt{p}$ is irrational for all positive integers p which are not a perfect square’.

EXERCISE A1.1

1. State whether the following statements are always true, always false or ambiguous. Justify your answers.
 - (i) All mathematics textbooks are interesting.
 - (ii) The distance from the Earth to the Sun is approximately 1.5×10^8 km.
 - (iii) All human beings grow old.
 - (iv) The journey from Uttarkashi to Harsil is tiring.
 - (v) The woman saw an elephant through a pair of binoculars.
2. State whether the following statements are true or false. Justify your answers.
 - (i) All hexagons are polygons.
 - (ii) Some polygons are pentagons.
 - (iii) Not all even numbers are divisible by 2.
 - (iv) Some real numbers are irrational.
 - (v) Not all real numbers are rational.
3. Let a and b be real numbers such that $ab \neq 0$. Then which of the following statements are true? Justify your answers.
 - (i) Both a and b must be zero.
 - (ii) Both a and b must be non-zero.
 - (iii) Either a or b must be non-zero.
4. Restate the following statements with appropriate conditions, so that they become true.
 - (i) If $a^2 > b^2$, then $a > b$.
 - (ii) If $x^2 = y^2$, then $x = y$.
 - (iii) If $(x + y)^2 = x^2 + y^2$, then $x = 0$.
 - (iv) The diagonals of a quadrilateral bisect each other.

A1.3 Deductive Reasoning

In Class IX, you were introduced to the idea of deductive reasoning. Here, we will work with many more examples which will illustrate how **deductive reasoning** is

used to **deduce** conclusions from given statements that we assume to be true. The given statements are called ‘premises’ or ‘hypotheses’. We begin with some examples.

Example 5 : Given that Bijapur is in the state of Karnataka, and suppose Shabana lives in Bijapur. In which state does Shabana live?

Solution : Here we have two premises:

- (i) Bijapur is in the state of Karnataka (ii) Shabana lives in Bijapur

From these premises, we deduce that Shabana lives in the state of Karnataka.

Example 6 : Given that all mathematics textbooks are interesting, and suppose you are reading a mathematics textbook. What can we conclude about the textbook you are reading?

Solution : Using the two premises (or hypotheses), we can deduce that you are reading an interesting textbook.

Example 7 : Given that $y = -6x + 5$, and suppose $x = 3$. What is y ?

Solution : Given the two hypotheses, we get $y = -6(3) + 5 = -13$.

Example 8 : Given that ABCD is a parallelogram, and suppose $AD = 5$ cm, $AB = 7$ cm (see Fig. A1.1). What can you conclude about the lengths of DC and BC?

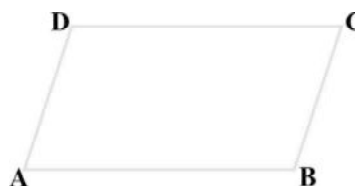


Fig. A1.1

Solution : We are given that ABCD is a parallelogram. So, we deduce that all the properties that hold for a parallelogram hold for ABCD. Therefore, in particular, the property that ‘the opposite sides of a parallelogram are equal to each other’, holds. Since we know $AD = 5$ cm, we can deduce that $BC = 5$ cm. Similarly, we deduce that $DC = 7$ cm.

Remark : In this example, we have seen how we will often need to find out and use properties hidden in a given premise.

Example 9 : Given that $\sqrt{p}$ is irrational for all primes p , and suppose that 19423 is a prime. What can you conclude about $\sqrt{19423}$?

Solution : We can conclude that $\sqrt{19423}$ is irrational.

In the examples above, you might have noticed that we do not know whether the hypotheses are true or not. We are **assuming** that they are true, and then applying deductive reasoning. For instance, in Example 9, we haven’t checked whether 19423

is a prime or not; we assume it to be a prime for the sake of our argument. What we are trying to emphasise in this section is that given a particular statement, how we use deductive reasoning to arrive at a conclusion. What really matters here is that we use the correct process of reasoning, and this process of reasoning does not depend on the trueness or falsity of the hypotheses. However, it must also be noted that if we start with an incorrect premise (or hypothesis), we may arrive at a wrong conclusion.

EXERCISE A1.2

1. Given that all women are mortal, and suppose that A is a woman, what can we conclude about A?
2. Given that the product of two rational numbers is rational, and suppose a and b are rationals, what can you conclude about ab ?
3. Given that the decimal expansion of irrational numbers is non-terminating, non-recurring, and $\sqrt{17}$ is irrational, what can we conclude about the decimal expansion of $\sqrt{17}$?
4. Given that $y = x^2 + 6$ and $x = -1$, what can we conclude about the value of y ?
5. Given that ABCD is a parallelogram and $\angle B = 80^\circ$. What can you conclude about the other angles of the parallelogram?
6. Given that PQRS is a cyclic quadrilateral and also its diagonals bisect each other. What can you conclude about the quadrilateral?
7. Given that $\sqrt{p}$ is irrational for all primes p and also suppose that 3721 is a prime. Can you conclude that $\sqrt{3721}$ is an irrational number? Is your conclusion correct? Why or why not?

A1.4 Conjectures, Theorems, Proofs and Mathematical Reasoning

Consider the Fig. A1.2. The first circle has one point on it, the second two points, the third three, and so on. All possible lines connecting the points are drawn in each case.

The lines divide the circle into mutually exclusive regions (having no common portion). We can count these and tabulate our results as shown :

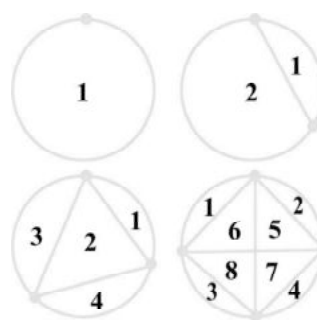


Fig. A1.2

Number of points	Number of regions
1	1
2	2
3	4
4	8
5	
6	
7	

Some of you might have come up with a formula predicting the number of regions given the number of points. From Class IX, you may remember that this intelligent guess is called a '*conjecture*'.

Suppose your conjecture is that given ' n ' points on a circle, there are 2^{n-1} mutually exclusive regions, created by joining the points with all possible lines. This seems an extremely sensible guess, and one can check that if $n = 5$, we do get 16 regions. So, having verified this formula for 5 points, are you satisfied that for any n points there are 2^{n-1} regions? If so, how would you respond, if someone asked you, how you can be sure about this for $n = 25$, say? To deal with such questions, you would need a proof which shows beyond doubt that this result is true, or a counter-example to show that this result fails for some ' n '. Actually, if you are patient and try it out for $n = 6$, you will find that there are 31 regions, and for $n = 7$ there are 57 regions. So, $n = 6$, is a counter-example to the conjecture above. This demonstrates the power of a counter-example. You may recall that in the Class IX we discussed that to **disprove a statement, it is enough to come up with a single counter-example**.

You may have noticed that we insisted on a proof regarding the number of regions in spite of verifying the result for $n = 1, 2, 3, 4$ and 5 . Let us consider a few more examples. You are familiar with the following result (given in Chapter 1):

$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$. To establish its validity, it is not enough to verify the

result for $n = 1, 2, 3$, and so on, because there may be some ' n ' for which this result is not true (just as in the example above, the result failed for $n = 6$). What we need is a proof which establishes its truth beyond doubt. You shall learn a proof for the same in higher classes.

Now, consider Fig. A1.3, where PQ and PR are tangents to the circle drawn from P.

You have proved that $PQ = PR$ (Theorem 10.2). You were not satisfied by only drawing several such figures, measuring the lengths of the respective tangents, and verifying for yourselves that the result was true in each case.

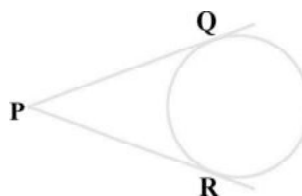


Fig. A1.3

Do you remember what did the proof consist of? It consisted of a sequence of statements (called **valid arguments**), each following from the earlier statements in the proof, or from previously proved (and known) results independent from the result to be proved, or from axioms, or from definitions, or from the assumptions you had made. And you concluded your proof with the statement $PQ = PR$, i.e., the statement you wanted to prove. This is the way any proof is constructed.

We shall now look at some examples and theorems and analyse their proofs to help us in getting a better understanding of how they are constructed.

We begin by using the so-called ‘direct’ or ‘deductive’ method of proof. In this method, we make several statements. Each is **based on previous statements**. If each statement is logically correct (i.e., a valid argument), it leads to a logically correct conclusion.

Example 10 : The sum of two rational numbers is a rational number.

Solution :

S.No.	Statements	Analysis/Comments
1.	Let x and y be rational numbers.	Since the result is about rationals, we start with x and y which are rational.
2.	Let $x = \frac{m}{n}$, $n \neq 0$ and $y = \frac{p}{q}$, $q \neq 0$ where m , n , p and q are integers.	Apply the definition of rationals.
3.	So, $x + y = \frac{m}{n} + \frac{p}{q} = \frac{mq + np}{nq}$	The result talks about the sum of rationals, so we look at $x + y$.

4.	Using the properties of integers, we see that $mq + np$ and nq are integers.	Using known properties of integers.
5.	Since $n \neq 0$ and $q \neq 0$, it follows that $nq \neq 0$.	Using known properties of integers.
6.	Therefore, $x + y = \frac{mq + np}{nq}$ is a rational number	Using the definition of a rational number.

Remark : Note that, each statement in the proof above is based on a previously established fact, or definition.

Example 11 : Every prime number greater than 3 is of the form $6k + 1$ or $6k + 5$, where k is some integer.

Solution :

S.No.	Statements	Analysis/Comments
1.	Let p be a prime number greater than 3.	Since the result has to do with a prime number greater than 3, we start with such a number.
2.	Dividing p by 6, we find that p can be of the form $6k$, $6k + 1$, $6k + 2$, $6k + 3$, $6k + 4$, or $6k + 5$, where k is an integer.	Using Euclid's division lemma.
3.	But $6k = 2(3k)$, $6k + 2 = 2(3k + 1)$, $6k + 4 = 2(3k + 2)$, and $6k + 3 = 3(2k + 1)$. So, they are not primes.	We now analyse the remainders given that p is prime.
4.	So, p is forced to be of the form $6k + 1$ or $6k + 5$, for some integer k .	We arrive at this conclusion having eliminated the other options.

Remark : In the above example, we have arrived at the conclusion by eliminating different options. This method is sometimes referred to as the **Proof by Exhaustion**.

Theorem A1.1 (Converse of the Pythagoras Theorem) : *If in a triangle the square of the length of one side is equal to the sum of the squares of the other two sides, then the angle opposite the first side is a right angle.*

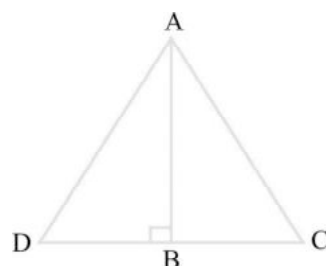


Fig. A1.4

Proof :

S.No.	Statements	Analysis
1.	Let $\triangle ABC$ satisfy the hypothesis $AC^2 = AB^2 + BC^2$.	Since we are proving a statement about such a triangle, we begin by taking this.
2.	Construct line BD perpendicular to AB, such that $BD = BC$, and join A to D.	This is the intuitive step we have talked about that we often need to take for proving theorems.
3.	By construction, $\triangle ABD$ is a right triangle, and from the Pythagoras Theorem, we have $AD^2 = AB^2 + BD^2$.	We use the Pythagoras theorem, which is already proved.
4.	By construction, $BD = BC$. Therefore, we have $AD^2 = AB^2 + BC^2$.	Logical deduction.
5.	Therefore, $AC^2 = AB^2 + BC^2 = AD^2$.	Using assumption, and previous statement.
6.	Since AC and AD are positive, we have $AC = AD$.	Using known property of numbers.
7.	We have just shown $AC = AD$. Also $BC = BD$ by construction, and AB is common. Therefore, by SSS, $\triangle ABC \cong \triangle ABD$.	Using known theorem.
8.	Since $\triangle ABC \cong \triangle ABD$, we get $\angle ABC = \angle ABD$, which is a right angle.	Logical deduction, based on previously established fact.

Remark : Each of the results above has been proved by a sequence of steps, all linked together. Their order is important. Each step in the proof follows from previous steps and earlier known results. (Also see Theorem 2.9.)

EXERCISE A1.3

In each of the following questions, we ask you to prove a statement. List all the steps in each proof, and give the reason for each step.

1. Prove that the sum of two consecutive odd numbers is divisible by 4.
2. Take two consecutive odd numbers. Find the sum of their squares, and then add 6 to the result. Prove that the new number is always divisible by 8.
3. If $p \geq 5$ is a prime number, show that $p^2 + 2$ is divisible by 3.
[Hint: Use Example 11].
4. Let x and y be rational numbers. Show that xy is a rational number.
5. If a and b are positive integers, then you know that $a = bq + r$, $0 \leq r < b$, where q is a whole number. Prove that $\text{HCF}(a, b) = \text{HCF}(b, r)$.
[Hint : Let $\text{HCF}(b, r) = h$. So, $b = k_1 h$ and $r = k_2 h$, where k_1 and k_2 are coprime.]
6. A line parallel to side BC of a triangle ABC, intersects AB and AC at D and E respectively.

Prove that $\frac{AD}{DB} = \frac{AE}{EC}$.

A1.5 Negation of a Statement

In this section, we discuss what it means to ‘negate’ a statement. Before we start, we would like to introduce some notation, which will make it easy for us to understand these concepts. To start with, let us look at a statement as a single unit, and give it a name. For example, we can denote the statement ‘It rained in Delhi on 1 September 2005’ by p . We can also write this by

p : It rained in Delhi on 1 September 2005.

Similarly, let us write

q : All teachers are female.

r : Mike’s dog has a black tail.

s : $2 + 2 = 4$.

t : Triangle ABC is equilateral.

This notation now helps us to discuss properties of statements, and also to see how we can combine them. In the beginning we will be working with what we call ‘simple’ statements, and will then move onto ‘compound’ statements.

Now consider the following table in which we make a new statement from each of the given statements.

Original statement	New statement
p : It rained in Delhi on 1 September 2005	$\sim p$: It is false that it rained in Delhi on 1 September 2005.
q : All teachers are female.	$\sim q$: It is false that all teachers are female.
r : Mike's dog has a black tail.	$\sim r$: It is false that Mike's dog has a black tail.
s : $2 + 2 = 4$.	$\sim s$: It is false that $2 + 2 = 4$.
t : Triangle ABC is equilateral.	$\sim t$: It is false that triangle ABC is equilateral.

Each new statement in the table is a *negation* of the corresponding old statement. That is, $\sim p$, $\sim q$, $\sim r$, $\sim s$ and $\sim t$ are negations of the statements p , q , r , s and t , respectively. Here, $\sim p$ is read as 'not p '. The statement $\sim p$ negates the assertion that the statement p makes. Notice that in our usual talk we would simply mean $\sim p$ as 'It did not rain in Delhi on 1 September 2005.' However, we need to be careful while doing so. You might think that one can obtain the negation of a statement by simply inserting the word 'not' in the given statement at a suitable place. While this works in the case of p , the difficulty comes when we have a statement that begins with 'all'. Consider, for example, the statement q : All teachers are female. We said the negation of this statement is $\sim q$: It is false that all teachers are female. This is the same as the statement 'There are some teachers who are males.' Now let us see what happens if we simply insert 'not' in q . We obtain the statement: 'All teachers are not female', or we can obtain the statement: 'Not all teachers are female.' The first statement can confuse people. It could imply (if we lay emphasis on the word 'All') that all teachers are male! This is certainly not the negation of q . However, the second statement gives the meaning of $\sim q$, i.e., that there is at least one teacher who is not a female. So, be careful when writing the negation of a statement!

So, how do we decide that we have the correct negation? We use the following criterion.

Let p be a statement and $\sim p$ its negation. Then $\sim p$ is false whenever p is true, and $\sim p$ is true whenever p is false.

For example, if it is true that Mike's dog has a black tail, then it is false that Mike's dog does not have a black tail. If it is false that 'Mike's dog has a black tail', then it is true that 'Mike's dog does not have a black tail'.

Similarly, the negations for the statements s and t are:

s : $2 + 2 = 4$; negation, $\sim s$: $2 + 2 \neq 4$.

t : Triangle ABC is equilateral; negation, $\sim t$: Triangle ABC is not equilateral.

Now, what is $\sim(\sim s)$? It would be $2 + 2 = 4$, which is s . And what is $\sim(\sim t)$? This would be 'the triangle ABC is equilateral', i.e., t . In fact, **for any statement p , $\sim(\sim p)$ is p .**

Example 12 : State the negations for the following statements:

- (i) Mike's dog does not have a black tail.
- (ii) All irrational numbers are real numbers.
- (iii) $\sqrt{2}$ is irrational.
- (iv) Some rational numbers are integers.
- (v) Not all teachers are males.
- (vi) Some horses are not brown.
- (vii) There is no real number x , such that $x^2 = -1$.

Solution :

- (i) It is false that Mike's dog does not have a black tail, i.e., Mike's dog has a black tail.
- (ii) It is false that all irrational numbers are real numbers, i.e., some (at least one) irrational numbers are not real numbers. One can also write this as, 'Not all irrational numbers are real numbers.'
- (iii) It is false that $\sqrt{2}$ is irrational, i.e., $\sqrt{2}$ is not irrational.
- (iv) It is false that some rational numbers are integers, i.e., no rational number is an integer.
- (v) It is false that not all teachers are males, i.e., all teachers are males.
- (vi) It is false that some horses are not brown, i.e., all horses are brown.
- (vii) It is false that there is no real number x , such that $x^2 = -1$, i.e., there is at least one real number x , such that $x^2 = -1$.

Remark : From the above discussion, you may arrive at the following **Working Rule** for obtaining the negation of a statement :

- (i) First write the statement with a 'not'.
- (ii) If there is any confusion, make suitable modification, specially in the statements involving 'All' or 'Some'.

EXERCISE A1.4

1. State the negations for the following statements :

(i) Man is mortal.	(ii) Line l is parallel to line m .
(iii) This chapter has many exercises.	(iv) All integers are rational numbers.
(v) Some prime numbers are odd.	(vi) No student is lazy.
(vii) Some cats are not black.	
(viii) There is no real number x , such that $\sqrt{x} = -1$.	
(ix) 2 divides the positive integer a .	(x) Integers a and b are coprime.
2. In each of the following questions, there are two statements. State if the second is the negation of the first or not.

(i) Mumtaz is hungry. Mumtaz is not hungry.	(ii) Some cats are black. Some cats are brown.
(iii) All elephants are huge. One elephant is not huge.	(iv) All fire engines are red. All fire engines are not red.
(v) No man is a cow. Some men are cows.	

A1.6 Converse of a Statement

We now investigate the notion of the converse of a statement. For this, we need the notion of a 'compound' statement, that is, a statement which is a combination of one or more 'simple' statements. There are many ways of creating compound statements, but we will focus on those that are created by connecting two simple statements with the use of the words 'if' and 'then'. For example, the statement 'If it is raining, then it is difficult to go on a bicycle', is made up of two statements:

p : It is raining

q : It is difficult to go on a bicycle.

Using our previous notation we can say: If p , then q . We can also say ' p implies q ', and denote it by $p \Rightarrow q$.

Now, suppose you have the statement 'If the water tank is black, then it contains potable water.' This is of the form $p \Rightarrow q$, where the hypothesis is p (the water tank is black) and the conclusion is q (the tank contains potable water). Suppose we interchange the hypothesis and the conclusion, what do we get? We get $q \Rightarrow p$, i.e., if the water in the tank is potable, then the tank must be black. This statement is called the **converse** of the statement $p \Rightarrow q$.

In general, the **converse** of the statement $p \Rightarrow q$ is $q \Rightarrow p$, where p and q are statements. Note that $p \Rightarrow q$ and $q \Rightarrow p$ are the converses of each other.

Example 13 : Write the converses of the following statements :

- (i) If Jamila is riding a bicycle, then 17 August falls on a Sunday.
- (ii) If 17 August is a Sunday, then Jamila is riding a bicycle.
- (iii) If Pauline is angry, then her face turns red.
- (iv) If a person has a degree in education, then she is allowed to teach.
- (v) If a person has a viral infection, then he runs a high temperature.
- (vi) If Ahmad is in Mumbai, then he is in India.
- (vii) If triangle ABC is equilateral, then all its interior angles are equal.
- (viii) If x is an irrational number, then the decimal expansion of x is non-terminating non-recurring.
- (ix) If $x - a$ is a factor of the polynomial $p(x)$, then $p(a) = 0$.

Solution : Each statement above is of the form $p \Rightarrow q$. So, to find the converse, we first identify p and q , and then write $q \Rightarrow p$.

- (i) p : Jamila is riding a bicycle, and q : 17 August falls on a Sunday. Therefore, the converse is: If 17 August falls on a Sunday, then Jamila is riding a bicycle.
- (ii) This is the converse of (i). Therefore, its converse is the statement given in (i) above.
- (iii) If Pauline's face turns red, then she is angry.
- (iv) If a person is allowed to teach, then she has a degree in education.
- (v) If a person runs a high temperature, then he has a viral infection.
- (vi) If Ahmad is in India, then he is in Mumbai.
- (vii) If all the interior angles of triangle ABC are equal, then it is equilateral.
- (viii) If the decimal expansion of x is non-terminating non-recurring, then x is an irrational number.
- (ix) If $p(a) = 0$, then $x - a$ is a factor of the polynomial $p(x)$.

Notice that we have simply written the converse of each of the statements above without worrying if they are true or false. For example, consider the following statement: If Ahmad is in Mumbai, then he is in India. This statement is true. Now consider the converse: If Ahmad is in India, then he is in Mumbai. This need not be true always – he could be in any other part of India.

In mathematics, especially in geometry, you will come across many situations where $p \Rightarrow q$ is true, and you will have to decide if the converse, i.e., $q \Rightarrow p$, is also true.

Example 14 : State the converses of the following statements. In each case, also decide whether the converse is true or false.

- (i) If n is an even integer, then $2n + 1$ is an odd integer.
- (ii) If the decimal expansion of a real number is terminating, then the number is rational.
- (iii) If a transversal intersects two parallel lines, then each pair of corresponding angles is equal.
- (iv) If each pair of opposite sides of a quadrilateral is equal, then the quadrilateral is a parallelogram.
- (v) If two triangles are congruent, then their corresponding angles are equal.

Solution :

- (i) The converse is 'If $2n + 1$ is an odd integer, then n is an even integer.' This is a false statement (for example, $15 = 2(7) + 1$, and 7 is odd).
- (ii) 'If a real number is rational, then its decimal expansion is terminating', is the converse. This is a false statement, because a rational number can also have a non-terminating recurring decimal expansion.
- (iii) The converse is 'If a transversal intersects two lines in such a way that each pair of corresponding angles are equal, then the two lines are parallel.' We have assumed, by Axiom 3.4 of your Class IX textbook, that this statement is true.
- (iv) 'If a quadrilateral is a parallelogram, then each pair of its opposite sides is equal', is the converse. This is true (Theorem 7.1, Class IX).
- (v) 'If the corresponding angles in two triangles are equal, then they are congruent', is the converse. This statement is false. We leave it to you to find suitable counter-examples.

EXERCISE A1.5

1. Write the converses of the following statements.
 - (i) If it is hot in Tokyo, then Sharan sweats a lot.
 - (ii) If Shalini is hungry, then her stomach grumbles.
 - (iii) If Jaswant has a scholarship, then she can get a degree.
 - (iv) If a plant has flowers, then it is alive.
 - (v) If an animal is a cat, then it has a tail.

2. Write the converses of the following statements. Also, decide in each case whether the converse is true or false.
- (i) If triangle ABC is isosceles, then its base angles are equal.
 - (ii) If an integer is odd, then its square is an odd integer.
 - (iii) If $x^2 = 1$, then $x = 1$.
 - (iv) If ABCD is a parallelogram, then AC and BD bisect each other.
 - (v) If a, b and c , are whole numbers, then $a + (b + c) = (a + b) + c$.
 - (vi) If x and y are two odd numbers, then $x + y$ is an even number.
 - (vii) If vertices of a parallelogram lie on a circle, then it is a rectangle.

A1.7 Proof by Contradiction

So far, in all our examples, we used direct arguments to establish the truth of the results. We now explore ‘indirect’ arguments, in particular, a very powerful tool in mathematics known as ‘proof by contradiction’. We have already used this method in Chapter 1 to establish the irrationality of several numbers and also in other chapters to prove some theorems. Here, we do several more examples to illustrate the idea.

Before we proceed, let us explain what a *contradiction* is. In mathematics, a contradiction occurs when we get a statement p such that p is true and $\sim p$, its negation, is also true. For example,

p : $x = \frac{a}{b}$, where a and b are coprime.

q : 2 divides both ‘ a ’ and ‘ b ’.

If we assume that p is true and also manage to show that q is true, then we have arrived at a contradiction, because q implies that the negation of p is true. If you remember, this is exactly what happened when we tried to prove that $\sqrt{2}$ is irrational (see Chapter 1).

How does proof by contradiction work? Let us see this through a specific example.

Suppose we are given the following :

All women are mortal. A is a woman. Prove that A is mortal.

Even though this is a rather easy example, let us see how we can prove this by contradiction.

- Let us assume that we want to establish the truth of a statement p (here we want to show that p : ‘A is mortal’ is true).

- So, we begin by assuming that the statement is not true, that is, we assume that the negation of p is true (i.e., A is not mortal).
- We then proceed to carry out a series of logical deductions based on the truth of the negation of p . (Since A is not mortal, we have a counter-example to the statement 'All women are mortal.' Hence, it is false that all women are mortal.)
- If this leads to a contradiction, then the contradiction arises because of our faulty assumption that p is not true. (We have a contradiction, since we have shown that the statement 'All women are mortal' and its negation, 'Not all women are mortal' is true at the same time. This contradiction arose, because we assumed that A is not mortal.)
- Therefore, our assumption is wrong, i.e., p has to be true. (So, A is mortal.)

Let us now look at examples from mathematics.

Example 15 : The product of a non-zero rational number and an irrational number is irrational.

Solution :

Statements	Analysis/Comment
We will use proof by contradiction. Let r be a non-zero rational number and x be an irrational number. Let $r = \frac{m}{n}$, where m, n are integers and $m \neq 0$, $n \neq 0$. We need to prove that rx is irrational.	
Assume rx is rational.	Here, we are assuming the negation of the statement that we need to prove.
Then $rx = \frac{p}{q}$, $q \neq 0$, where p and q are integers.	This follow from the previous statement and the definition of a rational number.
Rearranging the equation $rx = \frac{p}{q}$, $q \neq 0$, and using the fact that $r = \frac{m}{n}$, we get $x = \frac{p}{rq} = \frac{np}{mq}$.	

Since np and mq are integers and $mq \neq 0$, x is a rational number.	Using properties of integers, and definition of a rational number.
This is a contradiction, because we have shown x to be rational, but by our hypothesis, we have x is irrational.	This is what we were looking for — a contradiction.
The contradiction has arisen because of the faulty assumption that rx is rational. Therefore, rx is irrational.	Logical deduction.

We now prove Example 11, but this time using proof by contradiction. The proof is given below:

Statements	Analysis/Comment
Let us assume that the statement is not true.	As we saw earlier, this is the starting point for an argument using 'proof by contradiction'.
So we suppose that there exists a prime number $p > 3$, which is not of the form $6n + 1$ or $6n + 5$, where n is a whole number.	This is the negation of the statement in the result.
Using Euclid's division lemma on division by 6, and using the fact that p is not of the form $6n + 1$ or $6n + 5$, we get $p = 6n$ or $6n + 2$ or $6n + 3$ or $6n + 4$.	Using earlier proved results.
Therefore, p is divisible by either 2 or 3.	Logical deduction.
So, p is not a prime.	Logical deduction.
This is a contradiction, because by our hypothesis p is prime.	Precisely what we want!
The contradiction has arisen, because we assumed that there exists a prime number $p > 3$ which is not of the form $6n + 1$ or $6n + 5$.	
Hence, every prime number greater than 3 is of the form $6n + 1$ or $6n + 5$.	We reach the conclusion.

Remark : The example of the proof above shows you, yet again, that there can be several ways of proving a result.

Theorem A1.2 : *Out of all the line segments, drawn from a point to points of a line not passing through the point, the smallest is the perpendicular to the line.*

Proof :

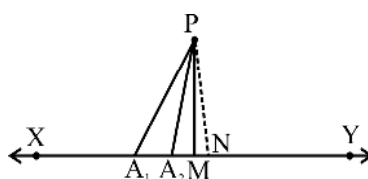


Fig. A1.5

Statements	Analysis/Comment
Let XY be the given line, P a point not lying on XY and PM, PA_1 , PA_2 , . . . etc., be the line segments drawn from P to the points of the line XY, out of which PM is the smallest (see Fig. A1.5).	Since we have to prove that out of all PM, PA_1 , PA_2 , . . . etc., the smallest is perpendicular to XY, we start by taking these line segments.
Let PM be not perpendicular to XY	This is the negation of the statement to be proved by contradiction.
Draw a perpendicular PN on the line XY, shown by dotted lines in Fig. A1.5.	We often need constructions to prove our results.
PN is the smallest of all the line segments PM, PA_1 , PA_2 , . . . etc., which means $PN < PM$.	Side of right triangle is less than the hypotenuse and known property of numbers.
This contradicts our hypothesis that PM is the smallest of all such line segments.	Precisely what we want!
Therefore, the line segment PM is perpendicular to XY.	We reach the conclusion.

EXERCISE A1.6

1. Suppose $a + b = c + d$, and $a < c$. Use proof by contradiction to show $b > d$.
2. Let r be a rational number and x be an irrational number. Use proof by contradiction to show that $r + x$ is an irrational number.
3. Use proof by contradiction to prove that if for an integer a , a^2 is even, then so is a .
[**Hint** : Assume a is not even, that is, it is of the form $2n + 1$, for some integer n , and then proceed.]
4. Use proof by contradiction to prove that if for an integer a , a^2 is divisible by 3, then a is divisible by 3.
5. Use proof by contradiction to show that there is no value of n for which 6^n ends with the digit zero.
6. Prove by contradiction that two distinct lines in a plane cannot intersect in more than one point.

A1.8 Summary

In this Appendix, you have studied the following points :

1. Different ingredients of a proof and other related concepts learnt in Class IX.
2. The negation of a statement.
3. The converse of a statement.
4. Proof by contradiction.

MATHEMATICAL MODELLING

A2

A2.1 Introduction

- An adult human body contains approximately 1,50,000 km of arteries and veins that carry blood.
- The human heart pumps 5 to 6 litres of blood in the body every 60 seconds.
- The temperature at the surface of the Sun is about $6,000^{\circ}\text{C}$.

Have you ever wondered how our scientists and mathematicians could possibly have estimated these results? Did they pull out the veins and arteries from some adult dead bodies and measure them? Did they drain out the blood to arrive at these results? Did they travel to the Sun with a thermometer to get the temperature of the Sun? Surely not. Then how did they get these figures?

Well, the answer lies in **mathematical modelling**, which we introduced to you in Class IX. Recall that a mathematical model is a mathematical description of some real-life situation. Also, recall that mathematical modelling is the process of creating a mathematical model of a problem, and using it to analyse and solve the problem.

So, in mathematical modelling, we take a real-world problem and convert it to an equivalent mathematical problem. We then solve the mathematical problem, and interpret its solution in the situation of the real-world problem. And then, it is important to see that the solution, we have obtained, ‘makes sense’, which is the stage of validating the model. Some examples, where mathematical modelling is of great importance, are:

- (i) Finding the width and depth of a river at an unreachable place.
- (ii) Estimating the mass of the Earth and other planets.
- (iii) Estimating the distance between Earth and any other planet.
- (iv) Predicting the arrival of the monsoon in a country.

- (v) Predicting the trend of the stock market.
- (vi) Estimating the volume of blood inside the body of a person.
- (vii) Predicting the population of a city after 10 years.
- (viii) Estimating the number of leaves in a tree.
- (ix) Estimating the ppm of different pollutants in the atmosphere of a city.
- (x) Estimating the effect of pollutants on the environment.
- (xi) Estimating the temperature on the Sun's surface.

In this chapter we shall revisit the process of mathematical modelling, and take examples from the world around us to illustrate this. In Section A2.2 we take you through all the stages of building a model. In Section A2.3, we discuss a variety of examples. In Section A2.4, we look at reasons for the importance of mathematical modelling.

A point to remember is that here we aim to make you aware of an important way in which mathematics helps to solve real-world problems. However, you need to know some more mathematics to really appreciate the power of mathematical modelling. In higher classes some examples giving this flavour will be found.

A2.2 Stages in Mathematical Modelling

In Class IX, we considered some examples of the use of modelling. Did they give you an insight into the process and the steps involved in it? Let us quickly revisit the main steps in mathematical modelling.

Step 1 (Understanding the problem) : Define the real problem, and if working in a team, discuss the issues that you wish to understand. Simplify by making assumptions and ignoring certain factors so that the problem is manageable.

For example, suppose our problem is to estimate the number of fishes in a lake. It is not possible to capture each of these fishes and count them. We could possibly capture a sample and from it try and estimate the total number of fishes in the lake.

Step 2 (Mathematical description and formulation) : Describe, in mathematical terms, the different aspects of the problem. Some ways to describe the features mathematically, include:

- define variables
- write equations or inequalities
- gather data and organise into tables
- make graphs
- calculate probabilities

For example, having taken a sample, as stated in Step 1, how do we estimate the entire population? We would have to then mark the sampled fishes, allow them to mix with the remaining ones in the lake, again draw a sample from the lake, and see how many of the previously marked ones are present in the new sample. Then, using ratio and proportion, we can come up with an estimate of the total population. For instance, let us take a sample of 20 fishes from the lake and mark them, and then release them in the same lake, so as to mix with the remaining fishes. We then take another sample (say 50), from the mixed population and see how many are marked. So, we gather our data and analyse it.

One major assumption we are making is that the marked fishes mix uniformly with the remaining fishes, and the sample we take is a good representative of the entire population.

Step 3 (Solving the mathematical problem) : The simplified mathematical problem developed in Step 2 is then solved using various mathematical techniques.

For instance, suppose in the second sample in the example in Step 2, 5 fishes are marked. So, $\frac{5}{50}$, i.e., $\frac{1}{10}$, of the population is marked. If this is typical of the whole population, then $\frac{1}{10}$ th of the population = 20.

So, the whole population = $20 \times 10 = 200$.

Step 4 (Interpreting the solution) : The solution obtained in the previous step is now looked at, in the context of the real-life situation that we had started with in Step 1.

For instance, our solution in the problem in Step 3 gives us the population of fishes as 200.

Step 5 (Validating the model) : We go back to the original situation and see if the results of the mathematical work make sense. If so, we use the model until new information becomes available or assumptions change.

Sometimes, because of the simplification assumptions we make, we may lose essential aspects of the real problem while giving its mathematical description. In such cases, the solution could very often be off the mark, and not make sense in the real situation. If this happens, we reconsider the assumptions made in Step 1 and revise them to be more realistic, possibly by including some factors which were not considered earlier.

For instance, in Step 3 we had obtained an estimate of the entire population of fishes. It may not be the actual number of fishes in the pond. We next see whether this is a good estimate of the population by repeating Steps 2 and 3 a few times, and taking the mean of the results obtained. This would give a closer estimate of the population.

Another way of visualising **the process of mathematical modelling** is shown in Fig. A2.1.

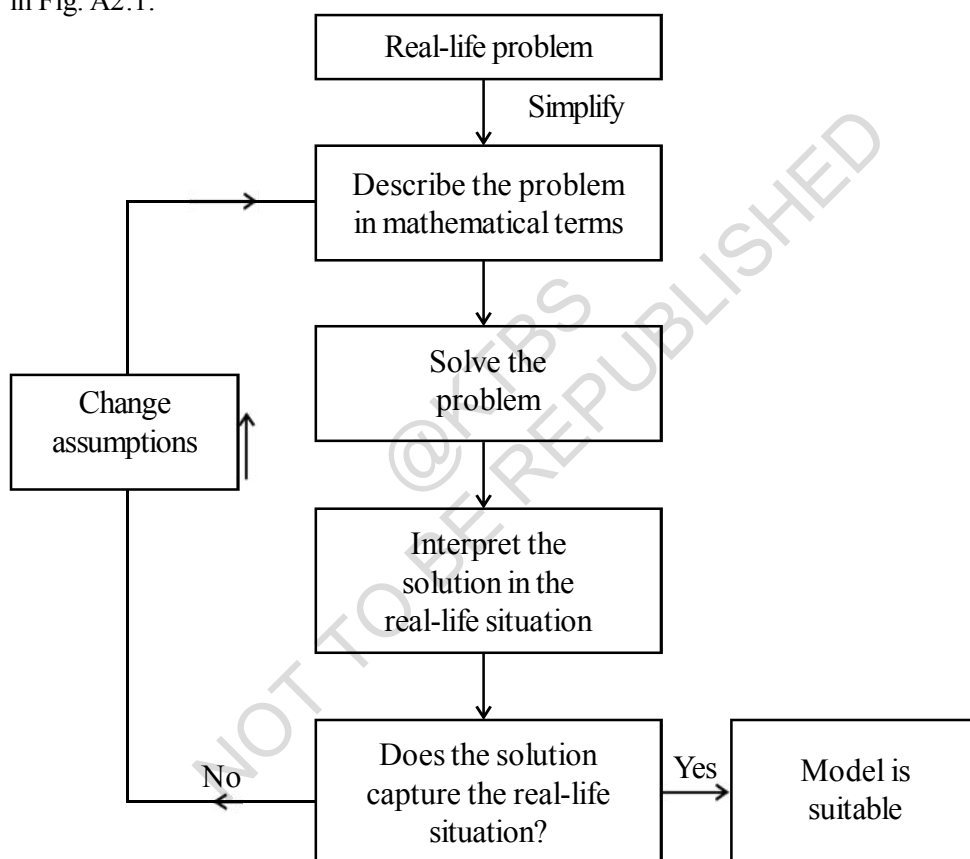


Fig. A2.1

Modellers look for a balance between simplification (for ease of solution) and accuracy. They hope to approximate reality closely enough to make some progress. The best outcome is to be able to predict what will happen, or estimate an outcome, with reasonable accuracy. Remember that different assumptions we use for simplifying the problem can lead to different models. So, there are no perfect models. There are good ones and yet better ones.

EXERCISE A2.1

1. Consider the following situation.

A problem dating back to the early 13th century, posed by Leonardo Fibonacci asks how many rabbits you would have if you started with just two and let them reproduce. Assume that a pair of rabbits produces a pair of offspring each month and that each pair of rabbits produces their first offspring at the age of 2 months. Month by month the number of pairs of rabbits is given by the sum of the rabbits in the two preceding months, except for the 0th and the 1st months.

Month	Pairs of Rabbits
0	1
1	1
2	2
3	3
4	5
5	8
6	13
7	21
8	34
9	55
10	89
11	144
12	233
13	377
14	610
15	987
16	1597

After just 16 months, you have nearly 1600 pairs of rabbits!

Clearly state the problem and the different stages of mathematical modelling in this situation.

A2.3 Some Illustrations

Let us now consider some examples of mathematical modelling.

Example 1 (Rolling of a pair of dice) : Suppose your teacher challenges you to the following guessing game: She would throw a pair of dice. Before that you need to guess the sum of the numbers that show up on the dice. For every correct answer, you get two points and for every wrong guess you lose two points. What numbers would be the best guess?

Solution :

Step 1 (Understanding the problem) : You need to know a few numbers which have higher chances of showing up.

Step 2 (Mathematical description) : In mathematical terms, the problem translates to finding out the probabilities of the various possible sums of numbers that the dice could show.

We can model the situation very simply by representing a roll of the dice as a random choice of one of the following thirty six pairs of numbers.

(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

The first number in each pair represents the number showing on the first die, and the second number is the number showing on the second die.

Step 3 (Solving the mathematical problem) : Summing the numbers in each pair above, we find that possible sums are 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 and 12. We have to find the probability for each of them, assuming all 36 pairs are equally likely.

We do this in the following table.

Sum	2	3	4	5	6	7	8	9	10	11	12
Probability	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Observe that the chance of getting a sum of a seven is $\frac{1}{6}$, which is larger than the chances of getting other numbers as sums.

Step 4 (Interpreting the solution) : Since the probability of getting the sum 7 is the highest, you should repeatedly guess the number seven.

Step 5 (Validating the model) : Toss a pair of dice a large number of times and prepare a relative frequency table. Compare the relative frequencies with the corresponding probabilities. If these are not close, then possibly the dice are biased. Then, we could obtain data to evaluate the number towards which the bias is.

Before going to the next example, you may need some background.

Not having the money you want when you need it, is a common experience for many people. Whether it is having enough money for buying essentials for daily living, or for buying comforts, we always require money. To enable the customers with limited funds to purchase goods like scooters, refrigerators, televisions, cars, etc., a scheme known as an *instalment scheme (or plan)* is introduced by traders.

Sometimes a trader introduces an *instalment* scheme as a marketing strategy to allure customers to purchase these articles. Under the instalment scheme, the customer is not required to make full payment of the article at the time of buying it. She/he is allowed to pay a part of it at the time of purchase, and the rest can be paid in instalments, which could be monthly, quarterly, half-yearly, or even yearly. Of course, the buyer will have to pay more in the instalment plan, because the seller is going to charge some interest on account of the payment made at a later date (called *deferred payment*).

Before we take a few examples to understand the instalment scheme, let us understand the most frequently used terms related to this concept.

The *cash price* of an article is the amount which a customer has to pay as full payment of the article at the time it is purchased. *Cash down payment* is the amount which a customer has to pay as part payment of the price of an article at the time of purchase.

Remark : If the instalment scheme is such that the remaining payment is completely made within one year of the purchase of the article, then simple interest is charged on the deferred payment.

In the past, charging interest on borrowed money was often considered evil, and, in particular, was long prohibited. One way people got around the law against paying interest was to borrow in one currency and repay in another, the interest being disguised in the exchange rate.

Let us now come to a related mathematical modelling problem.

Example 2 : Juhi wants to buy a bicycle. She goes to the market and finds that the bicycle she likes is available for ₹ 1800. Juhi has ₹ 600 with her. So, she tells the shopkeeper that she would not be able to buy it. The shopkeeper, after a bit of calculation, makes the following offer. He tells Juhi that she could take the bicycle by making a payment of ₹ 600 cash down and the remaining money could be made in two monthly instalments of ₹ 610 each. Juhi has two options one is to go for instalment scheme or to make cash payment by taking loan from a bank which is available at the rate of 10% per annum simple interest. Which option is more economical to her?

Solution :

Step 1 (Understanding the problem) : What Juhi needs to determine is whether she should take the offer made by the shopkeeper or not. For this, she should know the two rates of interest—one charged in the instalment scheme and the other charged by the bank (i.e., 10%).

Step 2 (Mathematical description) : In order to accept or reject the scheme, she needs to determine the interest that the shopkeeper is charging in comparison to the bank. Observe that since the entire money shall be paid in less than a year, simple interest shall be charged.

We know that the cash price of the bicycle = ₹ 1800.

Also, the cashdown payment under the instalment scheme = ₹ 600.

So, the balance price that needs to be paid in the instalment scheme = ₹ (1800 – 600) = ₹ 1200.

Let r % per annum be the rate of interest charged by the shopkeeper.

Amount of each instalment = ₹ 610

Amount paid in instalments = ₹ 610 + ₹ 610 = ₹ 1220

Interest paid in instalment scheme = ₹ 1220 – ₹ 1200 = ₹ 20 (1)

Since, Juhi kept a sum of ₹ 1200 for one month, therefore,

Principal for the first month = ₹ 1200

Principal for the second month = ₹ (1200 – 610) = ₹ 590

Balance of the second principal ₹ 590 + interest charged (₹ 20) = monthly instalment (₹ 610) = 2nd instalment

So, the total principal for one month = ₹ 1200 + ₹ 590 = ₹ 1790

Now,
$$\text{interest} = ₹ \frac{1790 \times r \times 1}{100 \times 12} \quad (2)$$

Step 3 (Solving the problem) : From (1) and (2)

$$\frac{1790 \times r \times 1}{100 \times 12} = 20$$

or
$$r = \frac{20 \times 1200}{1790} = 13.14 \text{ (approx.)}$$

Step 4 (Interpreting the solution) : The rate of interest charged in the instalment scheme = 13.14 %.

The rate of interest charged by the bank = 10%

So, she should prefer to borrow the money from the bank to buy the bicycle which is more economical.

Step 5 (Validating the model) : This stage in this case is not of much importance here as the numbers are fixed. However, if the formalities for taking loan from the bank such as cost of stamp paper, etc., which make the effective interest rate more than what it is the instalment scheme, then she may change her opinion.

Remark : Interest rate modelling is still at its early stages and validation is still a problem of financial markets. In case, different interest rates are incorporated in fixing instalments, validation becomes an important problem.

EXERCISE A2.2

In each of the problems below, show the different stages of mathematical modelling for solving the problems.

1. An ornithologist wants to estimate the number of parrots in a large field. She uses a net to catch some, and catches 32 parrots, which she rings and sets free. The following week she manages to net 40 parrots, of which 8 are ringed.
 - (i) What fraction of her second catch is ringed?
 - (ii) Find an estimate of the total number of parrots in the field.
2. Suppose the adjoining figure represents an aerial photograph of a forest with each dot representing a tree. Your purpose is to find the number of trees there are on this tract of land as part of an environmental census.



3. A T.V. can be purchased for ₹ 24000 cash or for ₹ 8000 cashdown payment and six monthly instalments of ₹ 2800 each. Ali goes to market to buy a T.V., and he has ₹ 8000 with him. He has now two options. One is to buy TV under instalment scheme or to make cash payment by taking loan from some financial society. The society charges simple interest at the rate of 18% per annum simple interest. Which option is better for Ali?

A2.4 Why is Mathematical Modelling Important?

As we have seen in the examples, mathematical modelling is an interdisciplinary subject. Mathematicians and specialists in other fields share their knowledge and expertise to improve existing products, develop better ones, or predict the behaviour of certain products.

There are, of course, many specific reasons for the importance of modelling, but most are related in some ways to the following :

- *To gain understanding.* If we have a mathematical model which reflects the essential behaviour of a real-world system of interest, we can understand that system better through an analysis of the model. Furthermore, in the process of building the model we find out which factors are most important in the system, and how the different aspects of the system are related.
- *To predict, or forecast, or simulate.* Very often, we wish to know what a real-world system will do in the future, but it is expensive, impractical or impossible to experiment directly with the system. For example, in weather prediction, to study drug efficacy in humans, finding an optimum design of a nuclear reactor, and so on.

Forecasting is very important in many types of organisations, since predictions of future events have to be incorporated into the decision-making process. For example:

In marketing departments, reliable forecasts of demand help in planning of the sale strategies.

A school board needs to be able to forecast the increase in the number of school going children in various districts so as to decide where and when to start new schools.

Most often, forecasters use the past data to predict the future. They first analyse the data in order to identify a pattern that can describe it. Then this data and pattern is extended into the future in order to prepare a forecast. This basic strategy is employed in most forecasting techniques, and is based on the assumption that the pattern that has been identified will continue in the future also.

- *To estimate.* Often, we need to estimate large values. You've seen examples of the trees in a forest, fish in a lake, etc. For another example, before elections, the contesting parties want to predict the probability of their party winning the elections. In particular, they want to estimate how many people in their constituency would vote for their party. Based on their predictions, they may want to decide on the campaign strategy. Exit polls have been used widely to predict the number of seats, a party is expected to get in elections.

EXERCISE A2.3

1. Based upon the data of the past five years, try and forecast the average percentage of marks in Mathematics that your school would obtain in the Class X board examination at the end of the year.

A2.5 Summary

In this Appendix, you have studied the following points :

1. A mathematical model is a mathematical description of a real-life situation. Mathematical modelling is the process of creating a mathematical model, solving it and using it to understand the real-life problem.
2. The various steps involved in modelling are : understanding the problem, formulating the mathematical model, solving it, interpreting it in the real-life situation, and, most importantly, validating the model.
3. Developed some mathematical models.
4. The importance of mathematical modelling.

ANSWERS/HINTS

Polynomials

EXERCISE 9.1

1. (i) No zeroes (ii) 1 (iii) 3 (iv) 2 (v) 4 (vi) 3

EXERCISE 9.2

1. (i) $-2, 4$ (ii) $\frac{1}{2}, \frac{1}{2}$ (iii) $-\frac{1}{3}, \frac{3}{2}$
(iv) $-2, 0$ (v) $-\sqrt{15}, \sqrt{15}$ (vi) $-1, \frac{4}{3}$
2. (i) $4x^2 - x - 4$ (ii) $3x^2 - 3\sqrt{2}x + 1$ (iii) $x^2 + \sqrt{5}$
(iv) $x^2 - x + 1$ (v) $4x^2 + x + 1$ (vi) $x^2 - 4x + 1$

EXERCISE 9.3

1. (i) Quotient $= x - 3$ and remainder $= 7x - 9$
(ii) Quotient $= x^2 + x - 3$ and remainder $= 8$
(iii) Quotient $= -x^2 - 2$ and remainder $= -5x + 10$
2. (i) Yes (ii) Yes (iii) No 3. $-1, -1$ 4. $g(x) = x^2 - x + 1$
5. (i) $p(x) = 2x^2 - 2x + 14, g(x) = 2, q(x) = x^2 - x + 7, r(x) = 0$
(ii) $p(x) = x^3 + x^2 + x + 1, g(x) = x^2 - 1, q(x) = x + 1, r(x) = 2x + 2$
(iii) $p(x) = x^3 + 2x^2 - x + 2, g(x) = x^2 - 1, q(x) = x + 2, r(x) = 4$

There can be several examples in each of (i), (ii) and (iii).

EXERCISE 9.4 (Optional)*

2. $x^3 - 2x^2 - 7x + 14$ 3. $a = 1, b = \pm\sqrt{2}$
4. $-5, 7$ 5. $k = 5$ and $a = -5$

Quadratic Equations **EXERCISE 10.1**

1. (i) Yes (ii) Yes (iii) No (iv) Yes
 (v) Yes (vi) No (vii) No (viii) Yes
2. (i) $2x^2 + x - 528 = 0$, where x is breadth (in metres) of the plot.
 (ii) $x^2 + x - 306 = 0$, where x is the smaller integer.
 (iii) $x^2 + 32x - 273 = 0$, where x (in years) is the present age of Rohan.
 (iv) $u^2 - 8u - 1280 = 0$, where u (in km/h) is the speed of the train.

EXERCISE 10.2

1. (i) $-2, 5$ (ii) $-2, \frac{3}{2}$ (iii) $-\frac{5}{\sqrt{2}}, -\sqrt{2}$
 (iv) $\frac{1}{4}, \frac{1}{4}$ (v) $\frac{1}{10}, \frac{1}{10}$
2. (i) $9, 36$ (ii) $25, 30$
3. Numbers are 13 and 14. 4. Positive integers are 13 and 14.
5. 5 cm and 12 cm 6. Number of articles = 6, Cost of each article = ₹ 15

EXERCISE 10.3

1. (i) $\frac{1}{2}, 3$ (ii) $\frac{-1-\sqrt{33}}{4}, \frac{-1+\sqrt{33}}{4}$ (iii) $-\frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}$
 (iv) Do not exist
2. Same as 1 3. (i) $\frac{3-\sqrt{13}}{2}, \frac{3+\sqrt{13}}{2}$ (ii) 1, 2 4. 7 years
5. Marks in mathematics = 12, marks in English = 18;
 or, Marks in mathematics = 13, marks in English = 17
6. 120 m, 90 m 7. 18, 12 or 18, -12
8. 40 km/h 9. 15 hours, 25 hours
10. Speed of the passenger train = 33 km/h, speed of express train = 44 km/h
11. 18 m, 12 m

EXERCISE 10.4

1. (i) Real roots do not exist (ii) Equal roots; $\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}}$ (iii) Distinct roots; $\frac{3 \pm \sqrt{3}}{2}$
 2. (i) $k = \pm 2\sqrt{6}$ (ii) $k = 6$
 3. Yes. 40 m, 20 m 4. No 5. Yes. 20 m, 20 m

Introduction to Trigonometry**EXERCISE 11.1**

1. (i) $\sin A = \frac{7}{25}, \cos A = \frac{24}{25}$ (ii) $\sin C = \frac{24}{25}, \cos C = \frac{7}{25}$
 2. 0 3. $\cos A = \frac{\sqrt{7}}{4}, \tan A = \frac{3}{\sqrt{7}}$ 4. $\sin A = \frac{15}{17}, \sec A = \frac{17}{8}$
 5. $\sin \theta = \frac{5}{13}, \cos \theta = \frac{12}{13}, \tan \theta = \frac{5}{12}, \cot \theta = \frac{12}{5}, \operatorname{cosec} \theta = \frac{13}{5}$
 7. (i) $\frac{49}{64}$ (ii) $\frac{49}{64}$ 8. Yes
 9. (i) 1 (ii) 0 10. $\sin P = \frac{12}{13}, \cos P = \frac{5}{13}, \tan P = \frac{12}{5}$
 11. (i) False (ii) True (iii) False (iv) False (v) False

EXERCISE 11.2

1. (i) 1 (ii) 2 (iii) $\frac{3\sqrt{2} - \sqrt{6}}{8}$ (iv) $\frac{43 - 24\sqrt{3}}{11}$ (v) $\frac{67}{12}$
 2. (i) A (ii) D (iii) A (iv) C 3. $\angle A = 45^\circ, \angle B = 15^\circ$
 4. (i) False (ii) True (iii) False (iv) False (v) True

EXERCISE 11.3

1. (i) 1 (ii) 1 (iii) 0 (iv) 0
 3. $\angle A = 36^\circ$ 5. $\angle A = 22^\circ$ 7. $\cos 23^\circ + \sin 15^\circ$

EXERCISE 11.4

$$1. \sin A = \frac{1}{\sqrt{1 + \cot^2 A}}, \tan A = \frac{1}{\cot A}, \sec A = \frac{\sqrt{1 + \cot^2 A}}{\cot A}$$

$$2. \sin A = \frac{\sqrt{\sec^2 A - 1}}{\sec A}, \cos A = \frac{1}{\sec A}, \tan A = \sqrt{\sec^2 A - 1}$$

$$\cot A = \frac{1}{\sqrt{\sec^2 A - 1}}, \operatorname{cosec} A = \frac{\sec A}{\sqrt{\sec^2 A - 1}}$$

3. (i) 1 (ii) 1 4. (i) B (ii) C (iii) D (iv) D

Some Applications Of Trigonometry**EXERCISE 12.1**

1. 10m 2. $8\sqrt{3}$ m 3. 3m, $2\sqrt{3}$ m 4. $10\sqrt{3}$ m
 5. $40\sqrt{3}$ m 6. $19\sqrt{3}$ m 7. $20(\sqrt{3} - 1)$ m 8. $0.8(\sqrt{3} + 1)$ m
 9. $16\frac{2}{3}$ m 10. $20\sqrt{3}$ m, 20m, 60m 11. $10\sqrt{3}$ m, 10m 12. $7(\sqrt{3} + 1)$ m
 13. $75(\sqrt{3} - 1)$ m 14. $58\sqrt{3}$ m 15. 3 seconds

Statistics**EXERCISE 13.1**

1. 8.1 plants. We have used direct method because numerical values of x_i and f_i are small.
 2. ₹ 545.20 3. $f = 20$ 4. 75.9
 5. 57.19 6. ₹ 211 7. 0.099 ppm
 8. 12.48 days 9. 69.43 %

EXERCISE 13.2

1. Mode = 36.8 years, Mean = 35.37 years. Maximum number of patients admitted in the hospital are of the age 36.8 years (approx.), while on an average the age of a patient admitted to the hospital is 35.37 years.

2. 65.625 hours
3. Modal monthly expenditure = ₹ 1847.83, Mean monthly expenditure = ₹ 2662.5.
4. Mode : 30.6, Mean = 29.2. Most states/U.T. have a student teacher ratio of 30.6 and on an average, this ratio is 29.2.
5. Mode = 4608.7 runs 6. Mode = 44.7 cars

EXERCISE 13.3

1. Median = 137 units, Mean = 137.05 units, Mode = 135.76 units.

The three measures are approximately the same in this case.

2. $x = 8, y = 7$
3. Median age = 35.76 years
4. Median length = 146.75 mm
5. Median life = 3406.98 hours
6. Median = 8.05, Mean = 8.32, Modal size = 7.88
7. Median weight = 56.67 kg

EXERCISE 13.4

1.

Daily income (in ₹)	Cumulative frequency
Less than 120	12
Less than 140	26
Less than 160	34
Less than 180	40
Less than 200	50

Draw ogive by plotting the points :
 (120, 12), (140, 26), (160, 34),
 (180, 40) and (200, 50)

2. Draw the ogive by plotting the points : (38, 0), (40, 3), (42, 5), (44, 9), (46, 14), (48, 28), (50, 32) and (52, 35). Here $\frac{n}{2} = 17.5$. Locate the point on the ogive whose ordinate is 17.5. The x-coordinate of this point will be the median.

3.

Production yield (kg/ha)	Cumulative frequency
More than or equal to 50	100
More than or equal to 55	98
More than or equal to 60	90
More than or equal to 65	78
More than or equal to 70	54
More than or equal to 75	16

Now, draw the ogive by plotting the points : (50, 100), (55, 98), (60, 90), (65, 78), (70, 54) and (75, 16).

Probability

EXERCISE 14.1

1. (i) 1 (ii) 0, impossible event (iii) 1, sure or certain event (iv) 1 (v) 0, 1
2. The experiments (iii) and (iv) have equally likely outcomes.
3. When we toss a coin, the outcomes head and tail are equally likely. So, the result of an individual coin toss is completely unpredictable.
4. B
5. 0.95
6. (i) 0 (ii) 1
7. 0.008
8. (i) $\frac{3}{8}$ (ii) $\frac{5}{8}$
9. (i) $\frac{5}{17}$ (ii) $\frac{8}{17}$ (iii) $\frac{13}{17}$
10. (i) $\frac{5}{9}$ (ii) $\frac{17}{18}$
11. $\frac{5}{13}$
12. (i) $\frac{1}{8}$ (ii) $\frac{1}{2}$ (iii) $\frac{3}{4}$ (iv) 1
13. (i) $\frac{1}{2}$ (ii) $\frac{1}{2}$ (iii) $\frac{1}{2}$
14. (i) $\frac{1}{26}$ (ii) $\frac{3}{13}$ (iii) $\frac{3}{26}$ (iv) $\frac{1}{52}$ (v) $\frac{1}{4}$ (vi) $\frac{1}{52}$
15. (i) $\frac{1}{5}$ (ii) (a) $\frac{1}{4}$ (b) 0
16. $\frac{11}{12}$
17. (i) $\frac{1}{5}$ (ii) $\frac{15}{19}$
18. (i) $\frac{9}{10}$ (ii) $\frac{1}{10}$ (iii) $\frac{1}{5}$

19. (i) $\frac{1}{3}$ (ii) $\frac{1}{6}$ 20. $\frac{\pi}{24}$ 21. (i) $\frac{31}{36}$ (ii) $\frac{5}{36}$

22. (i)

Sum on 2 dice	2	3	4	5	6	7	8	9	10	11	12
Probability	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

(ii) No. The eleven sums are not equally likely.

23. $\frac{3}{4}$; Possible outcomes are : HHH, TTT, HHT, HTH, HTT, THH, THT, TTH. Here, THH means tail in the first toss, head on the second toss and head on the third toss and so on.

24. (i) $\frac{25}{36}$ (ii) $\frac{11}{36}$

25. (i) Incorrect. We can classify the outcomes like this but they are not then 'equally likely'. Reason is that 'one of each' can result in two ways — from a head on first coin and tail on the second coin or from a tail on the first coin and head on the second coin. This makes it twice as likely as two heads (or two tails).
(ii) Correct. The two outcomes considered in the question are equally likely.

EXERCISE 14.2 (Optional)*

1. (i) $\frac{1}{5}$ (ii) $\frac{8}{25}$ (iii) $\frac{4}{5}$

2.

	1	2	2	3	3	6
1	2	3	3	4	4	7
2	3	4	4	5	5	8
2	3	4	4	5	5	8
3	4	5	5	6	6	9
3	4	5	5	6	6	9
6	7	8	8	9	9	12

- (i) $\frac{1}{2}$ (ii) $\frac{1}{9}$ (iii) $\frac{5}{12}$ 3. 10 4. $\frac{x}{12}$, $x = 3$ 5. 8

Surface Areas And Volumes

EXERCISE 15.1

1. 160 cm^2
2. 572 cm^2
3. 214.5 cm^2
4. Greatest diameter = 7 cm, surface area = 332.5 cm^2
5. $\frac{1}{4}l^2 (\pi + 24)$
6. 220 mm^2
7. 44 m^2 , ₹ 22000
8. 18 cm^2
9. 374 cm^2

EXERCISE 15.2

1. $\pi \text{ cm}^3$
2. 66 cm^3 . Volume of the air inside the model = Volume of air inside (cone + cylinder + cone)
 $= \left(\frac{1}{3} \pi r^2 h_1 + \pi r^2 h_2 + \frac{1}{3} \pi r^2 h_1 \right)$, where r is the radius of the cone and the cylinder, h_1 is the height (length) of the cone and h_2 is the height (length) of the cylinder.

$$\text{Required Volume} = \frac{1}{3} \pi r^2 (h_1 + 3h_2 + h_1).$$

3. 338 cm^3
4. 523.53 cm^3
5. 100
6. 892.26 kg
7. 1.131 m^3 (approx.)
8. Not correct. Correct answer is 346.51 cm^3 .

EXERCISE 15.3

1. 2.74 cm
2. 12 cm
3. 2.5 m
4. 1.125 m
5. 10
6. 400
7. 36 cm; $12\sqrt{13} \text{ cm}$
8. 562500 m^2 or 56.25 hectares.
9. 100 minutes

EXERCISE 15.4

1. $102\frac{2}{3} \text{ cm}^3$
2. 48 cm^2
3. $710\frac{2}{7} \text{ cm}^2$
4. Cost of milk is ₹ 209 and cost of metal sheet is ₹ 156.75.
5. 7964.4 m

EXERCISE 15.5 (Optional)*

1. 1256 cm; 788g (approx)
2. 30.14 cm^3 ; 52.75 cm^2
3. 1792
5. $782\frac{4}{7} \text{ cm}^2$

EXERCISE A1.1

1. (i) Ambiguous (ii) True (iii) True (iv) Ambiguous (v) Ambiguous
2. (i) True (ii) True (iii) False (iv) True (v) True
3. Only (ii) is true.
4. (i) If $a > 0$ and $a^2 > b^2$, then $a > b$.
(ii) If $xy \geq 0$ and $x^2 = y^2$, then $x = y$.
(iii) If $(x + y)^2 = x^2 + y^2$ and $y \neq 0$, then $x = 0$.
(iv) The diagonals of a parallelogram bisect each other.

EXERCISE A1.2

1. A is mortal
2. ab is rational
3. Decimal expansion of $\sqrt{17}$ is non-terminating non-recurring.
4. $y = 7$
5. $\angle A = 100^\circ$, $\angle C = 100^\circ$, $\angle D = 180^\circ$
6. PQRS is a rectangle.
7. Yes, because of the premise. No, because $\sqrt{3721} = 61$ which is not irrational. Since the premise was wrong, the conclusion is false.

EXERCISE A1.3

1. Take two consecutive odd numbers as $2n + 1$ and $2n + 3$ for some integer n .

EXERCISE A1.4

1. (i) Man is not mortal.
(ii) Line l is not parallel to line m .
(iii) The chapter does not have many exercises.
(iv) Not all integers are rational numbers.
(v) All prime numbers are not odd.
(vi) Some students are lazy.
(vii) All cats are black.
(viii) There is at least one real number x , such that $\sqrt{x} = -1$.
(ix) 2 does not divide the positive integer a .
(x) Integers a and b are not coprime.
2. (i) Yes (ii) No (iii) No (iv) No (v) Yes

EXERCISE A1.5

1. (i) If Sharan sweats a lot, then it is hot in Tokyo.
(ii) If Shalini's stomach grumbles, then she is hungry.
(iii) If Jaswant can get a degree, then she has a scholarship.
(iv) If a plant is alive, then it has flowers.
(v) If an animal has a tail, then it is a cat.
2. (i) If the base angles of triangle ABC are equal, then it is isosceles. True.
(ii) If the square of an integer is odd, then the integer is odd. True.
(iii) If $x = 1$, then $x^2 = 1$. True.
(iv) If AC and BD bisect each other, then ABCD is a parallelogram. True.
(v) If $a + (b + c) = (a + b) + c$, then a , b and c are whole numbers. False.
(vi) If $x + y$ is an even number, then x and y are odd. False.
(vii) If a parallelogram is a rectangle, its vertices lie on a circle. True.

EXERCISE A1.6

1. Suppose to the contrary $b \leq d$.
3. See Example 10 of Chapter 8.
6. See Theorem 2.1 of Class IX Mathematics Textbook.

EXERCISE A2.2

1. (i) $\frac{1}{5}$ (ii) 160
2. Take 1 cm² area and count the number of dots in it. Total number of trees will be the product of this number and the area (in cm²).
3. Rate of interest in instalment scheme is 17.74 %, which is less than 18 %.

EXERCISE A2.3

1. Students find their own answers.
